

A Beginner's Guide to Astrophysics and Cosmology

What the universe is, how we know, and why any of it bends light

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Contents

1. The scale ladder	1
2. Stars, and why old means red	9
3. Galaxies, and which ones bend light	16
4. Clusters and the cosmic web	23
5. Light is the only messenger	30
6. Telescopes, surveys, and the three observatories	36
7. Spectra — the fingerprint of everything	43
8. Redshift — what it is and why it means distance	48
9. The distance ladder	56
10. The expanding universe	63
11. The Big Bang and the oldest light	70
12. Dark matter	77
13. Dark energy and how it ends	85
14. The Hubble tension, and why lensing arbitrates it	92
15. Gravity bends light	99
16. What a strong lens is, and where to go next	105

1. The scale ladder

This chapter buys you a ruler. By the end of it you will carry two numbers — the Milky Way holds roughly 10^{11} stars and is roughly 10^5 light-years across — and every mass, size and distance in the rest of this book gets quoted against them at least once. That is not decoration. The main guide reproduces a galaxy cluster's mass to four significant figures and cannot tell you whether that is a lot, because the Milky Way appears nowhere in it. The ladder below is the fix: one rung at a time, from the ground under you to the edge of what light has had time to reach, with the units changing underfoot because no single unit survives the climb.

What you can skip

You own logarithms, orders of magnitude and log axes; nothing here explains a log scale, it only uses one. If you have read [Ch. 9 of the main guide](#) you also own the parsec's *arithmetic*, which it derives from one triangle and the small-angle approximation. What that derivation withholds, and this chapter supplies, is the ladder the parsec is a rung of — and the fact that a parsec is not a large distance. The one section to read slowly is [Our galaxy](#); everything later refers back to it.

Powers of ten

The physics in this chapter is nothing. The difficulty is entirely that the numbers span more decades than intuition carries, and intuition does not degrade gracefully at the edge of its range — it fails silently and returns a confident answer that is wrong by orders of magnitude.

So the ladder gets climbed rung by rung, each one quoted as a ratio to the one below it, with the unit swapped whenever the digits get out of hand. Astronomy’s pile of distance units is not historical clutter: each exists because the previous one produced numbers nobody could hold, and the swap points are the rungs of this ladder.

You already know this

The unit ladder is exponent normalisation. km, AU, ly, pc and Mpc are not five physical ideas; they are five choices of exponent, each defined from its neighbours by a fixed constant, picked so the mantissa stays short. Astronomy renormalises its units for the same reason a floating-point format carries a separate exponent field: the alternative is dragging a long tail of zeros through every statement, most of them digits you do not know. When a paper quotes a distance in Mpc, read it as scientific notation with the exponent hidden in the unit name.

The solar system, and where the AU runs out

Start with the Earth: the only rung you have direct physical access to, and the last one you can picture honestly — a ball you could walk around. The next rung up is the Earth–Sun distance, which astronomy names the **astronomical unit** (AU):

$$1 \text{ AU} \approx 1.496 \times 10^8 \text{ km},$$

the mean radius of the Earth’s orbit. This is the first rung where distance and time stop being separate statements. Light covers that gap in

$$\frac{1 \text{ AU}}{c} \approx 8.32 \text{ minutes},$$

so the Sun you observe is always the Sun as it was eight minutes ago. [Ch. 5](#) runs that fact out to its limit.

The AU is the right unit for the solar system and for nothing else. The planets all sit within a modest multiple of it — Neptune, the outermost, is the last rung on Figure 1.1 inside our own system. Then the planets stop, and there is a gap, and the gap is the point of this section.

The nearest star to the Sun is Proxima Centauri, at

$$4.24 \text{ light-years} = 268,142 \text{ AU}.$$

Sit with the ratio, not the two numbers. The distance to the *nearest other star* is a quarter of a million times the distance from here to the Sun. The solar system is not a scaled-down galaxy; it is a speck with an enormous amount of nothing around it. That emptiness is why two galaxies almost never line up well enough to make a lens ([Ch. 16](#)).

The AU has run out. Expressing interstellar distances in AU is exactly the failure mode the previous section named: correct, useless, and six digits long.

The light-year and the parsec

The next unit up is the **light-year**: the distance light covers in one year,

$$1 \text{ light-year} \approx 9.461 \times 10^{12} \text{ km.}$$

It is a distance, not a duration, despite the name — the most common misreading of the unit, and worth killing on sight. But the name is not an accident: since the unit is defined *from* a travel time, an object’s distance in light-years and the age of its arriving light are numerically the same. One number does two jobs, and [Ch. 5](#) is where that stops being a convenience and becomes the foundation of every claim in the book.

Now the awkward part. Professional astronomy does not publish in light-years. It publishes in **parsecs**, and

$$1 \text{ pc} \approx 3.2616 \text{ light-years.}$$

Both units survive because they are built from different things. The light-year is built from a physical constant, the speed of light. The parsec is built from an *instrument*: the distance at which one AU subtends one arcsecond, which is exactly the geometry of parallax — the only distance measurement that requires no astrophysical assumptions at all, using the Earth’s orbit as a baseline. [Ch. 9 of the main guide](#) derives the parsec from that triangle in two lines and reproduces astropy’s constant to the last bit, but never mentions parallax and never says what the unit is *for*. [Ch. 9 of this book](#) supplies the measurement the guide’s triangle is quietly a picture of.

Read one thing off Figure 1.1 here: the dot for “1 parsec” and the dot for the nearest star sit almost on top of each other. Nothing in the parsec’s definition knows about Proxima Centauri, so that is a coincidence — and it is the coincidence the distance ladder rests on. “Stars are spaced about a parsec apart” is the same statement as “the nearest stars have parallaxes of order an arcsecond”, which is the same statement as “the measurement is barely achievable with nineteenth-century optics”. It was, barely. Had the geometry landed the other side of that, the ladder would have had no first rung.

Above the parsec the ladder only adds prefixes: kiloparsec inside a galaxy, **megaparsec** between galaxies, gigaparsec for the cosmological scale. The main guide names the AU, the parsec and the Mpc, but only ever as definitions to convert with; it never attaches a distance to any of them. That is why you can read all 29 of its chapters and still not know whether a megaparsec is far.

Our galaxy — the ruler

The Sun is one star in the **Milky Way**, a barred spiral galaxy: a flattened rotating disk of stars and gas with a central bulge, seen edge-on from inside, which is why it looks like a band across the sky rather than a pinwheel. The Sun sits in the disk, roughly halfway out — a bad seat, since half the galaxy is behind dust from here.

Two numbers. This is the ruler, and it is built to be memorable.

$$N_{\star} \approx 10^{11} \text{ stars}, \quad D \approx 10^5 \text{ light-years across.}$$

A hundred billion stars, a hundred thousand light-years wide. One significant figure each, both an exact power of ten, which is exactly the precision this book needs from them. Light crosses the galaxy in

$$t_{\text{cross}} = 10^5 \text{ years,}$$

the same number as the diameter, because a light-year is a light-year. The reflex worth keeping: any picture of the far side of the Milky Way is a hundred thousand years out of date, and no instrument will fix that.

Both numbers are softer than the notation suggests, and pretending otherwise would be a simplification you would later have to unlearn. Nobody has counted the stars. 10^{11} is an estimate: measure the galaxy’s mass dynamically, subtract your best guess at the gas and the dark matter (Ch. 12), divide by an average stellar mass drawn from a fitted distribution. The published range spans a factor of a few, and that spread is a modelling choice, not a counting error. The diameter is softer still: the stellar disk has no edge, it fades, so “ 10^5 light-years across” encodes a convention about where you stop caring, with roughly a factor of two of slack. And the *matter* reaches much further than the light does — the dark halo is several times the visible disk, which is Ch. 12’s whole subject.

None of that damages the ruler, because a ruler has to be *stable and shared*, not accurate. A cluster mass quoted in Milky Ways carries exactly the precision of the comparison and no more — and it is still far more informative than the same mass quoted in bare solar masses, which is what the main guide does.

Beyond the galaxy

The Milky Way is not alone, and the pattern from the last rung repeats: things come in bound groups, separated by far more emptiness than the groups occupy.

The **Local Group** is our galaxy’s own neighbourhood: the Milky Way, Andromeda (the nearest large galaxy, and visible to an unaided eye from a dark northern site), and several dozen much smaller dwarf galaxies orbiting the two of them. Ch. 5 puts a number on how far Andromeda is, and therefore how old its photograph is; for the ladder it is enough that the Local Group sits a couple of decades above the ruler.

Above that are **clusters** — bound collections of galaxies, far more crowded than the Local Group and far heavier. This is the rung the main guide does much of its work on without ever introducing it, so Ch. 4 takes it whole rather than a paragraph here. What matters for the ladder is that a cluster sits *above* a galaxy and is not a synonym for one. The main guide quotes an Einstein radius for a galaxy and one for a cluster, chapters apart, and they differ for precisely that reason — a fact no sentence in it states and Ch. 4 exists to state.

Above clusters is the **cosmic web**: filaments of galaxies strung between dense knots, wrapped around mostly empty voids. This is not an artist’s impression or a simulation passed off as an observation — the web is measured, by taking redshifts for enormous numbers of galaxies and turning each into a distance (Ch. 8’s job). Its largest structures are the largest structures there are.

Past the web’s own scale the universe stops having features and becomes, statistically, the same everywhere.

The whole thing

The top rung needs a distinction most accounts blur, so take it first: the **observable** universe and the universe are different claims. The observable universe is the region light has had time to reach us from since the beginning (Ch. 11); it is bounded because the universe has a finite age, not because space stops. What lies past that boundary, and whether the universe is finite at all, is not known and probably not knowable. This book will not pretend otherwise.

For the size of the observable part, this program’s own cosmology hands you a number: the **Hubble distance**, c/H_0 — the speed of light over the expansion rate (Ch. 10). Using the H_0 this repository asserts throughout, and which Ch. 14 will show is actively disputed:

$$D_H = \frac{c}{H_0} \approx 4283 \text{ Mpc.}$$

Be careful with what that is. c/H_0 is a characteristic scale for the observable universe, not its radius. Light emitted at the beginning has been travelling through *expanding* space the whole time, so whatever emitted it is now considerably further away than c times the travel time; the true comoving radius is several times D_H . Ch. 15 of the main guide is where that becomes an integral and three rival definitions of “distance”; Ch. 10 of this book is where it becomes a picture. Quote D_H as the scale of the thing, not the edge of it.

Now convert it into the only unit you now have a feel for:

$$\frac{D_H}{D_{\text{Milky Way}}} = 139,685 \text{ Milky Ways.}$$

That is the ladder’s punchline: the observable universe, measured in galaxies laid end to end, is of order 10^5 of them — the same exponent as the Milky Way’s diameter in light-years. The repetition is a coincidence of unit choice and means nothing physical, but it is a free mnemonic to carry.

Unlocks

What you can now read

- **Ch. 9 — Arcseconds, magnitudes, and the units of the sky** opens with the right sentence — “*Nothing astronomical comes with a ruler. Every telescope measures a direction, not a distance*” — and then derives the parsec from “*the distance at which one astronomical unit — Earth’s orbital radius, roughly 1.496×10^8 km — subtends an angle of exactly $1''$.*” That is the whole treatment. It never says why anyone would build a unit out of that triangle (the answer is parallax, a word absent from all of the main guide), and it never says how far a parsec gets you. You now have both: the AU as a physical gap light crosses in 8.32 minutes, and the parsec as 3.2616 light-years — a distance that does not reach the nearest star.
- **The same chapter’s closing move** — “*You need the parsec to read an abstract, not to run this repo’s own fits*” — now reads as what it is: the guide stepping over the ladder because its likelihood is “arcsec in, arcsec out” and genuinely does not need it. A fair

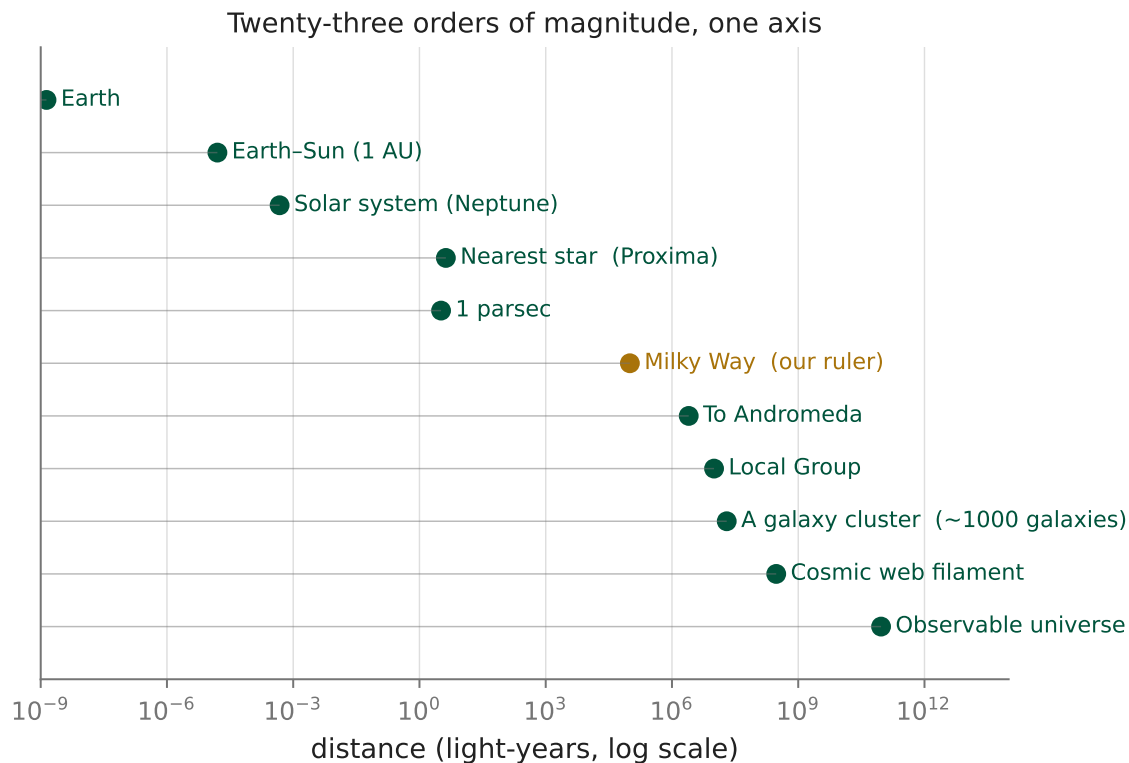


Figure 1: The whole ladder on one axis: distance in light-years, log scale, from the Earth’s own diameter to the observable universe. Labelled ticks step three decades at a time, so a rung a finger-width from its neighbour is a thousandfold away from it. The highlighted rung is the Milky Way, marked *our ruler*, and it is not centred: about fourteen decades of ladder run below it and six above. Read the gaps rather than the dots: “Nearest star” is a quarter of a million AU from “Earth–Sun”, and “1 parsec” lands almost on top of it, which is why parallax was measurable at all.

engineering call, and a poor way to learn what the numbers mean. When its Part III starts quoting Mpc, you know 4283 Mpc is roughly the scale of everything observable.

Exercises

Exercise 1.1 — where the AU runs out

Using $1 \text{ light-year} \approx 9.461 \times 10^{12} \text{ km}$ and $1 \text{ AU} \approx 1.496 \times 10^8 \text{ km}$, convert Proxima Centauri’s distance of 4.24 light-years into AU. Then answer the question the number is actually about: is the solar system a reasonable scale model of the galaxy?

Solution

One multiplication:

$$4.24 \times \frac{9.461 \times 10^{12}}{1.496 \times 10^8} \approx 268,142 \text{ AU}.$$

No. A scale model of the galaxy built from the solar system would put the next star a quarter of a million Earth–Sun distances away, and then 10^{11} of them at comparable spacing. Every diagram of “the Sun’s neighbourhood” compresses that gap by orders of magnitude, because the honest version is a blank page — which is why lensing alignments are rare (Ch. 16).

Exercise 1.2 — the galaxy is old news

The Milky Way is about 10^5 light-years across. How long does light take to cross it? Why is that answer the same number as the diameter, and what does it imply about any image of the far side of the galaxy?

Solution

10^5 years. It is the same number as the diameter in light-years by construction, not by coincidence: the light-year is *defined* as the distance light covers in a year, so distance-in-light-years and travel-time-in-years are one quantity read in two units. So a photograph of the far side of our own galaxy shows it as it was 10^5 years ago, and no instrument can change that — the delay is a property of the distance, not the telescope (Ch. 5).

Exercise 1.3 — the universe, in Milky Ways

Take the Hubble distance $D_H = c/H_0 \approx 4283 \text{ Mpc}$, with $1 \text{ pc} \approx 3.2616 \text{ light-years}$ and a Milky Way 10^5 light-years across. How many Milky Ways across is D_H ? Then state, in one sentence, why this is *not* the radius of the observable universe.

Solution

Convert Mpc to pc, pc to light-years, then divide by the ruler:

$$\frac{4283 \times 10^6 \times 3.2616}{10^5} \approx 139,685 \text{ Milky Ways.}$$

c/H_0 is a characteristic scale, not a radius: space expanded while the light was in transit, so whatever emitted the oldest light we receive is now several times D_H away, and “distance” itself splits into several inequivalent definitions ([Ch. 15 of the main guide](#)). Note what the answer does *not* hinge on: the two best measurements of H_0 disagree ([Ch. 14](#)) by a few percent, which moves this number by a few percent and the exponent not at all.

Exercise 1.4 — why two units for one distance

A paper reports a distance in Mpc; a press release reports the same distance in light-years, and $1 \text{ pc} \approx 3.2616 \text{ light-years}$ converts between them. Why does the field carry both?

Solution

Each is built from something different: the light-year from a physical constant, reporting a distance and the age of the light in one number ([Ex. 1.2](#)); the parsec from an *instrument*, being the distance at which 1 AU subtends $1''$. A field publishes in the unit its measurement is natively in, so the parsec wins in papers and the light-year wins in prose.

2. Stars, and why old means red

The main guide describes every lens galaxy it touches, in one clause, as having “an old and red stellar population,” and never returns to it. Chapter 12 then builds a measurement on that unexplained premise, reading the lens’s redshift off a calcium line and the source’s off an oxygen line: two galaxies, one sightline, two atomic transitions, and the reason the choice works is nowhere in the book. It comes down to a single exponent, derived here in one line of algebra.

What you can skip

You do not need nuclear physics: no cross-sections, no reaction networks, no Gamow peak. Take “four hydrogen nuclei fuse into one helium nucleus and release energy” as given. You do not need the equations of stellar structure either — this chapter states the one result it needs from them, the mass–luminosity relation, as an empirical power law. If you already know why the main sequence is a locus and not a track, skip to [Why massive means short](#).

What a star is

A star is a ball of plasma massive enough that its own gravity would crush it, held up by the pressure of its own heat. The tension *is* the star: at every radius, the weight of everything above presses down and the pressure of the gas below pushes back, exactly hard enough to cancel it. Astronomers call that balance **hydrostatic equilibrium**. Writing it down takes calculus, so this book does not: [Ch. 10](#) of the main guide has the equation. Its content is the sentence above — pressure falls off outward at whatever rate the enclosed weight demands — and Ch. 10 reuses that same statement for an entire galaxy, with the random motion of stars playing the role gas pressure plays inside a star. Same balance both places; only the source of the pressure changes.

Pressure requires heat, and heat leaks out — a star’s luminosity *is* heat escaping. So the balance is not free: something must resupply the heat, or the star cools, the pressure falls, and gravity wins. The resupply is **fusion**. In the core, at temperatures of order ten million kelvin, hydrogen nuclei collide hard enough to overcome their electrostatic repulsion and stick; four end up as one helium nucleus weighing slightly less than the four did, and the missing mass leaves as energy at the rate $E = mc^2$. That rate is favourable enough that a small mass deficit funds a very long time — the only reason “how long does a star live?” has an interesting answer.

You already know this

A main-sequence star is a negative-feedback loop, stable for the reason any such loop is. Perturb the fusion rate upward: the core heats, hot gas expands, expansion cools it, and the fusion rate — steeply temperature-dependent — falls back. Gravity sets the gain. Worth stating rather than waving at, because the loop can be **broken**, and the break is the plot of the [last section](#): in matter whose pressure does not depend on temperature, heating no longer causes expansion, the negative feedback term vanishes, and the loop runs away. Stars do not merely run out of fuel. They die of control failures.

The main sequence

Plot a few thousand stars with luminosity on one axis and surface temperature on the other, and the great majority fall along a single diagonal band: the **main sequence**. Be precise about what that band is, because the name invites a wrong reading. It is not a track a star travels along. It

is a *locus* — the set of points occupied, at any instant, by stars fusing hydrogen in their cores. A star arrives at one point, sits essentially still for nearly all of its life, then leaves the band entirely when the core hydrogen runs out. Nothing slides down it. The band is crowded for that reason: it is where a star spends almost all its time, so it is where a star is whenever you happen to look.

What sets the point is mass, and to a good approximation only mass. Hand the equations of stellar structure a birth mass and they return luminosity, radius, surface temperature, lifetime and eventual fate. The one relation this chapter needs is the empirical **mass–luminosity relation**:

$$\frac{L}{L_{\odot}} \approx \left(\frac{M}{M_{\odot}} \right)^{3.5}.$$

The exponent is superlinear for a mechanical reason. More mass means more weight bearing down on the core, so the core must run at higher pressure and temperature to hold it up — and the fusion rate is a very steep function of temperature. A modest increase in mass buys a large increase in burn rate. The star pays for its extra fuel by consuming it out of all proportion.

Be honest about the 3.5: it is a mid-range fit, not a law — nearer 4 below a solar mass, flattening toward 3 and below at the top end where radiation pressure starts to matter. This chapter uses it because the argument it supports is an order-of-magnitude argument, and [Exercise 2.1](#) shows that argument gets *stronger*, not weaker, under the accurate exponents.

Now colour. Surface temperature sets a star’s colour through Wien’s law ([Ch. 5](#)): hot peaks blue, cool peaks red. Massive stars are hot at the surface as well as the core, so the main sequence runs from hot, luminous, blue stars at the massive end to cool, faint, red ones at the low-mass end. **Blue and massive and luminous** are one thing; **red and low-mass and faint** are another. The rest of this chapter is that fact combined with a lifetime.

Why massive means short

A star’s main-sequence lifetime is the time it takes to burn its fuel:

$$t \sim \frac{\text{fuel}}{\text{burn rate}}.$$

The fuel is the hydrogen the star can get into its core, a roughly fixed fraction of its total mass, so $\text{fuel} \propto M$. The burn rate is the luminosity, because the luminosity *is* the rate at which fusion energy leaves. So $\text{burn rate} \propto M^{3.5}$, and

$$t \propto \frac{M}{M^{3.5}} = M^{-2.5}.$$

That is the whole derivation. That exponent carries this book’s entire account of galaxy colour, and two chapters of the main guide that never mention it. Calibrate on the one star whose lifetime we have an independent handle on — the Sun’s main-sequence lifetime is about 10 Gyr:

$$t(M) \approx 10 \text{ Gyr} \times \left(\frac{M}{M_{\odot}} \right)^{-2.5}.$$

Measure the results against the age of the universe, which this book quotes as its *own* cosmology returns it — the FlatLambdaCDM(70, 0.3) every campaign in this repository asserts — 13.47 Gyr. That is not the figure you will find in a textbook; the gap is real, is not a rounding error, and Ch. 11 explains it rather than smoothing it over. Figure 2.1 draws its line from the identical number.

A star of 10 $M_{\odot}$. Ten times the Sun’s mass, so $10^{-2.5}$ times its lifetime:

$$t = 10 \text{ Gyr} \times 10^{-2.5} = 0.0316 \text{ Gyr} = 31.62 \text{ Myr}.$$

That is a fraction 0.00316 of the Sun’s life, and 0.00235 of cosmic history — two-tenths of one percent. Read it as a statement about evidence rather than about stars: if you are looking at a 10 $M_{\odot}$ star, it was made essentially *now*. Blue light from a galaxy is not a fact about its past; it is a report on its present.

A star of 0.5 $M_{\odot}$. Half the Sun’s mass, so $0.5^{-2.5}$ times its lifetime:

$$t = 10 \text{ Gyr} \times 0.5^{-2.5} = 56.57 \text{ Gyr},$$

which is 4.2 times the age of the universe. Read that one plainly: **no half-solar-mass star has ever died**. Not one, anywhere — the universe has not run long enough for the first one to finish. Every low-mass star ever formed, and they are the overwhelming majority of the Milky Way’s 10^{11} (Ch. 1), is on the main sequence right now, still burning, still red.

Treat 56.57 Gyr as a lower bound rather than a measurement. Two independent corrections both push it upward, and Exercise 2.1 is where you work out which. The conclusion does not depend on the number being right — only on its being enormous, and both corrections make it more so.

You already know this

A lens galaxy is never resolved into stars. What a spectrograph gets is one integrated spectrum — Ch. 12 calls it “the light of a few hundred billion stellar photospheres, each close to a blackbody” — a mixture weighted by each component’s luminosity. Notice what $L \propto M^{3.5}$ and $t \propto M^{-2.5}$ do to that mixture *jointly*: the massive components are wildly over-weighted, being the luminous ones, and wildly short-lived. A population’s integrated colour is dominated by exactly the components that disappear first. Stop feeding it new massive components and it does not fade uniformly — it **reddens**, fast, because the bluest and brightest terms drop out of the sum while the red ones stay, and will stay for longer than the universe has existed. That is the entire content of “old and red.”

This is also, directly, the main guide’s line selection. The [OII] doublet is an emission feature of ionized gas, and the only thing that ionizes gas on a galactic scale is ultraviolet from hot, massive, blue stars — the ones that last 31.62 Myr. So [OII] never says a galaxy *has* stars; it says a galaxy made stars within the last few tens of Myr. Ca II K is an absorption feature of calcium in cool stellar atmospheres, which is what an old population is made of. Chapter 12 asserts that split in two parentheses and moves on; the reason it holds is one exponent.

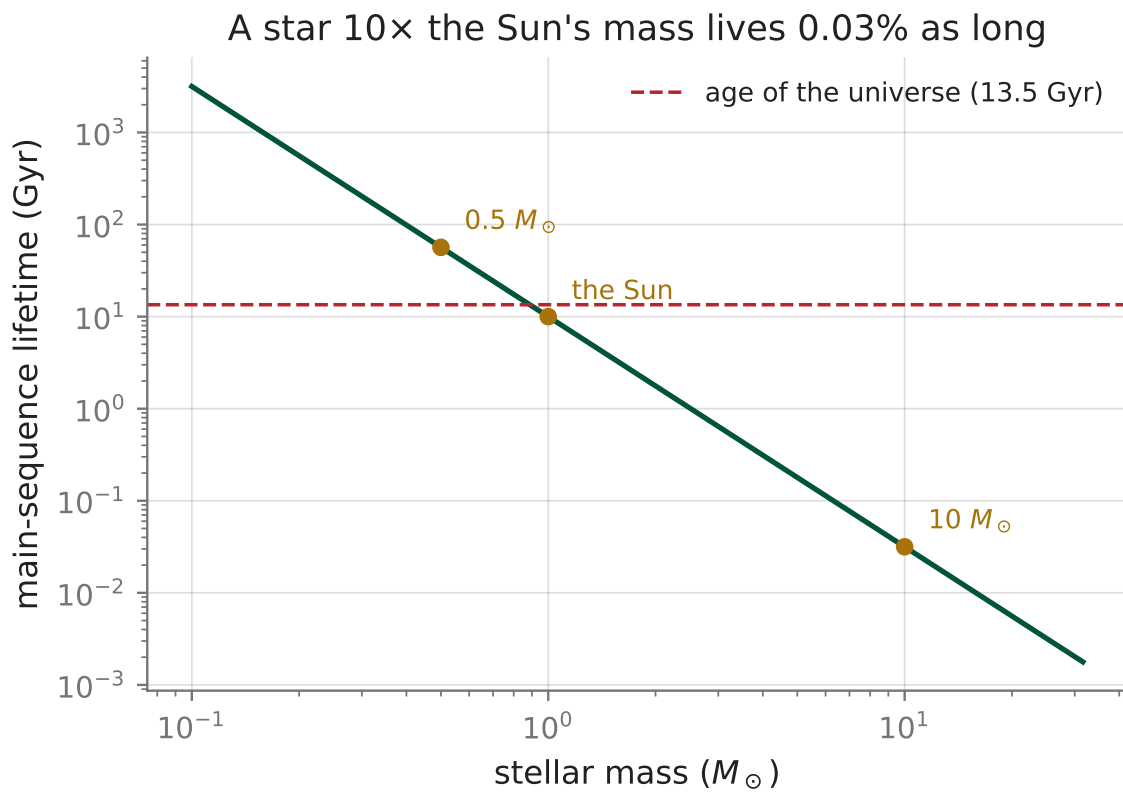


Figure 2: Main-sequence lifetime against birth mass, both axes logarithmic, so the power law $t = 10 \text{ Gyr} (M/M_{\odot})^{-2.5}$ plots as the straight line its exponent promises. The dashed horizontal line is the age of the universe on this book's own cosmology. Everything *above* that line has never happened: a star whose lifetime plots above it cannot have finished, whenever it formed. The Sun sits a little below the line; $0.5 M_{\odot}$ sits well above it; $10 M_{\odot}$ sits two and a half decades below the Sun, which is the exponent 2.5 read straight off the axes.

What is left behind

When the core hydrogen is gone, the loop from the first section loses its heat source. The core contracts, the outer layers expand and cool, and the star leaves the main sequence to become a giant: large, cool, and red for a reason that has nothing to do with being low-mass.

Flag that, because this chapter’s title is a trap taken too literally. An individual star is red either because it is small or because it is evolved — opposite ends of a life. It is *populations* that are red because they are old. A red dwarf is not an old star; it is a low-mass one.

What happens after the giant phase forks on birth mass, at roughly $8 M_{\odot}$.

Below the threshold. The star sheds its envelope and leaves its bare core behind: a **white dwarf**, fusing nothing, only cooling. A typical one carries about $0.6 M_{\odot}$ inside a radius of roughly 6371 km — the Earth’s radius, which is the point of quoting it that way — for a density of 1.10 tonne/cm^3 : a sugar cube of the stuff outweighs a person. Nothing holds it up but electron degeneracy pressure — a quantum-mechanical refusal of electrons to share a state — and that pressure does not depend on temperature. The thermostat is gone, which is why there is a ceiling: above the **Chandrasekhar mass**, $1.4 M_{\odot}$, degeneracy pressure cannot carry the weight and the object cannot exist as a white dwarf. That ceiling is what makes Type Ia supernovae *usable* as standard candles — an explosion triggered at a fixed mass should release a fixed energy — and so a rung of the distance ladder ([Ch. 9](#)) and the discovery of dark energy ([Ch. 13](#)). “Usable” is doing work there: real Type Ia’s are not identical, only alike enough that the residual spread calibrates out of the light-curve shape, and what tips a white dwarf over the ceiling — accretion from a companion, or a merger — is unsettled. [Ch. 9](#) inherits both; this chapter needs only the ceiling.

Above the threshold. The star fuses its way up the periodic table to iron, which yields no energy when fused. The heat source stops abruptly, the core collapses, and the rest is blown off as a supernova. What remains is a **neutron star** — $1.4 M_{\odot}$ inside a radius of 11 km, a sphere you could drive across — or, above some mass, a black hole. Its mean density is $4.99 \times 10^8 \text{ tonne/cm}^3$, which is 4.53×10^8 times the white dwarf’s. Both densities are computed from the masses and radii above, which are textbook benchmarks rather than repo measurements — [Exercise 2.2](#) is where that distinction earns its keep.

None of these remnants shines the way a main-sequence star does, and none makes [OII]. A galaxy that stopped forming stars long ago is left with exactly this: low-mass stars still on the main sequence, the giants some have become, and a growing population of dark remnants. Old, red, and — for a lens model — massive without being especially bright. [Ch. 3](#) takes up the question this chapter leaves open: what stops a galaxy making new blue stars at all.

Unlocks

What you can now read

- [Ch. 10 — Galaxies, Sersic profiles, and velocity dispersion](#) characterises every lens in the repository in one sentence: an elliptical’s light is “smooth, centrally concentrated, no spiral arms, no ongoing star formation, an old and red stellar population.” Five clauses presented as one observation — but “no ongoing star formation” is the *cause* and “old and red” is its *consequence*, and the mechanism joining them is this chapter’s $t \propto M^{-2.5}$: cut off the supply of new massive stars and the blue light drains out of the population within tens of Myr while the red light stays for longer than the universe has

existed. Ch. 10 spends its effort on the *dynamical* half of that story — disordered orbits, the Jeans equation, the isothermal conspiracy — and never says why the colour comes along with it.

- **Ch. 12 — Redshifts and what a spectrum tells you** measures two redshifts down one sightline using two different lines, and justifies the choice entirely inside two parentheses: Ca II K because “its depth traces old stars, so it shows up in a quiescent lens galaxy, not a star-forming background source,” and the [OII] doublet because it is “present in a star-forming galaxy, essentially absent in an old elliptical.” That is the whole argument, asserted twice and never derived. You now have the mechanism under it, and the line selection reads as what it is: the $M^{-2.5}$ law used as an instrument.

Exercises

Exercise 2.1 — How much does the 3.5 matter?

The mass–luminosity exponent is not really 3.5; below a solar mass it is closer to 4. Redo the $0.5 M_{\odot}$ lifetime with $L \propto M^4$ — you need the direction, not the arithmetic — and say whether the chapter’s conclusion survives. Then name the second, independent reason the 56.57 Gyr figure is too small.

Solution

With $L \propto M^4$, $t \propto M/L = M^{-3}$. For $M < M_{\odot}$ the base is below 1, so a more negative power makes the result *larger*: the lifetime goes up. The conclusion was already safe at 4.2 times the age of the universe, and gets safer.

The second reason is the fuel step, not the burn step. Fuel $\propto M$ counts only the core’s hydrogen; a fully convective $0.5 M_{\odot}$ star circulates essentially all of its hydrogen through the burning region, so its usable fuel fraction beats the Sun’s. Both corrections carry the same sign, which is the point worth keeping: the value of an order-of-magnitude estimate is not that it is accurate, it is that you can bound which way it is wrong.

Exercise 2.2 — A sugar cube of each

Using $\rho = M/(\frac{4}{3}\pi R^3)$ and $M_{\odot} = 1.989 \times 10^{30}$ kg, compute the mean density in tonnes per cm^3 of a $0.6 M_{\odot}$ white dwarf of radius 6371 km and a $1.4 M_{\odot}$ neutron star of radius 11 km, and take the ratio. Then: which of those four inputs does this repository actually stand behind?

Solution

White dwarf: $0.6 \times 1.989 \times 10^{30}$ kg over $\frac{4}{3}\pi(6.371 \times 10^8 \text{ cm})^3$ gives 1.10 tonne/ cm^3 . Neutron star: $1.4 \times 1.989 \times 10^{30}$ kg over $\frac{4}{3}\pi(1.1 \times 10^6 \text{ cm})^3$ gives 4.99×10^8 tonne/ cm^3 , a ratio of 4.53×10^8 .

None of the four inputs is repo-derived — they are textbook benchmarks, and the white dwarf’s “6371 km” is Earth’s radius standing in for “about Earth-sized.” Only the arithmetic on top of them is computed, so only the arithmetic is checked. That is the habit worth taking from this exercise: a number that reproduces is not the same as a number that is measured, and a build that verifies the first says nothing about the second.

Exercise 2.3 — Which one is the lens?

Two spectra come from the same arcsecond of sky. Spectrum A shows a strong [OII] emission doublet and weak calcium absorption. Spectrum B shows deep Ca II H & K absorption and essentially no [OII]. Which is the better candidate for the foreground lens galaxy? What timescale is spectrum A actually reporting on? And what would you need before concluding anything about which galaxy is in front?

Solution

B is the lens candidate: old population, no massive stars, therefore no ionizing ultraviolet, therefore no star formation for a long time — a quiescent elliptical, the class of galaxy massive enough to lens at all, the [deepest potential wells around](#). A's [OII] is present-tense: it reports on the last few tens of Myr specifically, because that is how long the stars supplying the ionizing photons last — 31.62 Myr for a $10 M_{\odot}$ star. It is not a history of the galaxy.

The trap is the last question. Colour and line type are not geometry. Nothing in either spectrum says which galaxy is *in front* — that is an inference from a prior about which galaxies lens, not a measurement. What settles it is the redshifts, one wavelength ratio per spectrum, and the fact that the higher- z object is the more distant one. That chain is [Ch. 8](#), the one the main guide never closes.

3. Galaxies, and which ones bend light

Chapter 10 of the main guide opens with the best one-line statement of what this program is: “Every lens in this repository is a galaxy first and a set of equations second.” It then spends the rest of the chapter on the equations, because it is entitled to assume you already hold the galaxy. This chapter is the galaxy. By the end you will know what one is, why the two families in any survey image look as different as they do, why one of them stopped making stars and went red, and why that same family — massive ellipticals — is what every campaign here points a telescope at. Chapter 2 gave you a star. A galaxy is 10^{11} of them, some gas, and one further ingredient that outweighs all of it.

What you can skip

You do not need galaxy-formation simulations, orbit integration, or the taxonomy of Hubble’s tuning fork. You need no stellar dynamics either, and this chapter gives you none: the *dynamical* distinction between the two families — ordered rotation versus disordered orbits — is handed off rather than explained, because Ch. 10 does it in its opening paragraph and better than a restatement would. What is left is the part Ch. 10 assumes and never states: what these objects are, what they look like, and why one kind is red.

What a galaxy is

A galaxy is a gravitationally bound system of stars, gas, dust and dark matter, separated from the next by a gap far larger than itself. That is the whole definition. Nothing enforces it and nothing maintains it; it is what a large enough overdensity of matter settles into, given time.

Take the Milky Way as the ruler (Ch. 1): 10^{11} stars in a disk about 10^5 light-years across. Ch. 1 gives no thickness, so take a further 10^3 light-years on that axis — a round figure rather than a measurement, and nothing below leans on its second digit. Those three numbers answer the first question worth asking, which is how full it is. Model the disk as a uniform cylinder, radius $R = 5 \times 10^4$ ly, thickness $h = 10^3$ ly. Its volume is

$$V = \pi R^2 h \approx 2.26 \times 10^{11} \text{ pc}^3,$$

so spreading 10^{11} stars through it uniformly gives a stellar number density of

$$n = \frac{10^{11}}{2.26 \times 10^{11} \text{ pc}^3} \approx 0.442 \text{ stars pc}^{-3}.$$

Roughly one star per two cubic parsecs. Invert that for a typical spacing — $n^{-1/3}$, the side of the cube each star gets to itself — and you get 1.313 pc, about 2.7×10^5 AU. The Sun is 0.0093 AU across, so the gap between neighbours is around

$$\frac{2.7 \times 10^5 \text{ AU}}{0.0093 \text{ AU}} \approx 2.9 \times 10^7$$

times the diameter of the object doing the occupying. A galaxy is not a crowd. It is 10^{11} objects separated by thirty million of their own diameters.

That number needs an immediate correction, because the model behind it is crude in a known direction. A real disk is not uniform: it is centrally concentrated, and it thins with radius and with height above the midplane. The Sun sits well out in it, where the measured density is about 0.1 stars per cubic parsec, making the uniform-disk average above too high by a factor of 4.42. That is a real error, not a rounding difference. It does not touch the conclusion, for one reason worth stating: it runs in the conservative direction. The true local density is *lower*, so the true spacing is *larger*, and the galaxy is emptier than the calculation above claims rather than fuller. That factor is the entire warranty on this density — an order-of-magnitude statement about emptiness, nothing more. Do not carry it into anything needing a second significant figure.

Emptiness has one consequence, and [the section below](#) turns on it. When two galaxies pass through each other — and they do, routinely — essentially no two stars collide. Gas fills a far larger fraction of the volume, so two galaxies’ reservoirs do run into each other, shock and heat.

You already know this

A galaxy is 10^{11} discrete point sources, and the main guide models its light with one smooth analytic function of radius — the Sersic profile, [Ch. 10](#). That is legitimate for the reason a kernel density estimate stops looking like a histogram once the sample is large enough: at these distances no telescope resolves an individual star, every pixel integrates millions, and the discreteness sits orders of magnitude below the photon noise. The smooth profile is not a concession to tractability; at that sampling density it is what the data are. It breaks where a KDE breaks — in the sparse outskirts, where the count per pixel gets small.

Spirals and ellipticals

Point a telescope at enough galaxies and they sort, imperfectly, into two families. The split is visible before it is understood, so it is worth separating what you *see* from what it *means*.

A **spiral** is a flattened rotating disk with a central bulge, traced by arms winding out from the middle. It holds cold gas and dust, often in dark lanes along the arms. Its light is comparatively blue, concentrated in the arms. The Milky Way is one.

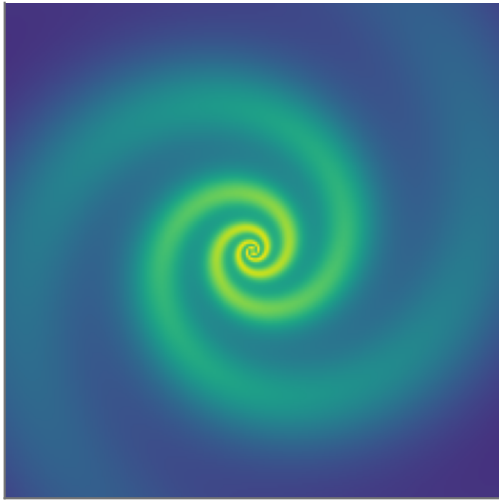
An **elliptical** is a smooth ellipse of light — no arms, no dust lanes, no disk, no visible structure at any radius, brightest in the middle and falling off monotonically outward. It is red, and it has little cold gas. [Ch. 10](#) describes exactly this and gives the reason in the same breath: “The light is correspondingly featureless — smooth, centrally concentrated, no spiral arms, no ongoing star formation, an old and red stellar population — because there is no organizing structure left to draw a pattern with.”

The colour is the informative part, and [Chapter 2](#) gave you why. Blue light means massive stars, and massive stars burn out fast enough that their presence is a statement in the present tense ([Ch. 2](#)): a galaxy shining blue is making stars *now*, because the blue ones cannot be old. Red means the massive stars are gone and only the long-lived low-mass ones remain — past tense, a galaxy that made its stars long ago and none since. Colour is a clock you can read off a single image. It is not a claim about composition; the elements are much the same either way.

Two honesty notes. First, the split is a continuum, not a partition: lenticulars sit between the two, irregulars sit off the diagram entirely, and mergers in progress belong to no class at all. “Spiral” and “elliptical” are the ends of a distribution, and the useful statements are about the ends. Second, morphology is what a telescope measures, and it is the *less* fundamental distinction. The one

Same class of object, two very different shapes

A spiral — disk, arms, still forming stars



An elliptical — smooth, compact, quiescent

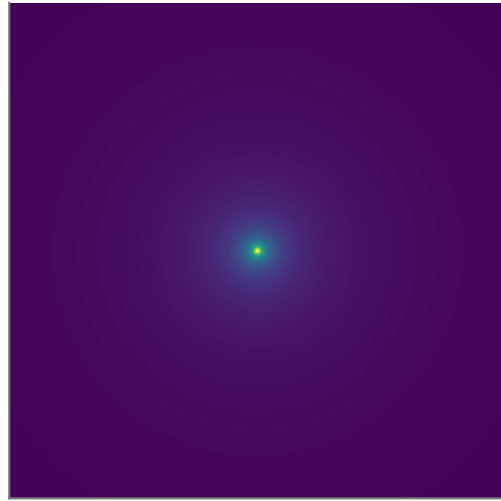


Figure 3: The visible split, drawn: an exponential disk traced by two logarithmic spiral arms (left) against a de Vaucouleurs profile — the Sersic index $n = 4$ of [Ch. 10](#) — with no structure at any radius (right). Both panels cover the same field of view, each normalized to its own peak under a square-root stretch, so the comparison to make is of *shape*, not brightness. Two things the figure does not say: the relative sizes are a choice of the illustration, not a law — real ellipticals span a wide range, and the most massive are far larger than a Milky Way-sized disk — and neither panel shows the dark matter, which is most of what is there and none of what is drawn.

that matters downstream — for the mass model, the Einstein radius, the whole apparatus — is dynamical: whether the system is held up against its own gravity by ordered rotation or by disordered motion. That is [Ch. 10’s opening section](#) — its first paragraph — and where you should read it rather than here.

Why star formation stops

Stars form out of cold, dense gas and out of nothing else. A galaxy with no cold gas makes no stars, and one that has run out fades from blue to red on the timescale its massive stars take to die — fast. “Elliptical,” “red,” “old stellar population” and “quiescent” are, to first order, four names for the same fact: the gas is gone.

Does the gas going away even need an explanation? No. The Milky Way’s cold gas reservoir is of order 10^{10} solar masses and it converts that gas into stars at roughly 2 solar masses per year. Divide:

$$t_{\text{depl}} = \frac{10^{10} M_{\odot}}{2 M_{\odot} \text{ yr}^{-1}} = 5 \times 10^9 \text{ yr} = 5 \text{ Gyr.}$$

Five billion years — less than the age of the universe ([Ch. 11](#)). Both inputs are order-of-magnitude benchmarks and the quotient inherits that, so read it as “a few Gyr,” not 5.0. The lesson survives the imprecision: a star-forming galaxy left alone shuts down within a time comparable to its own history, purely by consuming its supply. Star formation is not a steady state that needs breaking; it is a transient that needs *feeding*, and the Milky Way is still blue because gas keeps falling in to refill the tank.

So the question is not “why did ellipticals stop” but “why did their supply stop being replenished,” and the honest answer is that this is an open problem. Astronomers call the shutdown **quenching**. The candidate mechanisms are known, individually plausible, and collectively unsettled:

- **Mergers.** Two gas-rich spirals colliding destroy each other’s disks — each one’s ordered rotation scrambled into disordered orbits — and the gas, which unlike the stars *does* collide, is driven to the centre, burned in a violent burst of star formation, and largely used up or blown out. The remnant is a smooth, gas-poor, dispersion-supported blob: an elliptical. The emptiness argument licenses this picture — only because stars never hit each other can two galaxies interpenetrate and leave 10^{11} intact stars in a rearranged configuration.
- **Feedback.** An accreting supermassive black hole at the centre dumps enough energy into the surrounding gas to heat it or eject it outright.
- **Environment.** A galaxy falling into a cluster ([Ch. 4](#)) moves through hot intracluster gas that can strip its reservoir away, and its halo may be too hot for fresh gas to cool and settle in at all.

Which of these dominates, in which galaxies, at which epoch, is genuinely unsettled, and this book will not pretend otherwise. What *is* settled is the observation: massive galaxies are overwhelmingly red, quiescent and gas-poor, and have been for most of cosmic history. That observation is what the lens model consumes — the input, not the explanation.

Which ones lens

Now the payoff. Bending light into a visible arc takes mass, concentrated, in a small patch of sky. Ellipticals are characteristically the most massive galaxies in their environment — Ch. 10 calls them “the deepest potential wells around” and notes that their ubiquity among lenses “is close to a selection effect.” That is the sentence to hold. This repository does not study ellipticals because someone chose to, but because a galaxy that bends light detectably is, with high probability, an elliptical. The universe did the filtering before the sample was drawn.

The quantitative version is one line and it belongs to the main guide. An elliptical’s stars move on disordered orbits with a velocity spread σ_v — a standard deviation, measurable from the width of an absorption line in its spectrum — and the Einstein radius scales as $\theta_E \propto \sigma_v^2$. Quadratic, so the most massive ellipticals do not merely lens better than everything else; they dominate. Ch. 10 derives that relation from a single equilibrium condition and hands it to Ch. 19; Ch. 16 of this book draws the geometry it implies.

Three things follow, each of which the main guide uses without pausing:

- **The red, featureless light is a modelling asset.** A galaxy with no arms, no dust lanes and no star-forming knots has a surface brightness a smooth analytic profile describes well. That is why Ch. 10’s Sersic profile works at all, and why the campaign can subtract the lens’s own light cleanly enough to see the arcs behind it. A lensing program built around spirals would be fighting its own foreground.
- **The mass doing the bending is mostly not the stars.** The old red population is the visible marker of a potential well that is predominantly dark matter (Ch. 12). Everything this chapter said about stars describes the label on the well, not its depth.
- **Nobody fully understands the profile that results.** Stars and dark matter have unrelated distributions, and yet the *total* mass profile of a real massive elliptical comes out close to isothermal, over exactly the radii lensing probes, for reasons no one derives from first principles. Ch. 10 calls it the isothermal conspiracy, and it is no footnote: it is why this repository’s prior on the slope γ is centred where it is, and part of why the campaign’s headline number is worth being suspicious of.

Unlocks

What you can now read

- **Ch. 10 — Galaxies, Sersic profiles, and velocity dispersion** is the chapter this one exists for. It opens on “a massive elliptical lens galaxy,” describes “an old and red stellar population” with “no ongoing star formation,” and calls ellipticals “the most massive galaxies in their environment: the deepest potential wells around” — all correct, none explained, because its job is the mathematics that follows. You now hold each of those assumptions: what a galaxy is, why one family is red (its massive stars died and nothing replaced them, Ch. 2), why star formation stopped (the gas ran out within a depletion time and was never replenished), and why the lens sample is ellipticals by construction rather than by choice.
- Its skip box says “skip past the notation, not the physics.” That now costs you nothing: the physics it means is the dynamical distinction — ordered rotation versus disordered motion — which this chapter left for it, having given you the visible and historical halves first. And the one thing Ch. 10 does *not* resolve, the isothermal conspiracy, you can now

read as a real open problem rather than as notation.

Exercises

Exercise 3.1 — how empty is a galaxy?

Model the Milky Way's disk as a uniform cylinder, radius 5×10^4 ly, thickness 10^3 ly, containing 10^{11} stars. Compute the volume in cubic parsecs, the mean stellar number density, and the typical spacing $n^{-1/3}$ in units of the Sun's diameter. Then say what the answer implies about a collision between two galaxies.

Solution

$V = \pi R^2 h \approx 2.26 \times 10^{11} \text{ pc}^3$, so $n \approx 0.442$ stars per cubic parsec and $n^{-1/3} \approx 1.313 \text{ pc}$, or $2.7 \times 10^5 \text{ AU}$. Against the Sun's diameter of 0.0093 AU that is 2.9×10^7 solar diameters. At thirty million diameters of separation the probability any two stars meet is negligible, so the stellar components of two colliding galaxies pass straight through each other. Only the gas, not point-like at these scales, actually collides.

Exercise 3.2 — what the uniform disk gets wrong

The density above is a uniform-disk average. The measured density near the Sun is about 0.1 stars per cubic parsec. By what factor does the model miss, in which direction, and why? Then the part that matters: does the emptiness conclusion survive the error?

Solution

The ratio is $0.442/0.1 = 4.42$, so the uniform model is high by about four and a half. The model is exactly the one thing a real disk is not: uniform. Stellar density falls with distance from the centre and with height above the midplane, and the Sun sits far out in the disk — well below the average the model spreads evenly everywhere.

The conclusion survives because the bias runs the safe way. A *lower* true density means a *larger* true spacing, so the real neighbourhood is emptier than the uniform estimate says, not fuller. An error with a known sign, order-unity magnitude and an insensitive conclusion is fine to use provided you say so out loud. It would not be fine if anything downstream needed the density itself; nothing does.

Exercise 3.3 — the gas runs out

A galaxy holds $10^{10} M_\odot$ of cold gas and forms stars at $2 M_\odot$ per year, with no new gas arriving. How long until it stops? Compare to the age of the universe, and say what that implies about why the Milky Way is still blue.

Solution

$$t_{\text{depl}} = \frac{10^{10} M_{\odot}}{2 M_{\odot} \text{ yr}^{-1}} = 5 \text{ Gyr.}$$

That is shorter than the age of the universe (Ch. 11), which is the point: an unfed star-forming galaxy shuts itself down within a time comparable to its own history, with no external mechanism required. The Milky Way is still forming stars not because its original gas lasted but because gas keeps falling in to replace what was consumed. Quenching is therefore not about *destroying* a steady state but about *interrupting a resupply* — which is why the candidate mechanisms all concern cutting off or heating the inflow rather than the gas already present.

Exercise 3.4 — why the lens sample is what it is

Give the two independent reasons a lens-finding campaign’s lenses are mostly ellipticals — one about physics, one about measurement — and say which one Ch. 10 means by “close to a selection effect.”

Solution

Physics: a resolvable Einstein radius needs a deep, compact potential well, and $\theta_E \propto \sigma_v^2$ makes that steeply preferential — the ellipticals, being the most massive galaxies around, dominate the lensing cross-section. Measurement: a smooth, red, dust-free galaxy is fit well by a simple analytic profile (Ch. 10), so its light subtracts cleanly enough to reveal the arcs behind it.

“Close to a selection effect” means the first. Nobody chose these ellipticals — a detectable deflection selected them before any human made a decision, so “the lenses are ellipticals” is a property of the population that survives the filter, not an assumption the campaign makes.

4. Clusters and the cosmic web

The main guide quotes an Einstein radius in three different chapters and never joins them up. [Ch. 9](#) says a lensed ring is of order an arcsecond across. [Ch. 10](#) computes $\theta_E = 1.145''$ for its fiducial massive elliptical. [Ch. 15](#) computes $\theta_E = 13.03''$ for the Carousel and describes it, in passing, as “a cluster-scale strong lens” — without ever saying what a cluster is, or why its ring is ten times bigger than the elliptical’s. That factor of ten is neither a coincidence nor anything deep. A cluster has roughly a hundred times the mass inside its ring, and the ring grows as the *square root* of that mass. This chapter supplies what sits between Chapter 3’s galaxy and that square root: what a cluster is, what it is made of, what larger structure it sits in, and the one line of arithmetic that closes the gap the guide leaves open.

What you can skip

You do not need the virial theorem, the physics of the hot gas between cluster galaxies, or anything about how structure grows out of an initial density field. None of it is used downstream in this program, and each is a research field of its own. You also do not need to derive $M(< \theta_E) = \Sigma_{\text{cr}} \pi \ell_E^2$ — [Ch. 19](#) does that, and this chapter uses only its *shape*. If you already accept that the mass inside a ring scales as the ring’s radius squared, the third section is one square root: read the figure and move on.

Groups and clusters

Galaxies are not sprinkled at random. Gravity has unlimited range and no opposite sign to cancel it — electric charge comes in two signs and neutralizes itself over any large volume, so above the scale of a planet gravity is the only force with anything left to accumulate. Given time, a near-uniform distribution of matter with slight density excesses pulls itself into those excesses, and what starts as a ripple ends as a bound object. The result is a hierarchy, and every rung of it is real and named.

The Milky Way sits in a **group**: itself, Andromeda, and a few dozen much smaller dwarf galaxies, gravitationally bound and falling toward each other. Two large galaxies and a crowd of small ones. Groups are the common case — most galaxies live in one.

A **cluster** is the next rung and a different animal. Order 10^3 galaxies, bound together in a region a few megaparsecs across, with a total mass between

$$10^{14} M_{\odot} \quad \text{and} \quad 10^{15} M_{\odot}.$$

Against this book’s ruler — one Milky Way, total mass including its dark halo, which is most of it ([Ch. 12](#)) — that range is

$$66.7 \quad \text{to} \quad 667 \text{ Milky Ways.}$$

Both numbers deserve the same caveat: neither is a measurement of any particular object. “About a thousand galaxies” and “ 10^{14} – $10^{15} M_{\odot}$ ” are order-of-magnitude statements about a population with real scatter, and a cluster’s boundary is a convention, not a physical edge — the density falls off smoothly, so the galaxy count depends on where you stop counting. Groups and clusters are not two species. They are two regions of one continuum, and the objects in between get called “poor clusters” or “rich groups” depending on the author.

The second thing worth fixing early: those thousand galaxies are a *minority* of the mass. The hot, X-ray-emitting gas filling the space between them outweighs every star in every one of those galaxies combined, and dark matter outweighs the gas by a wide margin again. When you read “a cluster of galaxies,” the galaxies are the part that happens to be visible, not the part that does the gravitating. That is the same statement [Chapter 12](#) will make from four independent directions; cluster dynamics is one of the four, and historically it was the first.

The cosmic web

Zoom out past the cluster and the hierarchy does not continue into ever-larger blobs. It changes character. Map the positions of a million galaxies — which is what a redshift survey does, and what [Chapter 6](#) is about — and what appears is not a uniform scatter and not a set of nested spheres. It is a **web**: clusters sitting at the *nodes*, long thin **filaments** of galaxies running between them, flatter **sheets** spanning the filaments, and, occupying most of the volume, **voids**.

A void is a roughly spherical region with very few galaxies in it, and its characteristic size is about

$$100 \text{ Mpc.}$$

Put that against the ruler. The Milky Way’s disk is

$$0.03066 \text{ Mpc}$$

across, so one void spans

$$\frac{100}{0.03066} \approx 3262 \text{ Milky Way diameters}$$

of mostly nothing. Two honest corrections to that sentence. First, “very few galaxies” is not “empty”: voids hold dwarf galaxies and diffuse gas at much lower density than the filaments, and a void’s interior is a live research topic rather than a blank. Second, 100 Mpc is a *characteristic* scale — voids come in a distribution of sizes, and no single void was measured to produce that number.

The web is not an artist’s impression. It is what surveys measure, and it is also what you get if you take a nearly uniform early universe with tiny density fluctuations and let gravity run: overdense regions get denser and drain the underdense ones, fastest along whichever axis collapses first — which produces sheets, then filaments, then nodes, in that order. Simulations reproduce the observed web from initial conditions constrained by [Chapter 11](#)’s picture of the early universe. That agreement is one of cosmology’s strongest results, and it is *statistical*: the simulations reproduce the web’s characteristic scales and connectivity, not the position of any particular filament.

One consequence matters for the main guide. The web means the universe is lumpy on the scale of 100 Mpc — that lumpiness is what a void’s size records. Keep zooming out, though, and past a few hundred Mpc the lumps average away and one large volume looks statistically like any other. That is what makes [Ch. 14](#)’s FRW metric legal: it assumes a homogeneous, isotropic universe, which is *false* at the scale of this section and true only above it. The main guide states the cosmological principle as holding “on large scales” and stops there — it never says how large, which is the only part of the claim with any content. The answer is: larger than the web.

Why cluster rings are bigger

Now the arithmetic the guide leaves out. [Ch. 19](#) proves a fact this chapter will use and not re-derive: the mass inside the Einstein ring is

$$M(< \theta_E) = \Sigma_{\text{cr}} \pi \ell_E^2,$$

where ℓ_E is the ring’s *physical* radius and Σ_{cr} is a reference surface density fixed entirely by the lens and source distances ([Ch. 15](#)). Read only the shape: a constant density times the area of a circle. The physical radius is proportional to the angle, $\ell_E \propto \theta_E$, so with the distances held fixed,

$$M \propto \theta_E^2 \quad \Longleftrightarrow \quad \theta_E \propto \sqrt{M}.$$

That is the whole explanation. A hundred times the mass buys ten times the ring, because ten is the square root of a hundred. No calculus, no general relativity — the area of a circle, and one assumption (Σ_{cr} fixed) that the rest of this section will make you pay for.

Run it on the guide’s own two numbers. The ratio of the rings is

$$\frac{13.03''}{1.145''} = 11.38,$$

so the mass ratio the rings *imply* is that squared:

$$11.38^2 = 129.5.$$

The Carousel has roughly 130 times the mass inside its ring that the fiducial elliptical has inside its own. Going back the other way, $\sqrt{129.5} = 11.38$, recovers the ring ratio exactly — and that exactness is not evidence of anything. It is one relation read in two directions. The content is the exponent 2, not the arithmetic; the arithmetic cannot fail.

Use the ruler on the mass itself. [Ch. 15](#) computes, from two redshifts and nothing else,

$$M(< \theta_E) \approx 4.621 \times 10^{13} M_{\odot}$$

and then says nothing at all about whether that is a lot, because the main guide has no Milky Way in it to compare against. Divide by one — the Milky Way’s total mass, dark halo included, which this book takes as $1.5 \times 10^{12} M_{\odot}$ — and it is

30.8 Milky Ways

packed inside a circle whose physical radius [Ch. 15](#) works out to a small fraction of a megaparsec. Read that third digit as arithmetic, not precision: the halo has no edge ([Ch. 12](#)), so our Galaxy’s own total depends on where you stop counting, and published estimates spread over roughly a factor of two. Every “in Milky Ways” figure here inherits that. The ruler is good for orders of magnitude, which is all this chapter asks of it.

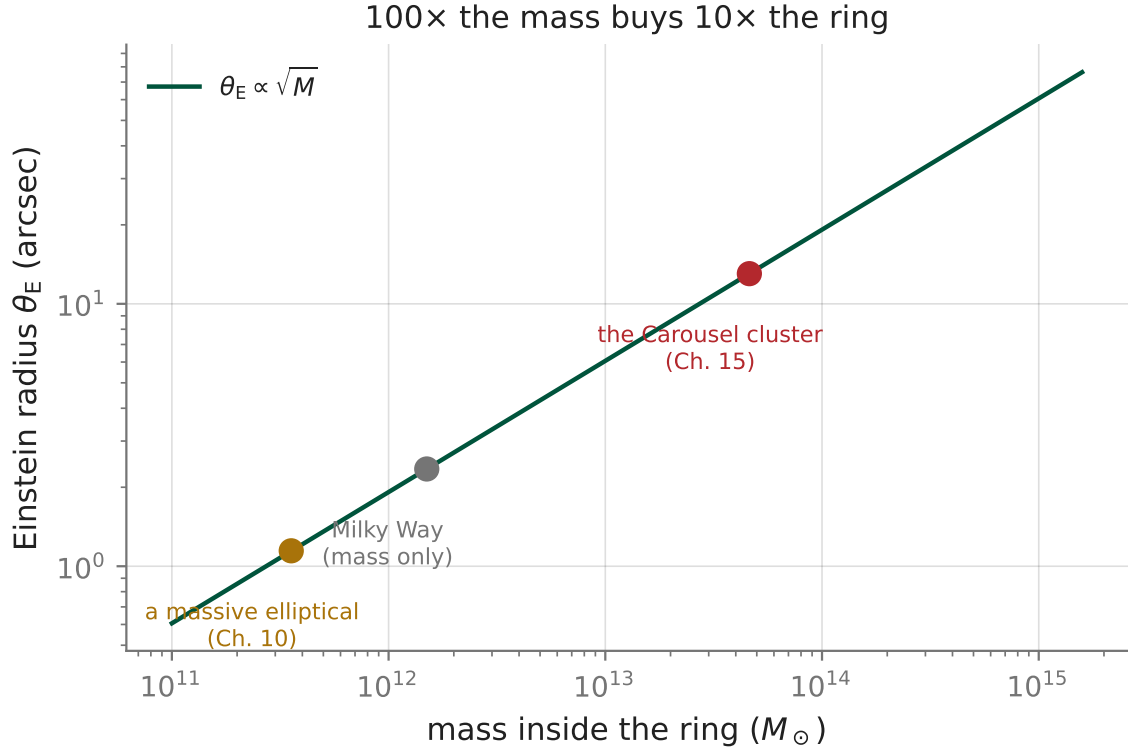


Figure 4: The relation $\theta_E \propto \sqrt{M}$, anchored on the Carousel’s measured pair, with the main guide’s two Einstein radii marked on it about two orders of magnitude in mass apart. The Milky Way’s point is labelled *mass only*: it marks where a Milky-Way-mass deflector would land on this line, not a ring anyone has measured — we sit inside our own Galaxy, which is the wrong side of it to be lensed by. The elliptical’s abscissa is not an independent measurement either; it is the Carousel’s mass divided by the mass ratio the two rings imply, which is why that point sits on the line by construction rather than by agreement.

You already know this

Figure 4.1 is a scaling plot, and it reads exactly like a runtime-versus-input-size plot: a power law is a straight line on log-log axes, and the exponent is the slope. Here the slope is $1/2$. The same caution transfers, too, and it is the reason the caption is worded the way it is: two points always fit a straight line. The slope in this figure is *imposed* by the relation above, and the labelled systems are placed on it. The figure illustrates the law. It does not test it, any more than two timing runs establish that your algorithm is $O(n \log n)$.

Two things about $4.621 \times 10^{13} M_{\odot}$ have to be said plainly, because the main guide says neither and a reader will otherwise walk away with a wrong number in his head.

It is the mass inside the ring, not the cluster. The Carousel’s total mass is not $4.621 \times 10^{13} M_{\odot}$; a cluster’s total is the 10^{14} – $10^{15} M_{\odot}$ of the first section, 66.7 to 667 Milky Ways. The ring at $13.03''$ encloses only the innermost part of the cluster, well inside its few-megaparsec extent — the Einstein radius is a probe of the core, which is precisely what makes it useful (it is a mass measurement that needs no assumption about the profile’s shape) and precisely what makes it partial. 30.8 Milky Ways is the core. How much more the whole cluster weighs is not something this ring can tell you: getting from the core out to the total needs a profile, and refusing to assume a profile is exactly what bought the core measurement its credibility.

The square root holds at fixed geometry, and the geometry is not fixed here. Σ_{cr} depends on the distances to the lens and the source (Ch. 15), and the two systems are not at the same redshifts: the fiducial elliptical sits at $z_l = 0.5$ with its source at $z_s = 2.0$, while the Carousel sits at $z_l = 0.49$ with the source plane used for this measurement at $z_s = 1.432$. The lens redshifts nearly match; the source redshifts do not. So the two Σ_{cr} values differ by an order-unity factor, and 129.5 is the mass ratio the rings imply *if you hold geometry fixed* — close to the true ratio, not equal to it. That is the honest status of this chapter’s headline: a scaling argument, correct about the exponent and about the order of magnitude, and not a substitute for Ch. 15’s machinery, which does the job exactly and is only a few lines longer.

What survives both caveats is the thing the guide never says: the ring is ten times bigger because the mass is a hundred times bigger, and rings go as the square root of mass.

Unlocks

What you can now read

- **Ch. 15 — Distances that do not add, and Σ_{cr}** opens its worked example with “The Carousel Lens ... is a cluster-scale strong lens at $z_l = 0.49$ ” and lets the phrase *cluster-scale* carry the entire argument for why this system’s numbers look nothing like Ch. 10’s. You now know what that phrase asserts: about 10^3 galaxies, 10^{14} – $10^{15} M_{\odot}$ in total, 66.7 to 667 Milky Ways. The chapter promises “two real, checkable numbers” and delivers $M(< \theta_E) \approx 4.621 \times 10^{13} M_{\odot}$ without a word on whether that is large. It is 30.8 Milky Ways — and it is the mass inside a $13.03''$ ring, which is the cluster’s core and not the cluster.
- **Ch. 15’s Σ_{cr} section** calls Σ_{cr} “the natural unit of surface density” and shows it is computable “the moment you have D_d , D_s , and D_{ds} in hand”. What it never says is why anyone would want it: because with it, $M(< \theta_E) = \Sigma_{\text{cr}} \pi \ell_E^2$ turns an angle you can

measure on an image into a mass, and that formula is the whole reason a 13.03'' ring and a 1.145'' ring are the same physics with 129.5 times the mass between them.

Exercises

Exercise 4.1 — The square root, both directions

The main guide gives $\theta_E = 1.145''$ for a fiducial massive elliptical and 13.03'' for the Carousel. Compute the ratio, and from it the mass ratio the two rings imply. Then say what the slogan “a hundred times the mass buys ten times the ring” gets right and what it hides.

Solution

The ring ratio is $13.03/1.145 = 11.38$, and since $M \propto \theta_E^2$ at fixed geometry, the implied mass ratio is $11.38^2 = 129.5$. The slogan gets the exponent right, which is the only part that matters for understanding why the two numbers differ by a factor of ten rather than a factor of a hundred. It hides two things. First, the rounding: the real numbers are 11.38 and 129.5, not 10 and 100. Second, and more seriously, “at fixed geometry” — Σ_{cr} depends on the lens and source redshifts, and the two systems do not share them ($z_s = 2.0$ against $z_s = 1.432$), so 129.5 is what the rings imply under an assumption that is only approximately true here.

Exercise 4.2 — Is the Carousel a lightweight cluster?

Chapter 15 computes $M(< \theta_E) \approx 4.621 \times 10^{13} M_\odot$ for the Carousel. Convert it to Milky Ways. Compare against this chapter’s cluster mass range. The comparison appears to say the Carousel is below the low end of the cluster range — is it a runt? Answer carefully.

Solution

$4.621 \times 10^{13} M_\odot$ divided by a Milky Way is 30.8, against a cluster range of 66.7 to 667 Milky Ways. It looks like the Carousel falls short of the low end by a factor of about two. It does not, and the reason is that the two quantities are not the same quantity. $4.621 \times 10^{13} M_\odot$ is the mass *inside the Einstein ring* — the mass in a circle of radius 13.03'' projected at the lens, which is the cluster’s core and a small fraction of its few-megaparsec extent. 10^{14} – $10^{15} M_\odot$ is a cluster’s *total*. Comparing them directly is a category error, and the fact that the core mass lands below the total range is exactly what you should expect, not evidence about this cluster’s rank. Note that the guide’s number is not wrong or misleading — it is explicitly written $M(< \theta_E)$. The trap is entirely in reading it as “the cluster’s mass”.

Exercise 4.3 — A void against the ruler

Voids are about 100 Mpc across; the Milky Way’s disk is about 0.03066 Mpc across. How many Milky Way diameters fit across a void? Then explain what is wrong with describing that volume as empty.

Solution

$100/0.03066 \approx 3262$ Milky Way diameters. “Empty” is wrong on two counts. Physically, voids are underdense, not vacant: they hold dwarf galaxies and diffuse gas, and what lives in them is an active research question rather than a settled nothing. Definitionally, 100 Mpc is a characteristic scale drawn from a distribution of void sizes, not a measurement of any particular object, so “a void is 3262 Milky Ways across” describes a population’s typical member and should be quoted with that qualifier attached.

Exercise 4.4 — Why the Milky Way’s point says *mass only*

Figure 4.1 marks the Milky Way on the θ_E – M line, but labels it *mass only* rather than giving it an Einstein radius like the other two points. Give two independent reasons why the Milky Way does not appear in this program’s lens catalogues.

Solution

First, geometry: strong lensing needs the deflector *between* you and a distant source, and we are inside the Milky Way. The Galaxy’s mass is not projected onto a sky position we can view from the far side, so there is no configuration in which our own Galaxy produces an Einstein ring for us. (Individual stars within it do produce microlensing, which is a different regime entirely — no resolvable ring.) Second, even for an observer well outside it, producing a ring requires a background source close to aligned on the line of sight and the right distance ratio to make Σ_{cr} small enough (Ch. 15); alignment that good is rare, which is Chapter 16’s subject. The point on the figure therefore answers a conditional — *if* a Milky-Way-mass deflector were placed in a lensing geometry, this is where its ring would sit — and the label says so rather than implying a measurement exists.

5. Light is the only messenger

Every number in the rest of this book — a distance, an age, a mass — is going to be extracted from light and nothing else. No probe has ever been sent to another galaxy; no sample has ever been returned from one. What follows is the physical vocabulary that makes that extraction possible: what light *is* across its full range, why an object’s temperature announces itself as a color, why brightness alone never tells you how far away something is, and the plain fact that sits underneath every redshift claim in this program — that light takes time to travel, so an image is never a report of *now*. Chapter 1 gave you a ruler for space; this chapter gives you the other half, a ruler for time, built out of the same unit.

What you can skip

You do not need Maxwell’s equations, quantum electrodynamics, or anything about *why* accelerating charges radiate. Take electromagnetic radiation as given, the way you’d take “matrix multiplication computes a linear map” as given without re-deriving it. You also do not need to derive the inverse-square law from a solid-angle integral; the one-line argument below is the whole content, and you already reason this way every time you think about how a sensor’s signal-to-noise falls off with range.

The electromagnetic spectrum

Light is an oscillating electric and magnetic field, and like any wave it has a wavelength λ and a frequency ν , related by

$$c = \lambda\nu,$$

where c is the speed of light, 299,792.458 km/s exactly (it is a defined constant, not a measured one — the meter is defined *from* it). “Light” in the everyday sense — what your eyes detect — is one narrow slice of a much wider **electromagnetic spectrum** that runs, by wavelength, from kilometers-long radio waves down to sub-picometer gamma rays, with microwaves, infrared, the visible band, ultraviolet, and X-rays in between. Shorter wavelength means higher frequency (fixed c), and light also comes in discrete energy packets called **photons**, whose energy per photon rises as wavelength falls — which is the whole reason gamma rays are dangerous and radio waves are not. Nothing later in this book needs that quantization explicitly; treat light as a wave characterized by λ and reach for “photon” only when someone hands you an energy in electron-volts instead of a wavelength.

The **visible band** — the sliver human eyes evolved to catch — runs roughly 380–750 nm, violet to red. Everything else in the spectrum is invisible to you but not to a telescope built to catch it: infrared cameras, radio dishes, and X-ray satellites are all reading the same electromagnetic field, just at wavelengths your retina cannot. One unit convention is worth fixing now because you will see it constantly: spectroscopy is conventionally quoted in **Ångströms** (1 Å = 0.1 nm = 10^{-10} m), a unit chosen when atomic spacings, not visible-light wavelengths, were the natural scale. [Ch. 12 of the main guide](#) will hand you rest wavelengths like “3933.66 Å” and “6564.61 Å” without ever saying which part of the spectrum those numbers are *in*. You can check now: 3933.66 Å = 393.366 nm sits just outside the visible band, in the near ultraviolet; 6564.61 Å = 656.461 nm sits at the red edge of visible, which is why that particular hydrogen line (Balmer-alpha, called H α) is the one every optical spectrograph is built to catch.

You already know this

Splitting light into a spectrum — passing it through a prism or a diffraction grating — is not merely *analogous* to a Fourier transform; it is a physical implementation of one. A grating imposes a path-length difference that varies with the sine of the exit angle, and the intensity pattern it produces is the squared magnitude of the Fourier transform of the incident wave, exactly the machinery of Ch. 7. A spectrograph is a Fourier transform built out of glass.

Why hot means blue

A dense, opaque body in thermal equilibrium — a stellar photosphere is the working example throughout this book — radiates a continuous spectrum whose *shape* depends on exactly one number, its temperature T . This is a **blackbody spectrum**, and its single most useful feature is **Wien’s displacement law**: the wavelength at which the emission peaks is inversely proportional to temperature,

$$\lambda_{\text{peak}} \approx \frac{2.898 \times 10^6 \text{ nm} \cdot \text{K}}{T}.$$

Hotter objects peak at *shorter* wavelengths — bluer light — and cooler objects peak at longer wavelengths — redder light. This is the entire physical content behind every “old and red” / “young and blue” statement this program makes about stars and galaxies, stated here in one formula instead of being taken on faith.

Put the Sun’s own surface temperature in:

$$\lambda_{\text{peak}} = \frac{2.898 \times 10^6}{5772} \approx 502.0 \text{ nm},$$

which lands squarely in the visible band — unsurprising, since human color vision evolved under exactly this light source. That peak wavelength corresponds to a frequency of

$$\nu = \frac{c}{\lambda} \approx 5.97 \times 10^{14} \text{ Hz},$$

roughly half a million times faster than a modern CPU’s clock — a useful sense of scale for how fast an electromagnetic oscillation actually is.

A much hotter star, 10,000 K, gives $\lambda_{\text{peak}} \approx 289.8 \text{ nm}$ — already past the blue end of the visible band, into the ultraviolet. This is a real trap worth naming: the star’s *peak* isn’t inside the visible window at all, so “the peak wavelength is the color you see” is not quite right. What you see is the shape of the curve *restricted* to 380–750 nm, and a 10,000 K blackbody’s visible tail still strongly favors blue over red relative to a cooler star’s — so the star looks blue-white, for a subtler reason than “the peak is blue.” A cool star at 3000 K peaks at $\lambda_{\text{peak}} \approx 965.9 \text{ nm}$, deep in the near-infrared, and the sliver of that curve inside the visible band is skewed red — hence “red dwarf.” An even hotter object, the afterglow of the Big Bang itself at 2.725 K, peaks at 1.06 mm — not visible light at all, but microwave. Ch. 11 is where that number stops being a curiosity and becomes the single most informative picture ever taken.

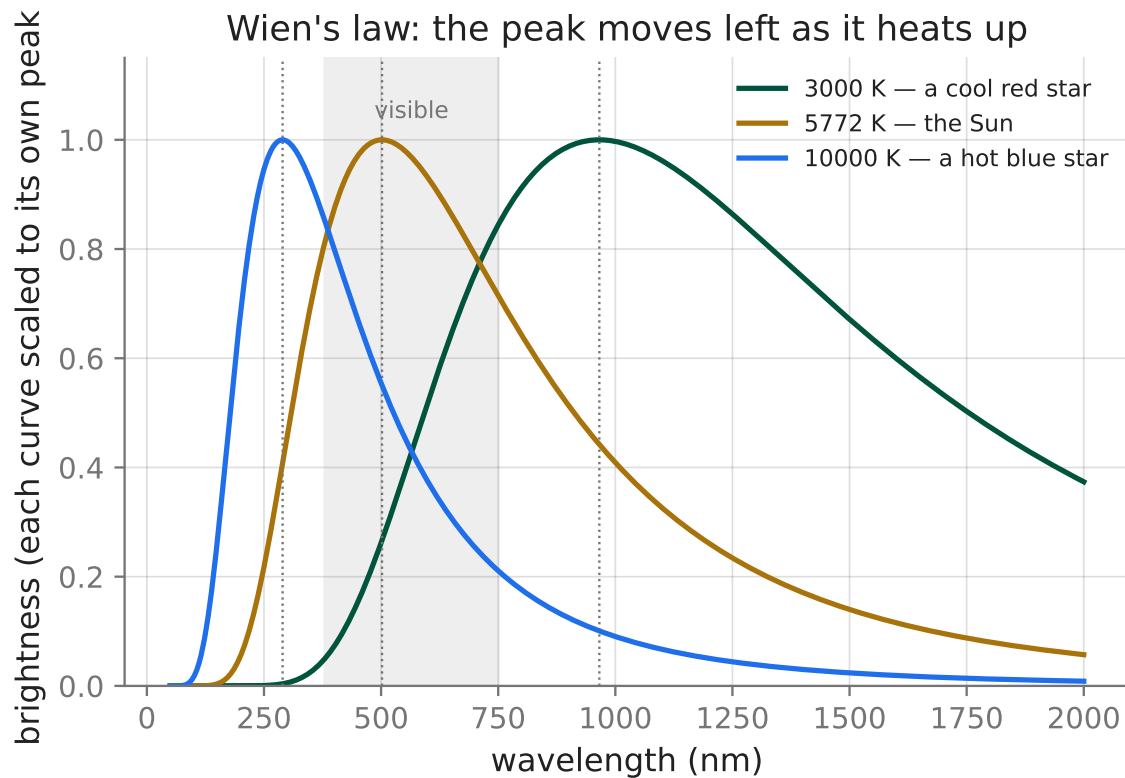


Figure 5: Blackbody curves at three temperatures, each normalized to its own peak, with the visible band (380–750 nm) shaded and each curve's Wien-law peak marked by a dotted line. The 10,000 K curve's peak falls to the *left* of the shaded band entirely — its light is still bluer than the Sun's where the two curves overlap in the visible, but the actual maximum of the curve is ultraviolet light your eye cannot see at all.

The inverse-square law

A source radiating total power (luminosity) L spreads that energy over the surface of an ever-expanding sphere as it travels outward. At distance d , that sphere has area $4\pi d^2$, so the power crossing a unit area — the **flux** F you actually measure — is

$$F = \frac{L}{4\pi d^2}.$$

Nothing about the source changed; the same total light is simply divided over a larger sphere the farther out you go. Double the distance and the sphere’s area quadruples, so the flux drops to a quarter:

$$\frac{F(2d)}{F(d)} = \left(\frac{d}{2d}\right)^2 = \frac{1}{4}.$$

Push it to ten times the distance and flux falls to one part in a hundred:

$$\frac{F(10d)}{F(d)} = \frac{1}{100}.$$

This is the reason apparent brightness, by itself, is ambiguous: a dim source nearby and a luminous source far away can produce the identical flux. [Ch. 9’s magnitude system](#) is a logarithmic re-encoding of exactly this quantity, chosen because real surveys must represent fluxes spanning ten or more orders of magnitude in one image — the inverse-square law is *why* that dynamic range is so wide in the first place, not an incidental fact about detectors. Breaking the distance/luminosity ambiguity is precisely the job Chapter 9 of this book takes on with **standard candles**: objects whose luminosity L is known independently, so a measured flux F can be inverted for d instead of read as an unresolvable mix of the two.

Looking back in time

Light travels fast, but not infinitely fast, and every distance this book has quoted so far converts directly into a *delay*. The Sun’s light takes

$$\frac{1 \text{ AU}}{c} \approx 8.3 \text{ minutes}$$

to reach Earth, so the Sun you see is never the Sun *now* — it is the Sun as it was 8.3 minutes ago. That gap only grows with distance, because a light-year is, by definition, the distance light covers in a year: the travel time in years and the distance in light-years are numerically the same number. Andromeda, the nearest large galaxy beyond our own, sits about 2.5 million light-years away, so its light took 2.5 million years to reach the telescope pointed at it tonight — the photons left before *Homo sapiens* existed, and what a picture of Andromeda shows is that departure, not any “now” on the other end.

There is no way around this, and no sense in which a more sensitive telescope could someday see Andromeda “as it is today”: simultaneity across that distance isn’t a measurement problem, it’s what “distance” *means* once you accept that nothing — light included — carries information faster

than c . Every image is old news, and the farther out you look, the older the news gets. For the systems this program actually studies — a lens galaxy at redshift $z = 0.5$, a source galaxy at $z = 2$ — that delay is not minutes or millions of years but billions: Chapter 8 turns this exact idea into a number for each one. This chapter’s job was only to establish that the delay is real and unavoidable; everything about *how big* it gets, and what redshift has to do with it, is what the rest of Part II builds toward.

Unlocks

What you can now read

- **Ch. 9 — Arcseconds, magnitudes, and the units of the sky** defines flux as “what a detector actually integrates” and builds an entire logarithmic magnitude system to compress it, but never explains *why* flux needs ten-plus orders of magnitude of dynamic range in the first place. You now have the answer: the inverse-square law, applied to real objects sitting at wildly different distances and luminosities, and the standard-candle idea (this book’s own Ch. 9) that inverts it back into a distance.
- **Ch. 12 — Redshifts and what a spectrum tells you** calls a galaxy’s continuum “the light of a few hundred billion stellar photospheres, each close to a blackbody” and moves on without ever defining a blackbody. It also hands you rest wavelengths in Ångströms — 3933.66 Å, 6564.61 Å — and later notes that one redshifted line lands “deep in the near-infrared, off the red end of a ground-based optical spectrograph entirely,” a claim you can now check yourself because you know where the visible band’s edges sit in nanometers.

Exercises

Exercise 5.1 — Wien’s law by hand

Using $\lambda_{\text{peak}} T \approx 2.898 \times 10^6 \text{ nm} \cdot \text{K}$, compute the peak wavelength for a 3000 K star and for a 10,000 K star. For each, say which part of the electromagnetic spectrum (infrared, visible, ultraviolet) the peak falls in, and explain why neither answer is the whole story about what color the star looks like to a human eye.

Solution

3000 K: $\lambda_{\text{peak}} = 2.898 \times 10^6 / 3000 \approx 965.9 \text{ nm}$, in the near-infrared — beyond the red edge of the visible band entirely. 10,000 K: $\lambda_{\text{peak}} = 2.898 \times 10^6 / 10,000 \approx 289.8 \text{ nm}$, in the ultraviolet — beyond the blue edge. Neither peak is inside the 380–750 nm band the eye responds to, so “peak wavelength” is not “perceived color”: what the eye sees is the *shape* of the whole curve restricted to the visible window, and a curve whose true peak lies just past either edge still tilts that visible sliver toward red (cool case) or blue (hot case) relative to a curve peaking inside the band.

Exercise 5.2 — Inverse square, both directions

Two galaxies have identical luminosity L . Galaxy B is 10× farther away than Galaxy A. What is the ratio of their fluxes? Now suppose instead that A and B are at the *same* distance but you measure B’s flux to be 1/100 of A’s — can you conclude B is 100× less luminous? What is the one piece of information missing that would let you conclude that?

Solution

Same luminosity, $10\times$ the distance: $F_B/F_A = (d_A/d_B)^2 = 1/100$. But flux alone never separates “farther” from “dimmer” — a source $100\times$ less luminous at the *same* distance produces an identical flux ratio to a source of identical luminosity $10\times$ farther away. Concluding B is $100\times$ less luminous requires knowing the *distance* independently, which is exactly what a standard candle supplies and a bare flux measurement never does on its own.

Exercise 5.3 — How old is the news?

Using $d = ct$, compute how many minutes ago the light left the Sun (use 1 AU / c), and how many years ago the light left Andromeda (use its distance in light-years directly). Then answer: is there any telescope, however sensitive, that could show you Andromeda “as it is right now”?

Solution

Sun: $1 \text{ AU}/c \approx 8.3$ minutes . Andromeda: because a light-year is defined as the distance light travels in a year, its distance in light-years, 2.5 million , *is* its light travel time in years . No telescope can show Andromeda “as it is now”: there is no signal that outruns light to tell you what is happening there today, so “now, at Andromeda” is not a measurement anyone could ever make — it isn’t that our instruments fall short, it’s that the question presupposes a simultaneity that a finite light speed does not allow.

Exercise 5.4 — Reading an Ångström

The main guide’s Ch. 12 quotes a rest wavelength of 3933.66 Å for the Ca II K absorption line. Convert this to nanometers and say whether it is inside the visible band. Do the same for 6564.61 Å (Balmer-alpha/ $H\alpha$).

Solution

$3933.66 \text{ Å} = 393.366 \text{ nm}$ — just below the 380–750 nm visible band’s blue edge is 380 nm, so this line sits right at the boundary, in the near ultraviolet/violet. $6564.61 \text{ Å} = 656.461 \text{ nm}$ — comfortably inside the visible band, near its red edge, which is why $H\alpha$ is a workhorse line for ground-based optical spectrographs: it stays observable out to larger redshifts before being stretched past 750 nm and out of the visible window entirely.

6. Telescopes, surveys, and the three observatories

Three instruments carry this program’s entire evidence base. The DESI Legacy Imaging Surveys find the lens candidates; the *Hubble Space Telescope* takes the sharp pictures the money number is fitted to; *Euclid* is the independent check. The main guide names all three across nine chapters and introduces none of them. This chapter is that introduction, and it reduces to two numbers per instrument: how finely it samples the sky, and how badly the light was blurred before it arrived. The second number decides things, and for the survey that finds the lenses it is bad enough to put a floor under what can be found at all.

What you can skip

You already own the sampling theory: pixel scale versus blur width is Nyquist–Shannon on a spatial signal, and the main guide says so rather than making you infer it. You do not need optical design, detector electronics or the CCD noise model — [Ch. 11](#) builds that from scratch. What is new is narrow: which physical effect limits which instrument, and what DESI, HST and Euclid refer to.

What a telescope collects

A telescope does two separate jobs, and keeping them apart matters, because they scale differently.

The first is **collecting**. Light arrives thinned out by the inverse-square law ([Ch. 5](#)), so how faint an object you can detect is set by how many photons you gather — the mirror’s *area*, growing as the square of its diameter. Here a telescope really is a bucket in the rain: a wider bucket catches proportionally more. The second is **resolving**: telling apart two directions on the sky that are nearly the same. This is where the bucket analogy breaks completely — a wider bucket tells you nothing more about *where* the rain fell; a wider mirror does, for reasons unrelated to how many photons it caught. Both jobs improve with diameter, through unrelated physics, and only one of them survives the atmosphere.

Angles on the sky are measured in **arcseconds**: one arcsecond is 1/3600 of a degree. Every number in this chapter is in arcseconds; the small-angle arithmetic converting one into a radian and then a physical size is [Ch. 9](#)’s job.

Two arcsecond quantities describe any camera, and they mean different things. **Pixel scale**, in arcsec per pixel, is how much sky one detector pixel covers: a unit conversion, fixed at design time, exact. **Point-spread function** width, quoted as FWHM in arcsec, is how wide a true point of light lands after everything between the source and the silicon has smeared it: a *measurement*, and it changes. Confusing the two, the main guide says, is “the single most common unit mistake in reading a lensing figure” ([Ch. 9](#)). It then gives both numbers for DESI and never says which instruments sit where.

Ground versus space

There are two ways to blur a point of light, and which one dominates is the entire difference between the three observatories.

Diffraction is unavoidable. Light is a wave ([Ch. 5](#)), and a wave passing through a finite aperture spreads. The angular scale of the spread goes as λ/D — wavelength over aperture diameter — so bigger mirrors are genuinely sharper, and redder light blurrier than bluer through the same optics.

For HST’s 2.4 m mirror at optical and near-infrared wavelengths this lands around a tenth of an arcsecond. No engineering removes it.

Seeing is the atmosphere. Pockets of air at slightly different temperatures have slightly different refractive indices, so each acts as a weak, wandering lens, and over an exposure of any real length a point source is smeared into a disk roughly an arcsecond across at a good site. Here is the fact that is not obvious until someone states it: **seeing does not care how big your telescope is**. An 8 m and a 4 m telescope on the same mountain on the same night have the *same* resolution. The larger sees fainter — twice the diameter, four times the photons — but not sharper. Aperture buys depth on the ground, not sharpness. That is why a 2.4 m telescope in orbit out-resolves every seeing-limited telescope ever built, however large its mirror. It is not a better mirror. It is above the air.

The main guide compresses all of this into one parenthesis, defining seeing as “the width of the point-spread function (PSF) imposed by the atmosphere (or, for a space telescope, by diffraction)” (Ch. 9). Five bracketed words carry the whole HST-versus-DESI split, and if you did not already know that ground telescopes are seeing-limited and space telescopes diffraction-limited, the parenthesis tells you nothing.

DESI’s imaging is taken through about 1.3” of atmosphere:

$$\text{FWHM}_{\text{DESI}} \approx 1.3''.$$

Treat that as a round number, not a constant. The main guide quotes the DESI Legacy Survey’s *measured g-band coadd PSF* at 1.35”, careful to say it was measured rather than assumed. The gap between the two figures is not sloppiness; it is the point. Seeing varies by site, by night, by band, and by how many exposures got stacked. Diffraction does not.

Where this simplification stops

Adaptive optics — sensing the distortion in real time and bending a small mirror to cancel it — genuinely works, and modern ground-based systems reach space-like sharpness with it. It is not a loophole here: it works best in the infrared, over a field far too small to survey the sky with, and needs a bright reference star nearby. Nothing in this program’s discovery half uses it. “Ground means blurry” is true of *wide-field survey imaging* — the only ground-based imaging this book cares about — not of ground-based astronomy in general.

Bands and filters

An imaging camera does not record a spectrum. It records one number per pixel: total photons, summed over whatever wavelengths the filter in front of it lets through. That filter’s transmission window is a **band**, and imaging in several bands gives a handful of coarse samples of each object’s spectrum instead of the thousands a spectrograph would give.

The DESI Legacy Surveys use three optical bands, named *g*, *r* and *z* — “hence *grz*”, as Ch. 27 puts it in passing. Roughly: *g* is blue-green, near 480 nm; *r* is red, near 640 nm; *z* is the far red running into the near-infrared, near 900 nm — past the red edge of what a human eye responds to (Ch. 5). Every finder in this program’s lineage consumes a three-channel cutout of exactly these bands, stacked like the channels of an RGB image and emphatically not one.

You already know this

A band is a lossy encoder: broadband imaging projects a spectrum of thousands of resolvable samples onto three basis functions — the filter transmission curves — and keeps the coefficients. Everything downstream is inference from a three-dimensional feature vector. A *photometric* redshift is regression from those three numbers; a *spectroscopic* redshift (Ch. 7) reads a line position off the uncompressed signal. The accuracy gap is a compression-ratio story.

Which band an object is brightest in is therefore a crude thermometer: an old, red stellar population (Ch. 2) puts more of its light in z than in g , and the difference of two bands’ magnitudes — a **colour** — is astronomy’s cheapest classifier.

Now the collision, which this corpus never flags.

The z -band is a filter. Redshift z is a dimensionless ratio. They are different things, spelled the same, and both appear constantly in the main guide — sometimes on the same page. Ch. 9, mid-derivation of the magnitude system, quotes the discovery sweep’s quality cut as “ $m_z < 20$ in the z -band”; that z is a piece of glass. Ch. 12 and Ch. 13 write z for redshift on nearly every line. The rule that separates them is mechanical: a z attached to a brightness (m_z , “the z -band flux”) is the filter; a z standing alone, or subscripted by an object (z_l for the lens, z_s for the source), is redshift.

The collision is not quite a coincidence, which makes it stickier. Stretching every wavelength by $1 + z$ (Ch. 8) drags a distant galaxy’s light out of the blue bands into the red ones, so a catalog of distant red galaxies — what lens hunters want — needs a filter out at 900 nm to catch them. The band is named z and it selects on z .

What a survey is

Classical observing is *pointed*: you have a target, you win telescope time, you observe it. A **survey** inverts that. You image a defined footprint to a defined depth in defined bands, with no target in mind, reduce it once with one pipeline, and publish catalogs that other people mine for science years later. The DESI Legacy Surveys’ data releases, DR7 through DR11 across this program’s finder lineage, are exactly this: public, uniform, pullable by anyone with a URL.

Surveys come in two kinds, and this program uses both. **Imaging surveys** photograph everything in the footprint: every object gets a picture and a few band magnitudes — cheap per object, weak per object. **Spectroscopic surveys** put a fiber optic on a pre-chosen target and disperse its light into a full spectrum: thousands of wavelength samples, one object at a time, and you must know where to aim *before* you start. So the imaging survey comes first, and its catalog tells the spectroscopic survey where to aim — which sets up the name collision below.

Why image the whole sky? Because a strong lens is an accident of alignment (Ch. 16), and you cannot point a telescope at an accident you have not found yet. The only way to find something that rare is to photograph everything and filter afterwards — which is why Ch. 27 opens with tens of millions of galaxies and warns that “a survey-scale finder is not a bigger version of a Kaggle classifier: the operating point has to be chosen against tens of millions of negatives, not a held-out test split.”

The three observatories

The DESI Legacy Imaging Surveys. Ground-based, imaging, public. Three bands (*grz*), about 1.3'' of seeing, and a pixel scale of

$$0.262''/\text{px}.$$

This is the discovery half.

The name is a trap the main guide walks into without comment. **DESI proper is a spectrograph** — a 5000-fiber instrument on a 4 m telescope, built to measure redshifts. **The DESI Legacy Imaging Surveys are imaging**, taken with *different telescopes* (DECam in the south, 90Prime and Mosaic-3 in the north), to choose which objects DESI’s fibers would later point at. When Ch. 27 says the surveys “photographed most of the extragalactic sky in three colors,” that is the cameras; when Ch. 12 says “two DESI fiber spectra, separated by under an arcsecond,” that is the spectrograph.

The Hubble Space Telescope. A 2.4 m mirror in low Earth orbit since 1990. Diffraction-limited, so about a tenth of an arcsecond, and its near-infrared detector’s own pixel is

$$0.128''/\text{px}.$$

Compare those two numbers. The PSF is roughly *one pixel* wide — and Nyquist asks for at least two samples across it. HST’s detector undersamples HST’s own optics: a single exposure discards resolution the mirror already delivered. This is a deliberate trade of sampling for field of view, fixed in software. Take several exposures, each offset by a fraction of a pixel — a **dither** — combine them onto one finer output grid, and you recover the sampling the optics were always capable of. That step is **drizzle**, and it is why Ch. 11 can state as bare habit that “*HST* rarely takes one exposure of a target; it takes several, each offset by a fraction of a pixel.” It is not a habit; it is the only way to get HST’s resolution out of HST’s pixels.

Euclid. A 1.2 m telescope launched in 2023 to L2, with a visible camera at

$$0.1''/\text{px}.$$

Euclid is the combination the other two do not offer: space-based sharpness over a survey-sized area of sky. HST is sharp and narrow; DESI is wide and blurry; Euclid is sharp and wide. That is why its first data release is what the main guide’s Chapters 27 and 28 re-grade their DESI candidates against.

Now put the survey’s two numbers against a real lens. A typical galaxy-scale Einstein ring has $\theta_E \approx 1.2''$: the value Figure 6.1 uses, and close to the fiducial elliptical the main guide builds in Ch. 19. Opposite sides of that ring sit a full diameter apart, $2\theta_E$, which in DESI pixels is

$$\frac{2 \times 1.2''}{0.262''/\text{px}} \approx 9.16 \text{ px}.$$

Nine pixels sounds workable. Now the blur:

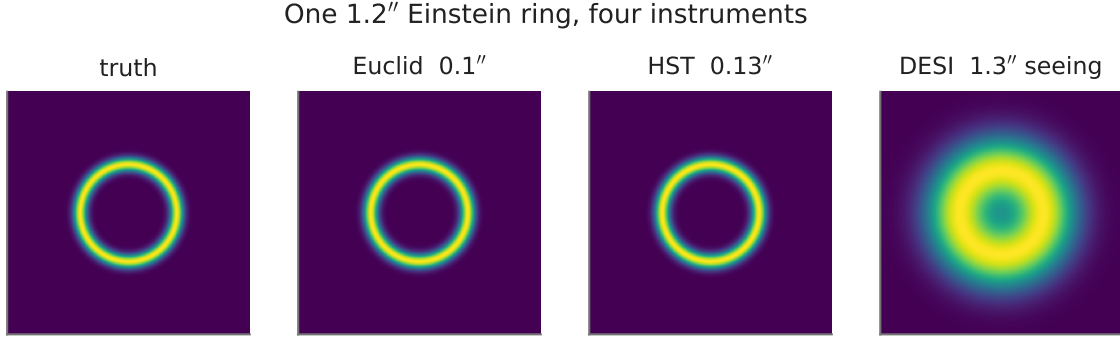


Figure 6: The same simulated 1.2'' Einstein ring, convolved with each instrument's characteristic resolution scale; the leftmost panel is the unconvolved truth. Euclid's and HST's panels are barely distinguishable from it. DESI's is a fat, soft annulus — and note what it is *not*: the hole is still open. This ring clears the resolution wall by about a factor of two, and the panel shows what clearing it by only that much buys you — a ring you could find, not a ring you could measure. The panels blur but do not pixelate: what is drawn is the PSF alone, with no detector grid, noise or sky background — all of which make the real DESI panel worse.

$$\frac{1.3''}{0.262''/\text{px}} \approx 4.96 \text{ px.}$$

The blur's full width at half maximum is more than half the ring's entire diameter. Each side of the ring is smeared across a distance comparable to its separation from the other — not enough to merge them, as Figure 6.1's DESI panel shows, but enough that what survives the atmosphere is the ring's position and almost none of its shape. That panel is a detection, not a picture. A ring much smaller than this one would not even be that.

This is the wall. [Ch. 27](#) turns the same comparison into a closed-form floor on the Einstein radius a ground-based survey can resolve at all, using a second-derivative sign test to decide when two blurred points stop being two. Then it confirms that floor from the other direction: when Euclid re-observed a small sample of DESI grade-C candidates, several jumped straight to grade A. The objects did not change; the wall moved.

How far? Against DESI's blur, Euclid's resolution element is smaller by

$$\frac{1.3''}{0.1''} = 13,$$

a comparison needing one caveat, unpacked in [Exercise 6.3](#): it divides a PSF width by a pixel scale, legitimate only because in space those two numbers converge by design.

One last calibration, because 1.3'' sounds small and is not. A galaxy the size of the Milky Way ([Ch. 1](#)), at the distance of the main guide's standard lens, covers only a few arcseconds of sky — the angle-to-size conversion is [Ch. 15's](#) job. DESI's blur is not a blemish on a picture of a galaxy. It is a good fraction of the galaxy.

Unlocks

What you can now read

- **Ch. 9 — Arcseconds, magnitudes, and the units of the sky** defines seeing as “the width of the point-spread function (PSF) imposed by the atmosphere (or, for a space telescope, by diffraction)” — hanging the entire ground-versus-space distinction on that parenthesis. You now have what it omits: two unrelated blur mechanisms, only one improving with aperture, and which observatory is limited by which. It also drops “ $m_z < 20$ in the z -band” (Ch. 9) into a book where z means redshift on almost every other page, and never flags it.
- **Ch. 11 — From photons to pixels: PSF, noise, and drizzle** states as bare fact that “*HST* rarely takes one exposure of a target; it takes several, each offset by a fraction of a pixel (a *dither*)” — and never says why. You now have the why: *HST*’s detector pixel undersamples *HST*’s own diffraction-limited PSF. That makes drizzle, and the correlated noise it produces — the first link in the guide’s Spine 1 — a consequence rather than a convention.
- **Ch. 27 — Finding lenses: the survey, the nets, and the resolution wall** opens on “the DESI Legacy Imaging Surveys photographed most of the extragalactic sky in three colors” and asks “how good does a picture have to be before a ring is even visible in it.” You now have the survey, the three colours, and the ring-versus-blur comparison in its own pixels; Ch. 27 turns that into a closed form, and Ch. 28 shows the same wall making human graders unreliable.

Exercises

Exercise 6.1 — The wall, in DESI pixels

A ring has $\theta_E = 1.2''$; DESI’s pixel scale is $0.262''/\text{px}$ and its seeing about $1.3''$ FWHM. Compute (a) the ring’s diameter and (b) the seeing FWHM, both in DESI pixels. Then: is DESI *undersampled* by its pixel grid, and is that the same question as whether it can see the ring?

Solution

- (a) The diameter is $2\theta_E$, so $2 \times 1.2/0.262 \approx 9.16$ pixels .
- (b) $1.3/0.262 \approx 4.96$ pixels .

DESI is *not* undersampled: (b) sits comfortably above the Nyquist floor of two pixels per FWHM. But that is a statement about the pixel grid, not about the sky — the pixels sample the blur faithfully; they cannot undo it. What decides whether the ring is visible is (b) against (a), and no finer pixel grid changes it. Sampling the blur and resolving the ring are different questions, which is why Ch. 9 insists pixel scale and seeing are not the same quantity.

Exercise 6.2 — Which z is which?

Classify each z as the filter or the redshift, without looking anything up.

1. “the faint-end quality cut is $m_z < 20$ ”
2. “a lens at $z_l = 0.5$ and a source at $z_s = 2.0$ ”

3. “ $1 + z = \lambda_{\text{obs}}/\lambda_{\text{emit}}$ ”

Solution

1. Filter — a subscript on a magnitude m , naming the band.
2. Redshift — subscripted *itself*, l for lens and s for source, which only makes sense for a per-object quantity.
3. Redshift — bare, inside a dimensionless ratio of wavelengths. A filter has no arithmetic.

Exercise 6.3 — What Euclid’s factor of 13 does and does not say

DESI’s seeing is about $1.3''$; Euclid’s visible pixel scale is $0.1''$. Compute the ratio, then state what is wrong with that calculation, and why the answer survives anyway.

Solution

$$1.3/0.1 = 13 .$$

What is wrong: it divides a PSF width (a measurement, in arcsec FWHM) by a pixel scale (a unit conversion, in arcsec per pixel) — the two quantities [Ch. 9](#) warns are most often confused.

Why it survives: for Euclid the two coincide near $0.1''$, and not by accident. In space the only blur is diffraction, so a camera’s pixels are sized to sample the diffraction limit and the two converge by construction — which is why the main guide uses $0.1''$ as Euclid’s FWHM. The ratio is really blur against blur, the comparison that matters. The same shortcut on DESI would be nonsense: its pixel scale and seeing differ by the factor from Exercise 6.1, because on the ground nothing forces them together.

7. Spectra — the fingerprint of everything

Ch. 5 treated light as a messenger carrying two numbers: how much (flux) and what color (temperature, via Wien’s law). A spectrograph — the instrument Ch. 6 described as a wavelength-binned camera rather than an ordinary one — throws away that single color and replaces it with hundreds or thousands of numbers: flux measured in narrow bins across the whole spectrum, not one broad-band average. What comes back is not smooth. Sitting on top of a smooth background are sharp spikes and dips at exact, repeatable wavelengths — and that pattern is the reason Ch. 12 of the main guide can open with the sentence “you do not need atomic physics” and mean it literally. This chapter is what that sentence is quietly resting on.

What you can skip

You do not need the quantum mechanics that fixes an atom’s energy levels — accept it as a postulate, exactly as Ch. 12 asks you to. You do not need spectrograph engineering (fibers, gratings, detectors); Ch. 6 already gave you the instrument, and this chapter only needs its output. You do not need a line list for every element in the periodic table — one worked example (hydrogen’s Balmer series) carries the whole argument, and two more lines are quoted only because Ch. 12 uses them by name without saying where they come from.

Continuum and lines

A star’s photosphere radiates close to a blackbody curve set by its temperature (Ch. 5) — smooth, one broad hump peaking wherever Wien’s law puts it. A galaxy is a few hundred billion of these photospheres summed together, at a spread of temperatures, so its overall spectrum is smooth for the same reason a histogram of many overlapping distributions is smooth: no single star’s signature survives the sum. That smooth hump is the **continuum**.

Real spectra are not just a hump, though. Superimposed on the continuum are narrow features — flux spikes above it or dips below it — sitting at specific wavelengths and nowhere else. These are **lines**, and unlike the continuum they are not a property of the mixture. They are a property of exactly which chemical elements and ions are present in the gas doing the radiating or the absorbing, which is what makes them worth the name “fingerprint.”

Two questions follow immediately. Why do the dips sit at those five wavelengths and nowhere else? And why dips, rather than spikes? The next two sections answer them in turn.

Why elements have fingerprints

An atom’s electrons occupy a fixed ladder of allowed energies. When an electron falls from a higher rung to a lower one, it emits a photon carrying exactly the energy difference between those two rungs, no more and no less. Because the rungs are fixed by quantum mechanics and not by the local environment, that energy — and therefore the wavelength — is identical wherever in the universe the same atom makes the same jump. Hydrogen’s version of this ladder is well characterized enough to fit in one formula, the Rydberg formula for the Balmer series (the family of jumps that land on hydrogen’s second rung):

Two notation flags before writing it, because this corpus does not let a Greek letter drift. Wavelength is written λ here — a chapter-local use, with nothing to do with the SMC tempering parameter of the same letter used elsewhere in this corpus (flagged explicitly in Ch. 12 too, the one place both books actually name a spectrum). And the lines themselves carry names fixed by

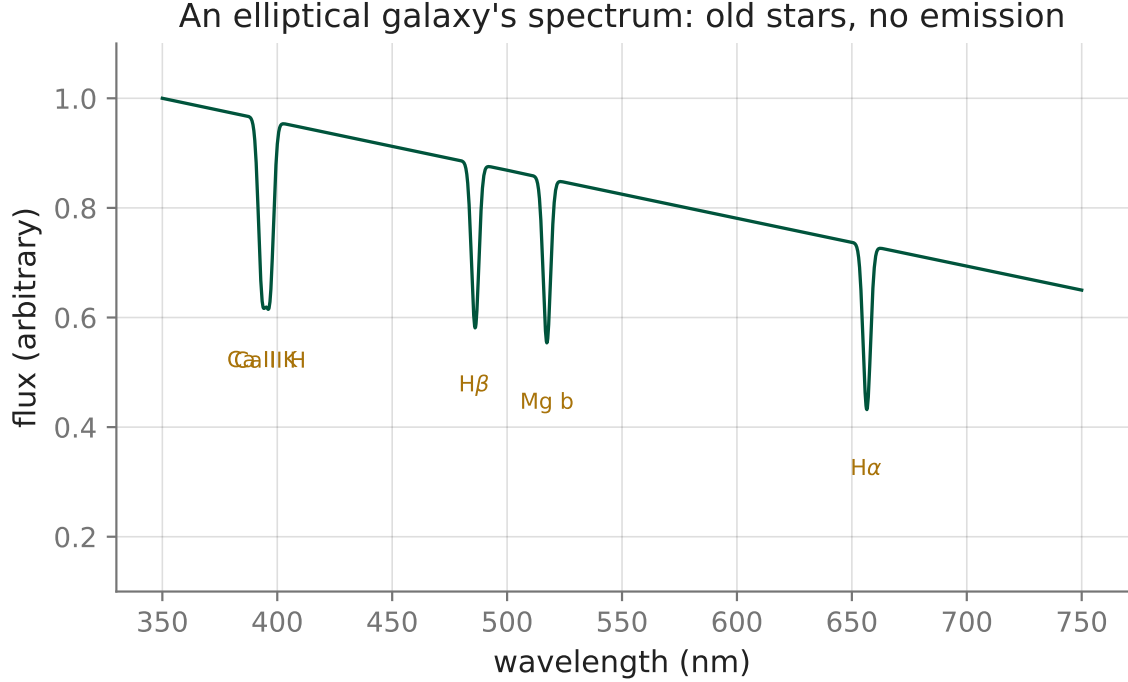


Figure 7: An elliptical galaxy’s spectrum: a smooth continuum (Ch. 5’s summed blackbodies) with five absorption dips at fixed wavelengths — Ca II K, Ca II H, H β , Mg b, H α — and no emission anywhere. Old stars, no ongoing star formation: exactly the population Ch. 3 already singled out as this program’s lenses.

nineteenth-century spectroscopists — H α , H β , H γ for the reddest, second-reddest and third-reddest lines in hydrogen’s series — which share no meaning with the deflection angle α , the source position β , or the density slope γ this corpus otherwise reserves for lensing. Every bare α , β , γ without an “H” in front of it means the lensing quantity, everywhere else in both books.

$$\frac{1}{\lambda} = R \left(\frac{1}{4} - \frac{1}{n^2} \right), \quad n = 3, 4, 5, \dots,$$

with $R = 1.0967758 \times 10^7 \text{ m}^{-1}$ (the Rydberg constant) and n the rung the electron falls from. Feed in $n = 3$, hydrogen’s reddest and brightest visible line:

$$\lambda_{\text{H}} = \frac{1}{R \left(\frac{1}{4} - \frac{1}{9} \right)} = 656.470 \text{ nm}.$$

One formula, one constant, an exact prediction (Exercise 7.1 gets two more). That reproducibility — not “hydrogen glows red,” but “hydrogen glows at *this* wavelength, computed from one number, everywhere” — is the whole content of “fingerprint.” Compare it to the continuum: Ch. 5’s Sun peaks near 502 nm, a wavelength set entirely by its 5772 K surface temperature — heat the photosphere and the peak moves. H α ’s 656.470 nm does not move. It is fixed by hydrogen’s internal structure, true whether the gas producing it sits in the Sun, a star-forming galaxy near the edge of the observable universe, or a lab on Earth.

That fixed number, though, hides one convention most line lists don't bother to state. The formula above gives $H\alpha$'s wavelength *in vacuum*. Precision spectroscopy is historically done through air, and older line lists — some still in circulation — quote instead the air wavelength, 656.288 nm, shorter because air's refractive index slows the light down and compresses the wave. The two conventions disagree by only 0.0277% of $H\alpha$'s own wavelength (Exercise 7.2 works the arithmetic), and that gap is exactly air's refractive index. It sounds negligible until you remember that [Ch. 8](#) defines a redshift as a wavelength *ratio*: divide by the wrong rest wavelength and every redshift measured against that line is wrong in its fourth digit, silently, with no flag raised anywhere in the pipeline.

Hydrogen is not the only fingerprint this program leans on. A singly-ionized calcium atom absorbs at 393.366 nm (called Ca II K — the element, its ionization state, and a spectroscopist's historical letter for this particular line; the main guide writes the same number as 3933.66 Å, since 1 nm is exactly 10 Å), and singly-ionized oxygen emits a close doublet just past 372 nm (written [OII], the brackets marking a transition that only happens in gas thin enough that nothing collides an excited ion back down before it can radiate). Both numbers are exact for the same reason $H\alpha$ is: a fixed quantum ladder, unique to that ion, reproducible anywhere in the universe. [Ch. 12](#) uses this exact pair, unexplained, to pull two redshifts out of one exposure. This section is where those two numbers come from.

Emission versus absorption

The fixed energy that names a fingerprint says nothing about whether it shows up as a spike above the continuum or a dip below it. That depends on which direction the electron jumps, and what is sitting where along the line of sight.

An **emission line** appears when an electron falls from a higher rung to a lower one and radiates a photon the instrument would not otherwise have collected: it *adds* light at that exact wavelength, on top of whatever continuum is already there. That needs a supply of atoms already sitting in the excited (higher) state, which needs something heating or ionizing the gas — hot young stars flooding a star-forming region with ultraviolet light, or an active galactic nucleus. The [OII] doublet (372.7 nm) is the classic marker of exactly this: gas ionized by ongoing star formation, radiating.

An **absorption line** appears when the opposite jump happens *in front of* a continuum source rather than beside it: a photon from the continuum, on its way to the observer, carries exactly the energy an electron in some cooler foreground gas needs to jump up a rung, and that electron takes it — removing one photon from the beam at that wavelength and leaving a dip. That needs cooler gas, or a stellar atmosphere, sitting between the continuum's source and the observer, with nothing re-ionizing it before the jump can happen. Ca II K (393.366 nm) is the classic marker of the opposite population: old stars, whose own cooler outer atmospheres absorb calcium's signature out of their own light, with no ongoing star formation anywhere in the galaxy to turn it into emission instead.

That is exactly the distinction [Ch. 2](#) and [Ch. 3](#) already drew without any spectroscopy: a spiral galaxy, still forming stars, carries ionized gas and shows emission; an elliptical, whose star formation stopped long ago ([Ch. 3](#)), has none left to ionize and shows absorption only. Figure 7.1 is a real instance of the second case — five absorption dips and not one emission spike, because there is no ionized gas left in this galaxy to radiate anything. That absence is not a detail; it is the entire reason [Ch. 3](#) can single out ellipticals as this program's lenses on sight, before a redshift is ever measured.

You already know this

A redshift pipeline does not search for one line; it slides a whole expected *pattern* of lines against the data and scores how well the pattern lines up at each trial redshift — matching a fixed multi-feature template against noisy data by sliding it and scoring the overlap. That is a cross-correlation, the identical operation whether the “template” is a galaxy spectrum or a convolutional kernel: the pattern, not any single feature, is what survives noise. [Ch. 12](#) runs exactly this, over a library of galaxy, quasar and star templates, at full DESI scale.

Unlocks

What you can now read

- [Ch. 12 — Redshifts and what a spectrum tells you](#) opens with a skip box that hands you this chapter’s entire content in one sentence: “take it as given that an electron dropping between two fixed energy levels emits or absorbs light at one specific, reproducible wavelength.” You now have the mechanism behind that sentence, why it can be a spike or a dip, and — concretely — where Ch. 12’s own two working numbers, Ca II K at 3933.66 Å for the lens and [OII] at 3727 Å for the source, come from and why those two lines, specifically, are the right diagnostic for an old galaxy and a star-forming one respectively.

Exercises

Exercise 7.1 — The rest of the Balmer series, by hand

Using $\frac{1}{\lambda} = R \left(\frac{1}{4} - \frac{1}{n^2} \right)$ with $R = 1.0967758 \times 10^7 \text{ m}^{-1}$, compute the wavelengths of H β ($n = 4$) and H γ ($n = 5$), and check them against the values below.

Solution

For H β : $\frac{1}{4} - \frac{1}{16} = 0.1875$, so $\lambda = 1/(R \times 0.1875) = 486.274 \text{ nm}$. For H γ : $\frac{1}{4} - \frac{1}{25} = 0.21$, so $\lambda = 1/(R \times 0.21) = 434.173 \text{ nm}$. Same formula, same constant, two more exact predictions — nothing about hydrogen changes between lines, only n does.

Exercise 7.2 — Quantifying the air/vacuum trap

Using the vacuum value 656.470 nm and the air value 656.288 nm, compute the fractional difference as a percentage. Then say, in one sentence, why a redshift pipeline needs to know which convention its line list uses.

Solution

$(656.470 - 656.288)/656.470 = 2.77 \times 10^{-4}$, i.e. 0.0277% of the wavelength itself — small, but a redshift is defined as a wavelength *ratio* ([Ch. 8](#)), so a 0.03% error in the rest wavelength you divide by becomes a 0.03% error in every redshift measured against that line, in the fourth significant digit, with no warning flag raised anywhere downstream.

Exercise 7.3 — Reading Figure 7.1

Figure 7.1 shows five absorption dips and no emission. If you were instead handed the spectrum of a star-forming spiral galaxy over the same wavelength range, which single line would you most expect to flip from absorption to emission, and which line, absent here entirely, would you expect to appear for the first time? Why do Ca II K and Mg b not flip the same way?

Solution

H α is the one most likely to flip: hot ionized gas around young stars emits strongly there, often overwhelming the much weaker absorption from the same transition in the galaxy's own stellar atmospheres. The line that would appear for the first time is [OII] at 372.7 nm, since a quiescent elliptical has no ionized gas anywhere to produce it. Ca II K and Mg b do not flip, because they trace absorption in the light of the whole stellar population, old and young alike — star formation elsewhere in the galaxy adds an emission channel without removing that underlying absorption.

8. Redshift — what it is and why it means distance

This is the chapter the book was built around. Every claim the main guide makes about a real system — which galaxy is in front, which is behind, how much mass sits inside a ring — rests on one measured quantity, z , and the guide never says what it means. Its Ch. 12 defines z and immediately withdraws the definition (“a definition, not yet a physical claim”); its Ch. 13 says what z is *not* (a velocity); its Ch. 15 hands you an integral. Nobody joins them up. This chapter does — including the sentence nothing in the corpus says out loud, why a bigger z means farther away — and closes by locating the guide’s standard test pair, $z = 0.5$ and $z = 2.0$.

What you can skip

You do not need the derivation of $1 + z = 1/a$ from the FRW metric; it is one line in Ch. 13 of the main guide, it needs a metric to write down, and this chapter states the result instead. You do not need the integral that turns z into a distance either — that is Ch. 15, and **astropy** evaluates it numerically because there is no closed form. What you *do* need is one sentence each from Ch. 5 (light takes time) and Ch. 7 (a line’s rest wavelength is a constant of the transition, not a property of the object).

What redshift measures

Chapter 7 established the fact that makes this measurement possible: the wavelength at which a given atomic transition emits or absorbs is fixed by quantum mechanics and nothing else. Every hydrogen atom anywhere emits Balmer-alpha at the same wavelength, whatever galaxy it sits in. So when you find that entire fixed pattern displaced in a real spectrum — every line moved by the *same* multiplicative factor — the displacement is a fact about the light’s journey, not about the atom.

Define that factor. With λ_{emit} the known rest wavelength and λ_{obs} where you actually find the line:

$$z \equiv \frac{\Delta\lambda}{\lambda_{\text{emit}}} = \frac{\lambda_{\text{obs}} - \lambda_{\text{emit}}}{\lambda_{\text{emit}}}, \quad \text{equivalently} \quad 1 + z = \frac{\lambda_{\text{obs}}}{\lambda_{\text{emit}}}.$$

z is dimensionless — a wavelength over a wavelength. It is not a distance, not a speed, and not, yet, anything about the universe. It is a number you read off a plot.

The arithmetic is as light as it looks. A redshift of $z = 0.5$ means every wavelength arrives 50% longer than it left; $z = 2.0$ means 200% longer, three times the emitted wavelength, since $1 + z = 3$. Take the line Ch. 7 derived from the Rydberg formula, Balmer-alpha at 656.47 nm, and put it in a galaxy at $z = 2.0$:

$$\lambda_{\text{obs}} = 656.47 \text{ nm} \times 3 = 1969.41 \text{ nm}.$$

That line left its galaxy at the red edge of the visible band (Ch. 5) and arrives at nearly two micrometres — near-infrared, invisible to the eye and off the red end of a ground-based optical spectrograph. Ch. 12 of the main guide makes the identical observation about its own source, and it is why a redshift pipeline cross-correlates a whole template library rather than hunting for one hard-coded line: which lines are even *observable* depends on the z you are trying to measure.

Two naming traps, cleared now. “Redshift” does not mean the light turns red; it means every wavelength gets longer. A galaxy’s ultraviolet arrives as visible blue, its visible red as infrared — “red” names the *direction* of the shift, not the colour of what lands in the detector. And the z in “ z -band” (Ch. 6) is an unrelated use of the same letter.

Everything above is bookkeeping. It says nothing about how far away the galaxy is — which is exactly why the main guide’s Ch. 12 defines z and then declines to interpret it. The rest of this chapter is the physical claim it deferred.

The Doppler picture

Everyone’s first explanation of redshift is the ambulance siren, and it is worth having provided you know how far it goes. A source moving away emits each successive wave crest from slightly farther off, so the crests arrive spaced wider than they were emitted: longer wavelength, lower pitch. For a source receding at $v \ll c$, the stretch is proportional to the speed:

$$\frac{\Delta\lambda}{\lambda} = \frac{v}{c}, \quad \text{so} \quad z \approx \frac{v}{c}, \quad v \approx cz.$$

This is a real effect, and this program measures it. The stars inside a lens galaxy orbit in every direction at once, so each one’s light picks up its own small Doppler shift and the summed light of the population has its lines *smeared* rather than displaced. That smear’s width is the velocity dispersion σ_v of Ch. 12 — a genuine Doppler measurement, at speeds three orders of magnitude below c where the formula above is excellent.

Now run the same formula on redshifts of the size this program catalogues, using the exact speed of light from Ch. 5, $c = 299,792.458$ km/s. At $z = 0.5$:

$$v = cz = 299,792.458 \times 0.5 = 149,896 \text{ km/s.}$$

Half the speed of light. Uncomfortable, but not yet impossible. At $z = 2.0$:

$$v = cz = 299,792.458 \times 2.0 = 599,585 \text{ km/s} = 2c.$$

Twice the speed of light. No care with the measurement would change that answer, because the answer is not a measurement — it is a formula reporting an impossibility. The impossibility arrives the moment z exceeds one, since $cz > c$ exactly when $z > 1$, and this program routinely works past $z = 1$.

So say it plainly, because the main guide will contradict you if you do not hear it here: **cosmological redshift is not a Doppler shift.** The Doppler picture is a serviceable intuition for what a stretched wave *is*, and at small z it returns the right number, but it is the low- z limit of something else, and the something else is what is true. Ch. 13 runs this identical trap on a real system — the Carousel cluster’s reference plane at $z = 1.432$ — gets $1.432c$, and concludes: “*The correct statement was never a velocity.*”

Name where the analogy breaks, since a simplification you have to unlearn is worse than none. The siren’s shift is imprinted at emission and set by how fast the ambulance moves *through the air* relative to you; nothing further happens to the sound in transit. The cosmological shift is the

opposite on both counts. It is not imprinted at emission — it accumulates over the entire journey — and no galaxy is moving through space at anything like these speeds. The galaxies are, to a good approximation, sitting still.

The cosmological picture

Here is the actual cause. While the light was in flight, space itself expanded, and the wave was stretched by exactly the same factor as everything else. [Chapter 10](#) paints that expansion properly — no centre, nothing moving *through* space, the ruler itself growing. The one piece of bookkeeping needed here is the **scale factor** a : a single dimensionless number for the size of the universe at a given moment, normalised so that $a = 1$ today. It has no meaning in isolation; only the *ratio* of a at two times means anything, and a measured redshift is exactly that ratio:

$$1 + z = \frac{\lambda_{\text{obs}}}{\lambda_{\text{emit}}} = \frac{1}{a(t_{\text{emit}})}, \quad \text{so} \quad a(t_{\text{emit}}) = \frac{1}{1 + z}.$$

That is the entire physical content of a redshift. It reports how large the universe was when the light left, relative to how large it is now. Not a speed — a size then, against a size now. For the pair this chapter is chasing:

$$a(z = 0.5) = \frac{1}{1.5} = 0.667, \quad a(z = 2.0) = \frac{1}{3} = 0.333.$$

The $z = 0.5$ light left when the universe was two-thirds its present size; the $z = 2.0$ light when it was one-third. Notice already that four times the redshift is not four times anything: $1/(1 + z)$ is not linear in z , and those two scale factors differ by a factor of two, not four.

Only the endpoints matter. The stretch accumulates continuously across a journey of billions of years, but the total is exactly the ratio of the size then to the size now. Whatever the expansion did in between — sped up, slowed down, both — leaves no trace in z . That is why z is such a clean thing to measure, and why a survey catalogue stores z and not a history. (The history comes back later: it is what makes the conversion from z to a distance an integral rather than a multiplication.)

And a caveat this book will not paper over. You will find working physicists who maintain that the cosmological redshift *can* be described as an accumulated Doppler shift — the total stretch decomposed into a chain of infinitesimal shifts between neighbouring observers along the photon’s path. In a strict technical sense they are not wrong; this is a live pedagogical argument, not a settled misconception. What that description cannot do is hand you a single recession velocity to put into $v = cz$. The expansion description gives one unambiguous number, $a = 1/(1 + z)$, at every redshift; the velocity description needs a convention for comparing velocities at two widely separated points in a curved spacetime, and general relativity does not supply one. That asymmetry is why the guide states the size ratio and refuses the speed.

Bigger z means farther

Here is the sentence this book exists to say, and that no chapter of the main guide ever says:

A bigger z means the universe was smaller when the light left, which means the light left longer ago, which means the source is farther away.

Three links, four quantities. Take the links one at a time, because exactly one of them is doing work you might not expect.

1. **Bigger $z \rightarrow$ smaller universe then.** Immediate from the previous section: $a = 1/(1+z)$ falls as z rises. No physics beyond the definition.
2. **Smaller then $\rightarrow$ longer ago.** This link needs a fact about the universe’s *history* rather than about light: a has been increasing throughout the entire span this book covers. Given that, “the universe was one-third its present size” names exactly one moment in time. Had the universe ever contracted and re-expanded, a given a would name two moments and the chain would break here. It has not, over any epoch we can see, and [Ch. 11](#) runs a back to zero.
3. **Longer ago $\rightarrow$ farther away.** Light covers c per unit time and no more, so more time in flight is more distance covered. This is [Ch. 5](#)’s point — looking far away *is* looking into the past — read in the other direction.

Now the numbers, for the pair the main guide uses as its standard test case and never once locates. The light from a $z = 0.5$ lens galaxy left 5.04 Gyr ago; the light from a $z = 2.0$ source left 10.24 Gyr ago. Set those against the age of the universe, which in the cosmology this repository fixes by fiat — `FlatLambdaCDM(H0=70, Om0=0.3)`, asserted everywhere — is 13.47 Gyr. (That is shorter than the age you will see quoted from CMB-derived parameters. The gap is not a rounding error: it is the Hubble tension in disguise, and [Ch. 14](#) is where it stops being a footnote.) So the lens light left 37.4% of the universe’s whole life ago, and the source light 76.0% of it ago. The source you are modelling is not a contemporary object photographed at distance; it is a galaxy as it existed before three-quarters of cosmic history had happened, and no telescope can show you what became of it.

Notice the compression: four times the redshift bought only about twice the lookback time. It has to. Lookback time is bounded above by the age of the universe, so as z climbs the curve flattens against that ceiling and never touches it. z is not a linear odometer.

You already know this

Each arrow in that chain is a deterministic function of the one before, so everything downstream is a function of z alone. That is why a catalogue of a million galaxies stores z and not four columns: z is the primary key, and scale factor, lookback time and every flavour of distance are derived views, computed on demand from that one number. A pipeline that reports z has already committed to all four, whether or not it says so. Note the limit of that framing, though: being a function of z is all a derived view needs, and it does not make a column *monotone* in z . The last section of this chapter exhibits one that is not.

z in real numbers

One honest complication first. Once space has been expanding while the light was in flight, “how far away” needs a qualifier, because the source has kept receding since its light left. There is no single “the distance,” and this is not a technicality — the main guide’s [Ch. 15](#) derives three distances from one metric because three different questions get three different answers. Two of them matter here.

Comoving distance is how far apart you and the source are *today*, measured on a grid that expands along with the universe. Getting it from z needs the integral this book hands off; here are its answers:

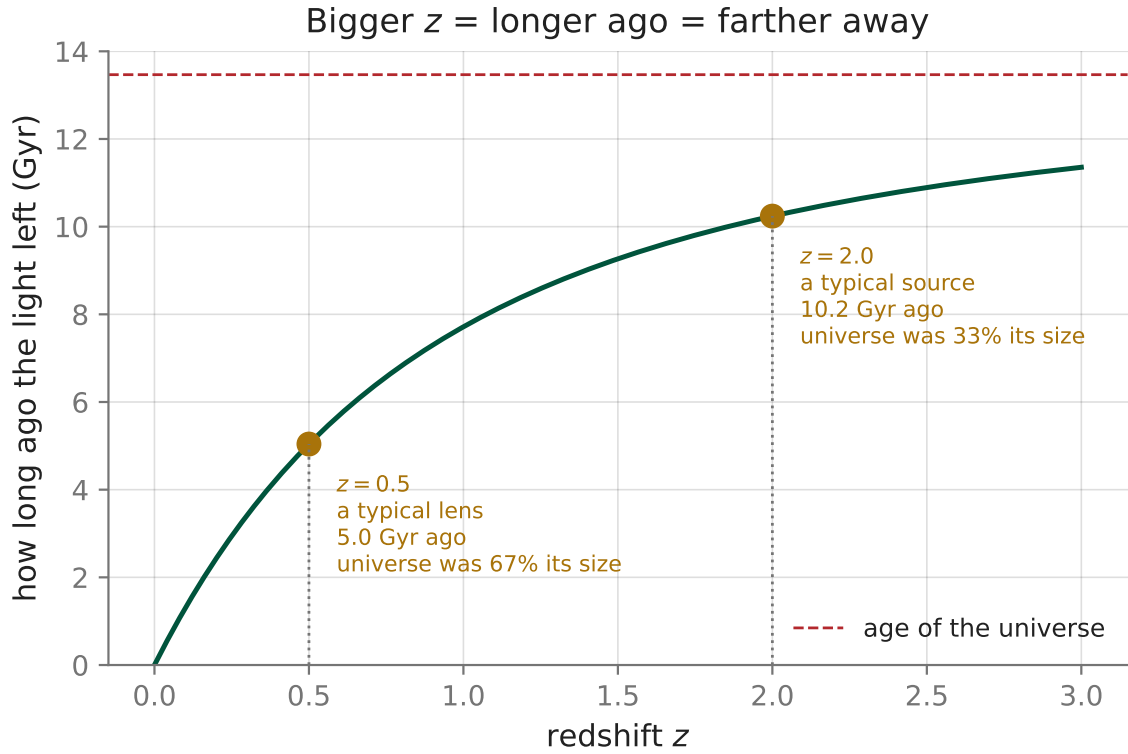


Figure 8: The chain on one axis. Horizontal: a redshift, the thing you actually measure. Vertical: how long ago the light left. The marked points are the main guide’s standard test pair — $z = 0.5$, a typical lens, and $z = 2.0$, a typical source — each annotated with its lookback time and with the size the universe had when that light departed (67% and 33% of today’s). The dashed line is the age of the universe in this repository’s assumed cosmology; the curve rises toward it and never reaches it, which is why a fourfold increase in redshift does not buy a fourfold increase in lookback time. Nothing here is drawn by hand: the curve is **astropy** integrating the expansion history, the calculation [Ch. 15](#) writes as an integral.

$$D_C(z = 0.5) = 1888.63 \text{ Mpc}, \quad D_C(z = 2.0) = 5179.86 \text{ Mpc}.$$

The second is, to the digit, the main guide’s own $D_C(2.0) = 5179.862 \text{ Mpc}$ — arrived at there by an integral.

A megaparsec is an abstraction; the Milky Way is not. Lay Milky Ways edge to edge and count:

$$z = 0.5 : 61,599 \text{ Milky Ways}, \quad z = 2.0 : 168,945 \text{ Milky Ways}.$$

Read those counts to one digit and no further. The distances are solid; the ruler is not. [Ch. 1](#) was explicit that the stellar disk has no edge — it fades — so “ 10^5 light-years across” is a convention carrying about a factor of two of slack. The trailing digits are the division’s, not the universe’s; what survives is the exponent, and the exponent is the point. It is still the picture the main guide never provides for the two numbers it leans on hardest: the lens sits some tens of thousands of Milky Ways away, the source a few times farther, in a direction that happens to have a galaxy in the way. None of that separation is visible in the data — both land in the same handful of pixels.

Angular-diameter distance is the other one, and it is what every lensing formula in the corpus actually consumes: the distance that makes ordinary Euclidean angle arithmetic come out right, physical size divided by observed angle. It is not the comoving distance:

$$D_A(z = 0.5) = 1259.08 \text{ Mpc}, \quad D_A(z = 2.0) = 1726.62 \text{ Mpc}.$$

Look at what happened to the gap. Measured comoving, the source is far more distant than the lens; measured by angle, the two are much closer. The relation is $D_A = D_C/(1+z)$, and the divisor is larger for the more distant object, so the angular ladder is squeezed relative to the comoving one. Physically: the source’s light left when everything was closer together, so the galaxy subtends a *larger* angle than its present-day distance would suggest.

Push that far enough and the squeeze becomes a reversal. Angular-diameter distance does not grow without bound with redshift — it rises, peaks, and then declines, so a galaxy at very high z can subtend a larger angle than a nearer one. This does not damage the chain: “bigger z means farther away” holds without exception for lookback time and comoving distance, the two senses in which anyone means it. But you do have to say which distance you mean, every time. [Ch. 15](#) is where that discipline gets enforced, including its sharpest consequence — that these distances do not subtract, so the lens-to-source distance is emphatically *not* $D_s - D_d$ ([Ch. 15](#)).

That is the whole chain. A spectrum gives one dimensionless ratio; the ratio gives the size of the universe at emission; the size gives the moment; the moment gives the distance. Four readings, one measurement, and nothing in between that needs a velocity.

Unlocks

What you can now read

- [Ch. 12 — Redshifts and what a spectrum tells you](#) opens by promising that z “tells you how far away something is” and then takes the promise straight back: “*This is a definition, not yet a physical claim... treat z as a wavelength ratio you read off a spectrum, full stop.*” It never returns to collect. You now have the physical claim it

deferred, and when it puts a redshifted Balmer-alpha “*deep in the near-infrared,*” you have already run that calculation yourself.

- **Ch. 13 — The expanding universe and redshift** states the conclusion — “*Not a velocity through space — a size, then, compared against a size, now*” — and runs the naive- cz trap on the Carousel cluster’s reference plane at $z = 1.432$ to get $1.432c$. You met that trap here at $z = 2.0$, where it returns $2c$, so the chapter’s punchline lands as confirmation rather than as a surprise to take on trust. Its Exercise 13.2 computes $a = 1/(1 + z)$ for $z = 0.5$ and $z = 2.0$ and gets 0.667 and 0.333 — your pair, your numbers.
- **Ch. 15 — Distances that do not add** derives comoving, angular-diameter and luminosity distance from the FRW metric and quotes $D_C(2.0) = 5179.862$ Mpc as the output of an integral. You already know what that number *is*: of order 10^5 Milky Ways, and light that left when the universe was a third its present size. You also know why the chapter needs three distances instead of one — the piece it assumes rather than motivates.

Exercises

Exercise 8.1 — Where does the line land?

Balmer-alpha’s rest wavelength is 656.47 nm. Compute where it arrives from a galaxy at $z = 2.0$, name the part of the spectrum it lands in, and say why a pipeline cannot hunt for one named line.

Solution

$1 + z = 3$, so $\lambda_{\text{obs}} = 656.47 \times 3 = 1969.41$ nm — near-infrared, past the red edge of the visible band and outside a ground-based optical spectrograph’s range. Which lines are observable depends on z , the very quantity being measured, so the search runs over a whole template pattern: at any z where one line has left the band, another has usually entered it.

Exercise 8.2 — The naive velocity, and where it dies

Read z as a recession speed via $v = cz$, with $c = 299,792.458$ km/s. Compute v at $z = 0.5$ and $z = 2.0$. At what redshift does this reading first predict something impossible, and what exactly is wrong — the measurement, the physics, or the formula?

Solution

At $z = 0.5$: $v = 299,792.458 \times 0.5 = 149,896$ km/s, half the speed of light. At $z = 2.0$: $v = 599,585$ km/s, which is exactly $2c$. The reading first predicts a superluminal speed at $z > 1$, since $cz > c$ exactly when $z > 1$ — no computation needed. What is wrong is the formula, and specifically the domain it is used on: $v \approx cz$ is only ever the small- v approximation to something else. Nothing is measured wrongly and no physics is violated, because no galaxy is moving through space at these speeds at all. The correct statement is $a = 1/(1 + z)$: exact at every z , and not a speed.

Exercise 8.3 — Close the chain yourself

For $z = 0.5$ and $z = 2.0$: compute the scale factor at emission, state in one sentence what each one says about the universe, and then read the lookback time for each off Figure 8.1. Finally, explain why quadrupling the redshift did not quadruple the lookback time.

Solution

$a = 1/(1+z)$ gives $1/1.5 = 0.667$ and $1/3 = 0.333$: the $z = 0.5$ light left when the universe was two-thirds its present size, the $z = 2.0$ light when it was one-third. The lookback times are 5.04 Gyr and 10.24 Gyr — that is 37.4% and 76.0% of the universe’s whole 13.47 Gyr life ago. Quadrupling z roughly doubled the lookback time because lookback time is bounded above by the age of the universe: the curve has a ceiling it approaches and cannot cross, so equal steps in z buy ever-smaller steps in time. $a = 1/(1+z)$ is not linear, and neither is anything downstream of it.

Exercise 8.4 — Two distances, one source

For the $z = 2.0$ source, the comoving distance is 5179.86 Mpc and the angular-diameter distance is 1726.62 Mpc. Compute their ratio and identify it. Then explain, without any integral, why the angular-diameter distance is the smaller of the two.

Solution

$5179.86/1726.62 = 3.000$, which is exactly $1+z$: the relation is $D_A = D_C/(1+z)$, and nothing else. D_A is the smaller one for the same reason z exists at all. The light left when the universe was a third its present size, so the source was physically much closer to us *then*, and the angle it subtends was set at that moment, not this one. An angular-diameter distance answers “what distance makes the observed angle come out right,” and that answer is anchored to the geometry at emission.

9. The distance ladder

Chapter 5 left an ambiguity it could not resolve: flux alone never separates “dim and nearby” from “luminous and far away.” Chapter 8 gave you redshift, read straight off a spectrum and needing no distance at all — but turning a redshift into a distance requires the expansion rate, and measuring the expansion rate requires a distance obtained some other way first. This chapter is the other way, and it is not one method but four, stacked: geometry for the nearest stars, a class of pulsating star for nearby galaxies, exploding white dwarfs for most of the observable universe, redshift for everything beyond. Each is calibrated by the one below it and reaches farther — which is what the word *ladder* is doing, and also the field’s central liability: every rung inherits the rung below’s zero point, and its mistakes along with it.

What you can skip

You do not need stellar-interior physics to accept that a certain kind of star pulsates with a period set by its luminosity, or nuclear physics to accept that white dwarfs detonate at a near-fixed mass. Take both as empirical, the way you’d take a calibration curve as given. You also already own everything structural here: a ladder is a bootstrap, each rung is an estimator fitted against labels the previous rung produced, and the failure mode follows from that description alone. The astronomy is only *which* four estimators, and what each one costs.

Nothing comes with a ruler

Ch. 9 of the main guide opens with the sentence this section is named for: “*Nothing astronomical comes with a ruler. Every telescope measures a direction, not a distance.*” That is not a limitation of current instruments. A detector records where on the sky a photon arrived and how many arrived — a direction and a brightness. Neither is a distance, and no improvement converts one into the other, because the information is not in the light: a sky image is a projection, and the coordinate you want is the one the projection threw away.

So distance is never measured directly beyond the solar system. It is *inferred*, from some other observable that distance controls, and only two are in wide use. The first is geometry: look at the same object from two places and its apparent direction shifts by an amount that depends on how far away it is. The second is the inverse-square law of Ch. 5: if you know an object’s intrinsic luminosity L independently, a measured flux F pins the distance, because $F = L/(4\pi d^2)$ then has one unknown left. Geometry is the honest one — it assumes nothing about the object — and it runs out almost immediately. Everything above it trades self-sufficiency for reach.

Parallax

Hold a finger at arm’s length and look at it with one eye, then the other. It jumps against the background, and the jump is bigger when the finger is closer. That is **parallax**, the only distance measurement in astronomy that is pure geometry.

You already know this

Parallax is stereo triangulation with a known baseline — structure from motion, where the camera motion is given rather than estimated. The depth solve is the one your rectified-stereo code does, with the same pathology: disparity goes as $1/d$, so depth error grows as d^2 and the estimator degrades catastrophically, not gracefully, with range.

The baseline is Earth’s orbital radius, one **astronomical unit** (1 AU): over six months Earth moves to the opposite side of its orbit, and a nearby star shifts against the far more distant background. Half that shift, as an angle, is the star’s **parallax angle** p , and because p is always minuscule the small-angle relation reduces the triangle to a division — p in radians is $(1 \text{ AU})/d$.

What astronomers did with that division is the origin story the main guide omits while deriving its arithmetic. Rather than carry radians, define the distance unit so the formula has no constant in it. A **parsec** — a contraction of *parallax* and *arcsecond* — is *defined* as the distance at which 1 AU subtends exactly one arcsecond ($1''$, one 3600th of a degree). Feed that back into the triangle and it returns itself:

$$d [\text{pc}] = \frac{1}{p [']}. \quad \text{or} \quad d [\text{pc}] = \frac{1}{p ['] \times 3600}.$$

A parsec is not a length someone chose; it is the reciprocal of the angle astronomers actually measure. The factor the main guide leans on so heavily, $1 \text{ rad} \approx 206,265''$, only restates that in radians and carries no physics.

Run it on Proxima Centauri, the nearest star to the Sun, which has the largest parallax of any star:

$$p = 0.7687'' \quad \Rightarrow \quad d = \frac{1}{0.7687} \approx 1.30 \text{ pc}.$$

In [Ch. 1](#)’s units that is 4.24 light-years — out of one measured angle and one definition, with nothing assumed about Proxima itself. Note how small the angle is: the *largest* stellar parallax in the sky is under one arcsecond, roughly what a coin subtends from several kilometers away.

That is why this rung ends where it does. Gaia measures parallaxes to about 20 millionths of an arcsecond for its best targets, which puts the ceiling at

$$d = \frac{1}{20 \times 10^{-6}} = 50,000 \text{ pc},$$

or about 163,000 light-years. Against this book’s ruler — a Milky Way 100,000 light-years across — pure geometry, pushed to the limit of the best instrument ever built for it, reaches 1.63 Milky Way diameters. Barely past our own galaxy’s far edge, and not one step toward any other — and that ceiling is generous, since it is where the parallax merely *equals* its own error bar. Every distance in this program therefore rests on something less trustworthy than geometry.

Standard candles

An object whose L is knowable independently — not from its brightness, which is the thing you are trying to explain, but from something else about it — is a **standard candle**. Finding them is most of what the ladder is.

First the unit needs fixing, because the main guide hands you half of it. [Ch. 9](#) defines the **apparent magnitude** m — Pogson’s backwards logarithmic flux scale, in which $m_1 - m_2 = -2.5 \log_{10}(F_1/F_2)$ and brighter means *smaller*. It describes arriving photons, so it mixes luminosity and distance together and is not a property of the object at all.

The fix is a convention, and it is the piece the main guide never states. Define the **absolute magnitude** M as the apparent magnitude the object *would* have if it sat at a fixed reference distance of exactly 10 parsecs. That makes M a restatement of L in magnitude units — a property of the object, distance divided out by fiat. Now put the two together. The flux at distance d and the flux at 10 pc are in the ratio $(10\text{ pc}/d)^2$ by inverse square, so Pogson’s definition gives

$$m - M = -2.5 \log_{10} \left(\frac{10\text{ pc}}{d} \right)^2 = 5 \log_{10} \left(\frac{d}{10\text{ pc}} \right).$$

That quantity, $m - M$, is the **distance modulus**: one number, in magnitudes, that *is* the distance. It is the formula [Ch. 15](#) drops on you fully formed, M and all, having defined neither — and there is nothing in it beyond the reference-distance convention above.

The first real candle was found by Henrietta Leavitt in 1912. **Cepheids** are stars that pulsate, brightening and dimming over days to months, and Leavitt noticed that in a population all at one distance, the slow pulsators were uniformly the bright ones. The period tracks the luminosity — and a period is a *clock*, immune to distance, to dust, and to everything else that corrupts a brightness. So a Cepheid announces its own L , hence its own M , and its measured m gives d .

Except that the period–luminosity relation yields only a *relative* luminosity until someone supplies the zero point, and that is where the ladder becomes a ladder: the zero point comes from Cepheids near enough for parallax. Rung 1 calibrates rung 2. It is also where honesty is owed — that zero point depends on the star’s chemical composition and on how much dust has reddened it, both imperfectly known, and arguments over exactly these corrections are a live part of why the numbers at the top are still disputed.

Supernovae

Cepheids are stars, so they run out too: past a few tens of megaparsecs no telescope resolves an individual one. The next rung has to be brighter, by a margin no ordinary star can supply.

A **Type Ia supernova** is a white dwarf — the burnt-out core of a star like the Sun, held up not by fusion but by electron degeneracy pressure — that accretes matter from a companion until it crosses a critical mass and detonates. That near-fixed detonation mass is the entire point: the fuel is the same every time, so the explosion is nearly the same brightness every time. At peak, a Type Ia reaches

$$M \approx -19.3,$$

which is a single exploding star briefly outshining all [10¹¹ ordinary stars](#) of its host galaxy. That is what buys the reach.

“Nearly the same” is carrying weight. Type Ia peak luminosities scatter, and the scatter is real. What rescues them is an empirical correlation — intrinsically brighter ones fade more slowly — so the light curve’s own decline rate predicts the offset and lets you correct it out. A Type Ia is not a standard candle so much as a *standardizable* one. And as with Cepheids the absolute scale is not self-supplied: $M \approx -19.3$ is known only because Cepheids were found in galaxies that also hosted Type Ia supernovae. Rung 2 calibrates rung 3.

Run it at this program’s fiducial lens redshift, $z = 0.5$, which [Ch. 8](#) put in physical terms. The luminosity distance there, in the cosmology this repository fixes by fiat, is

$$D_L(z = 0.5) \approx 2833 \text{ Mpc.}$$

Convert to parsecs, take the modulus, and add it to M :

$$m = M + 5 \log_{10} \left(\frac{2.833 \times 10^9 \text{ pc}}{10 \text{ pc}} \right) = -19.3 + 42.26 = 22.96.$$

The brightest standard object in astronomy, at the redshift of an ordinary lens galaxy, arrives at magnitude 22.96 — far below naked-eye visibility, and a fair reminder of what the inverse-square law does over 2833 Mpc. On this book’s ruler that distance is about 92,000 Milky Way diameters, against the 1.63 where geometry gave out (Exercise 9.4 makes you form the ratio). The entire cost of that factor is the phrase “we knew M from somewhere else.”

One caveat, not pedantic. D_L is a *luminosity* distance, not a length you could lay a tape along: by $z = 0.5$ the expansion has already made “the distance” ambiguous — the main guide’s [Ch. 15](#) carries three of them, disagreeing by factors of $(1 + z)$ — and the modulus consumes exactly the one that makes $F = L/(4\pi D_L^2)$ true. Exercise 9.3 takes up the second caveat: that number came *from* a cosmology.

Errors compound

Read the figure as a dependency graph and the liability is immediate. Each overlap is a zero-point transfer: parallax fixes the period–luminosity zero point, Cepheids in galaxies that hosted a Type Ia fix $M \approx -19.3$, and supernovae far enough out that expansion dominates their redshift fix the expansion rate. Nothing at the top was ever measured against a ruler. It was measured against something that was measured against something that was measured against Earth’s orbit.

Magnitudes at least make the propagation easy to see. M enters the modulus additively, so a zero-point error is a constant offset in magnitudes — and by $m - M = 5 \log_{10}(d/10 \text{ pc})$, that offset is a constant *multiplicative* error in every distance the rung reports. It does not average down and it does not show in the scatter (Exercise 9.4).

You already know this

This is self-training, exactly. Rung 1 is your labelled set — trustworthy, and far too small to cover the domain you care about. Each higher rung is a model trained on pseudo-labels the rung below emitted, then deployed on a distribution the labelled set never reached. You know what happens: variance shrinks with each generation and bias does not, because a systematic in the teacher is not noise the student can see. Every diagnostic the student has is computed against inherited labels, so it comes back clean. Only a measurement from outside the chain catches it.

That last sentence is what [Ch. 14](#) is about. A second, independent route reads the expansion rate out of the early universe and never touches this ladder — and the two answers disagree by more than either side’s quoted uncertainties allow. The ladder has been rebuilt end to end by people hunting for exactly the compounding error above, and it has not been found; that failure is why

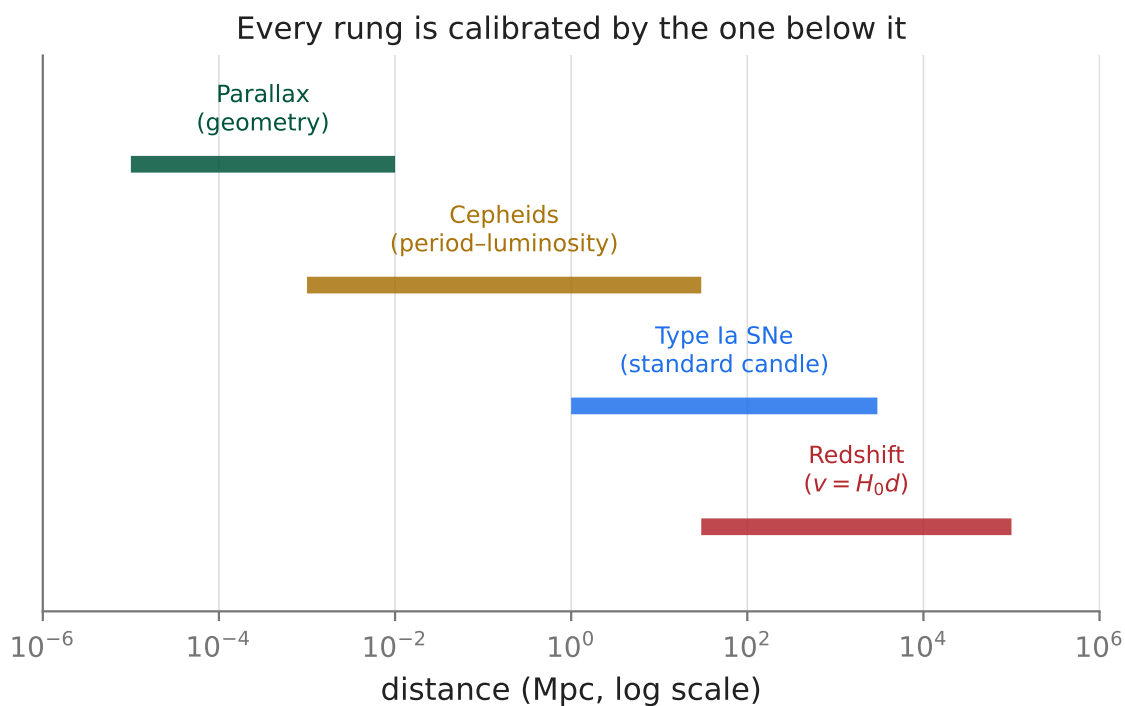


Figure 9: The four rungs and the distances each works over, on a logarithmic axis in megaparsecs. The ladder is drawn top-down — parallax, rung 1, is the *top* bar — so the panel title’s “the one below it” means earlier in the chain, not lower on the page. There is no vertical axis because there is no vertical quantity; the stacking is only the ladder’s order. What matters is that consecutive bars *overlap*: each method’s near end reaches back into territory the rung before it already covers. That overlap is not redundancy; it is where the calibration happens.

the disagreement is taken seriously rather than dismissed. The punchline for this reader is [Ch. 14](#): strong lensing gets a distance from geometry and a light-travel-time difference, borrowing nothing from Cepheids and nothing from supernovae — a third witness, outside both chains, which makes this program’s own field an arbiter of the argument rather than a spectator to it.

Unlocks

What you can now read

- [Ch. 9 — Arcseconds, magnitudes, and the units of the sky](#) defines the parsec as “the distance at which one astronomical unit — Earth’s orbital radius ... — subtends an angle of exactly 1””, and its own exercises call that “purely a statement about the geometry of one right triangle.” Both true, and neither says whose triangle, why anyone would measure that angle, or what the unit is named after — the word *parallax* does not occur in the chapter. You now have the missing half: the angle is a parallax, the baseline is Earth’s own orbit, and d in parsecs is $1/p$ in arcseconds by construction. Its [magnitudes section](#) has the same gap: it defines apparent magnitude m from Pogson’s law in full, and stops there.
- [Ch. 15 — Three distances, and why they do not add](#) states that “ D_L is what a Type Ia supernova’s distance modulus, $m - M = 5 \log_{10}(D_L/10 \text{ pc})$, actually consumes” — the only place an absolute magnitude appears in twenty-nine chapters, the guide’s M being a mass everywhere else, inside a formula whose other two ingredients it likewise never explains. You now have all three, plus the question that formula raises and does not touch: where a number like -19.3 comes from, and why the answer is a chain of calibrations rather than a measurement.

Exercises

Exercise 9.1 — parallax, from the nearest star to the wall

Proxima Centauri’s parallax is $0.7687''$; Gaia’s best precision is about 20 millionths of an arcsecond. Give Proxima’s distance in parsecs and light-years, then the distance at which a parallax angle equals Gaia’s precision, and compare that to the Milky Way’s 100,000-light-year diameter. Two follow-ups: why did the parsec answer need no astronomical constant while the light-year answer did, and why is the Gaia reach optimistic?

Solution

Proxima: $d = 1/p = 1/0.7687 \approx 1.30 \text{ pc}$, or 4.24 light-years. Gaia’s ceiling: $d = 1/(20 \times 10^{-6}) = 50,000 \text{ pc}$, or 163,000 light-years, which is 1.63 Milky Way diameters.

The parsec answer is free because the unit was *defined* to make it free: no physical quantity — not Earth’s orbit, not the speed of light — need be known numerically. Light-years and orbital radii are unrelated, so bridging them takes a measured constant. The Gaia reach is optimistic twice over: the parallax there merely equals its own uncertainty, and the quoted precision is a best case, not a typical one. Either way, geometry alone cannot leave the Milky Way.

Exercise 9.2 — derive the distance modulus

Starting only from Pogson's definition, $m_1 - m_2 = -2.5 \log_{10}(F_1/F_2)$, the inverse-square law, and the convention that M is the apparent magnitude an object would have at 10 pc, derive $m - M = 5 \log_{10}(d/10 \text{ pc})$. Then say what breaks if you pick a different reference distance.

Solution

Inverse square gives $F(d)/F(10 \text{ pc}) = (10 \text{ pc}/d)^2$. Pogson with $m_1 = m$, $m_2 = M$:

$$m - M = -2.5 \log_{10} \left(\frac{10 \text{ pc}}{d} \right)^2 = -5 \log_{10} \left(\frac{10 \text{ pc}}{d} \right) = 5 \log_{10} \left(\frac{d}{10 \text{ pc}} \right).$$

The -2.5 times the exponent 2 is where the 5 comes from; the sign flips when the ratio is inverted. Nothing breaks with a different reference distance — the derivation goes through for any d_0 , giving $m - M = 5 \log_{10}(d/d_0)$, shifting every published M by a constant. The 10 pc is convention, not physics.

Exercise 9.3 — a supernova at the guide's own lens redshift

A Type Ia peaks at $M \approx -19.3$. At $z = 0.5$ the luminosity distance is $D_L \approx 2833 \text{ Mpc}$. Compute the distance modulus and the apparent magnitude. Then explain why this calculation, unlike Exercise 9.1, is not independent of cosmology.

Solution

In parsecs, $D_L = 2.833 \times 10^9 \text{ pc}$, so $m - M = 5 \log_{10}(2.833 \times 10^9/10) \approx 42.26$ and $m = -19.3 + 42.26 \approx 22.96$. The dependence enters through D_L : a redshift is not a distance, and turning $z = 0.5$ into 2833 Mpc requires an assumed expansion history — here the guide's fixed $H_0 = 70$, $\Omega_m = 0.3$ model. Exercise 9.1 assumed nothing beyond one triangle; this one assumes a cosmology. Which is why supernovae get used in the *other* direction in practice: measure m , take M from the ladder below, and let D_L against z *determine* the expansion history rather than presuppose it.

Exercise 9.4 — what the hand-off buys, and what it costs

Parallax pushed to Gaia's precision reaches 1.63 Milky Way diameters; one Type Ia at $z = 0.5$ sits at 92,000. Form the ratio, then state exactly what had to be assumed to buy that factor — and why a zero-point error in that assumption would survive any amount of extra data.

Solution

The ratio is about 57,000, bought with exactly one assumption: that M was already known, from a chain of calibrations terminating at Earth's orbit. An error in that M is an offset, not noise — it moves every distance the rung reports by the same factor, so averaging cannot touch it and the residuals look exactly as they should.

10. The expanding universe

[Chapter 8](#) left you holding a claim without a mechanism: a distant galaxy’s spectrum comes back with every line shifted toward longer wavelengths, the shift grows with distance, and the reason is *not* that the galaxy is flying away from us through space. This chapter supplies the mechanism. It is one formula — $v = H_0 d$ — and one picture, and the picture does more work than the formula. By the end you should be able to say why a law under which everything recedes from us is not a claim that we sit anywhere special, what is expanding given that it is not the galaxies, and why no number this law returns ever breaks a speed limit. [Chapter 13 of the main guide](#) writes all of it down as a function $a(t)$ and extracts Hubble’s law from a one-line Taylor expansion. That derivation means nothing without this chapter.

What you can skip

No general relativity, no metric, no Friedmann equation — [Ch. 14 of the main guide](#) does all three and nothing here needs any of them. Skip the fitting machinery too: Figure 10.1 is a straight line through scattered points. What is worth your attention is what the slope *means*, why the scatter is there and is not instrument noise, and the line of algebra in “There is no centre”, which is the whole argument.

Hubble’s law

Measure a galaxy’s distance by one of the rungs in [Ch. 9](#), measure its redshift off its spectrum, read that redshift as a speed for the moment, and plot one against the other. The points fall on a line through the origin:

$$v = H_0 d.$$

That is **Hubble’s law** — named for Edwin Hubble, who measured it, though the IAU now calls it the Hubble–Lemaître law, since Georges Lemaître wrote it down first. Everywhere this program touches cosmology it fixes the slope at

$$H_0 = 70 \text{ km/s per Mpc},$$

megaparsecs being millions of parsecs, the unit [Ch. 1](#) built. Where that value comes from, and the fact that the two best ways of measuring it disagree by more than either one’s error bars, is [Ch. 14](#). Take it as given here.

Read the units before anything else, because they are the tell. Velocity over distance is 1/time. H_0 is a **rate, not a speed and not a length** — the fractional amount by which a distance grows per unit time. It has no direction and is attached to no object. That single observation is what the rest of this chapter unpacks. Three readings of it, all arithmetic:

- A galaxy at 100 Mpc recedes at $70 \times 100 = 7000 \text{ km/s}$.
- **Hubble time**, $1/H_0 = 13.97 \text{ Gyr}$ — invert a rate, get a time.
- **Hubble distance**, $c/H_0 = 4283 \text{ Mpc}$ — the km/s cancels exactly, leaving a pure length: the characteristic scale of the observable universe, the top rung of [Ch. 1](#)’s ladder, which [Ch. 1](#) converts into Milky Ways laid end to end so that it means something.

The Hubble time is the most over-read number in introductory cosmology. If the expansion had always run at today's rate, everything would have been on top of everything else exactly $1/H_0$ ago, and 13.97 Gyr would be the age of the universe. It lands within a few percent of the real answer, which is what makes it tempting. It is **not** the age. The rate has not been constant: for most of cosmic history gravity pulled the expansion back, and more recently something has been pushing it faster (Ch. 13). A rate that changes cannot be inverted into an age by dividing once. Ch. 11 does the honest version and gets a number close to 13.97 but not equal to it; the gap is physics, not rounding.

Real galaxies do not sit exactly on the line, and the reason is not measurement error. Each also carries a **peculiar velocity** — ordinary motion through space, the accumulated result of gravitational tugs from its neighbours — of order 900 km/s. That wobble is the same size no matter how far away the galaxy is, so its *fractional* importance collapses as d grows. Put the two side by side: 900 km/s of peculiar motion against 7000 km/s of Hubble flow at 100 Mpc. Nearby, the wobble wins outright.

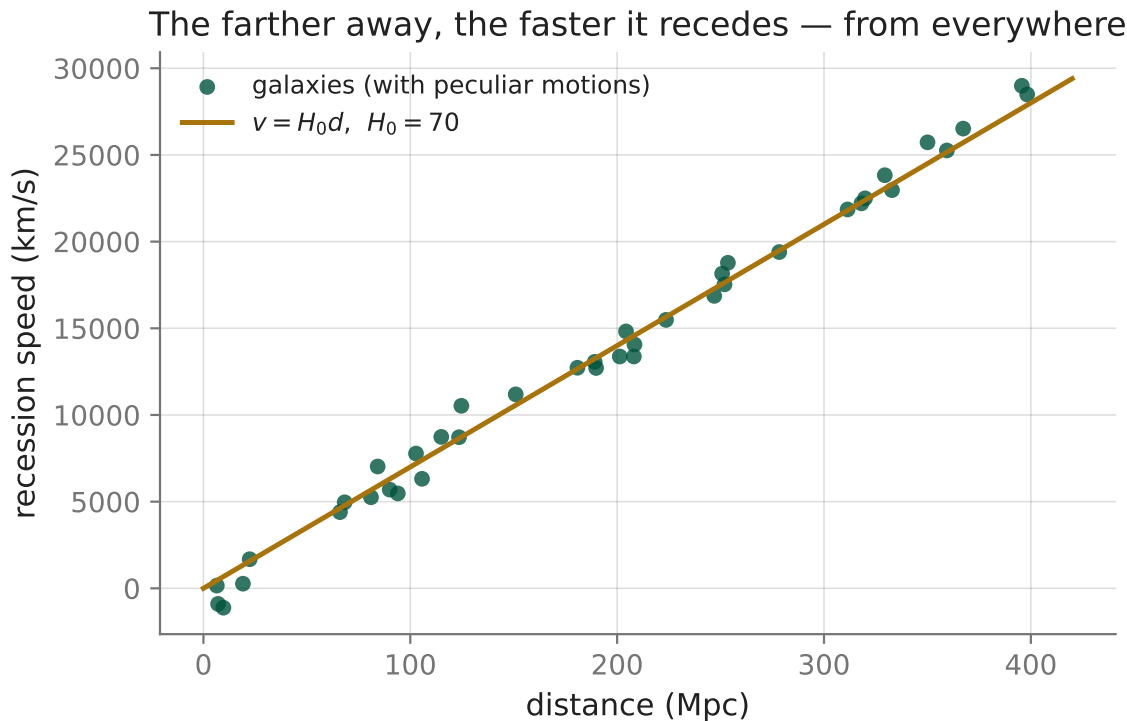


Figure 10: Forty mock galaxies: distances drawn uniformly across the plotted range, speeds computed as $H_0 d$ and then perturbed by a Gaussian peculiar motion of 900 km/s. A simulation, not a measurement — drawn *from* the law, not evidence *for* it. Its job is the scatter, which is the same size everywhere: invisible at the right-hand edge, larger than the signal itself at the left, where two points land below zero because those galaxies are approaching. The title is the claim this chapter has to earn — every galaxy plotted here would draw the same plot, with itself at the origin.

The left edge is not a defect of the figure. It is why you cannot measure H_0 from your own neighbourhood, and why the distance ladder has to reach as far as it does. Andromeda, the nearest large galaxy to the Milky Way, is *blueshifted* — falling toward us, because at that separation the Local Group's own gravity beats the expansion outright. If you believed the expansion were everything flying apart, that one fact would refute you.

The raisin bread

Here is the standard analogy, which is good as far as it goes. Put bread dough with raisins scattered through it in the oven. The dough rises: every part of it swells by the same fractional amount, all at once. Sit on any raisin and watch. Every other raisin moves away from you, and a raisin twice as far away moves away twice as fast — because there is twice as much dough between it and you, and every bit of that dough is swelling at the same rate. That is Hubble’s law, and the loaf has derived the *linearity* of it, the feature that matters most, from something you can hold in your hands.

Two things the analogy gets right, and they are the two worth keeping:

- **No raisin moves through the dough.** Each sits in the same place relative to the dough around it, start to finish. Separations grow because the dough between them grows. There is no propulsion anywhere in this story.
- **The raisins do not expand.** They are held together by something stronger than the dough’s swelling — which maps exactly onto the real thing: atoms, the Solar System, the Milky Way, the Local Group are all bound by gravity or by forces stronger still, and none of them grows.

Now the part every popular account leaves out. **The analogy breaks, and it breaks at the centre.**

A loaf has an edge, and it has a middle. The raisins near the crust really are moving outward through the oven’s air, away from a particular raisin near the middle — and a raisin could work out where that middle is by looking around, because raisins near the edge see dough on one side and empty oven on the other. That asymmetry is a compass. The loaf has a preferred point and a boundary, and its expansion is an expansion *into* something: the oven, already there.

The universe has none of that. No edge to find, no middle to be displaced from, no oven to grow into. The observable universe does have a horizon — a limit set by how far light has had time to reach us since the beginning — but that is a fact about *our vantage point*, not a wall, and every observer has their own. Do not read the Hubble distance as that horizon: it is a characteristic scale for the observable region, not its radius, because light has been crossing *expanding* space the whole way and whatever emitted it now sits several times c/H_0 from us (Ch. 1 draws the same line). Whether the universe as a whole is finite or infinite is a question astronomy genuinely cannot answer: the measurements are consistent with infinite, and equally consistent with finite but far larger than anything we can see. The loaf, by construction, has an answer. That is where you put the bread down.

There is no centre

If everything recedes from us, and the far things recede faster, why are we not at the centre? The answer is not philosophical and it is not an appeal to modesty. It is one line of algebra, and it works because the law is linear.

Put yourself at the origin. A galaxy at position $\mathbf{d}$ recedes at $\mathbf{v} = H_0\mathbf{d}$ — the law in vector form, H_0 acting as a scalar on the position. Now stand on a galaxy at $\mathbf{b}$, itself receding from you at $H_0\mathbf{b}$. What does an astronomer there measure looking at the first galaxy? Subtract:

$$\mathbf{v}' = H_0\mathbf{d} - H_0\mathbf{b} = H_0(\mathbf{d} - \mathbf{b}) = H_0\mathbf{d}',$$

where $\mathbf{d}'$ is the first galaxy's position *as measured from* $\mathbf{b}$. Same law, same H_0 , no trace of the origin you started from. Every observer recovers the identical constant of proportionality, so every one of them is equally entitled, and equally unentitled, to call themselves the centre: measuring Hubble's law from Earth tells you nothing about where Earth is. That is what Figure 10.1's title asserts, and the algebra above is the whole proof.

You already know this

The argument turns entirely on the law being **linear** — H_0 acting as a scalar on the position, with no other dependence on $\mathbf{d}$. Linearity is what makes the subtraction close: the same function, applied to the new separation, comes back. Try anything else — a law where speed went as distance *squared*, say — and it does not close: the law an observer at $\mathbf{b}$ writes down has a form that depends on $\mathbf{b}$, so the point where it takes its simplest form is a centre everybody can find and agree on. “The universe has no centre” is a claim about function composition, not about modesty. It is the translation-equivariance argument you already make when you reach for a convolution over a dense layer.

Two caveats, because this uniformity is an assumption doing real work. It holds only *on average and on large scales* — close up the universe is emphatically lumpy, which is what galaxies, clusters and the cosmic web (Ch. 4) *are*, and Figure 10.1's 900 km/s of peculiar motion is that lumpiness talking. And that it holds at all is measured, not assumed: the cosmic microwave background (Ch. 11) is the same temperature in every direction to a few parts in a hundred thousand.

What is actually expanding

Strip the bread away and here is what survives. **Nothing is moving through space.** What grows is the distance between things that are not moving: the ruler itself is getting longer, everywhere, at the same fractional rate, at once. Galaxies keep their addresses; the map's scale bar changes underneath them. This is why H_0 came out of the units as a rate — it describes nobody's motion, it describes how fast a distance inflates.

The main guide names this in one line. Give every galaxy a fixed **comoving** coordinate that never changes, and let a single global number $a(t)$ — the **scale factor** — convert comoving coordinates into physical distances at time t . All the time evolution lives in that one multiplier, and Hubble's law is what you get by differentiating it once. That is Ch. 13 of the main guide; it takes half a page, and you now know what it is a picture *of*. Two consequences follow, and the “everything is flying apart” picture gets both wrong.

Bound things do not expand. The raisins were right. Your desk, the Earth's orbit, the Milky Way's hundred billion stars, the Local Group — none is growing, because each is held together by something that beats a fractional stretch of H_0 per unit time. Expansion is not a force things resist; it is what distances between unheld objects do.

There is no speed limit on it. This is the trap Ch. 8 flagged, and the main guide works it on the same number, so work it once. This program's Carousel cluster has a background reference plane at redshift $z = 1.432$. Read that as a velocity the naive way — $v = cz$, the Doppler reading — and you get

$$v = c \times 1.432 = 429,303 \text{ km/s},$$

which is 1.432 times the speed of light. Nothing is wrong with the observation and nothing is wrong with relativity. What is wrong is the word *velocity*. Relativity bounds how fast anything moves **through** space; it says nothing about how fast a purely geometric distance can grow while nothing moves through the space between. $v = cz$ was only ever the low- z approximation — the first Taylor term ([Ch. 2 of the main guide](#)) — and pushing it to $z > 1$ breaks the approximation, not the physics. The always-true statement is not a velocity at all: $a = 1/(1+z)$, the universe was that fraction of its present size when the light left.

The same trap sits inside Hubble’s law itself, and the Hubble distance is where it surfaces. $c/H_0 = 4283$ Mpc is, by construction, the distance at which the linear law’s own prediction $H_0 d$ reaches c ; push d past it and $v = H_0 d$ hands you a recession faster than light. Not a wall, not an edge, not a bug: recession is not motion, so there is nothing there to forbid, and we do observe galaxies whose recession velocities exceed c on any consistent accounting. A number this law returns is geometry, never somebody’s speedometer.

Unlocks

What you can now read

- [Ch. 13 — The expanding universe and redshift](#) defines the scale factor $a(t)$ as a “dimensionless function of time” that “carries no units and no meaning in isolation.” It then lays “a fixed grid over the universe,” puts galaxies at fixed comoving coordinates, and tells you “nothing on this grid moves. What changes is the ruler.” That is this chapter’s loaf with the dough thrown away and the honest parts kept: you know what the grid is a picture of, and why the raisins stay put.
- The same chapter [derives Hubble’s law](#) as $a(t)$ ’s first Taylor term, then insists it “is not an empirical fit stapled onto the expansion after the fact.” That sentence is a rebuttal, and you have not until now been shown what it rebuts: galaxies flying apart through space with a line fitted to them. It also quotes $1/H_0 = 13.97$ Gyr as “roughly the age of the universe, though not exactly” and defers the *why* to [Ch. 14](#) — you have it already.
- [Ch. 13’s naive- \$cz\$ trap](#) runs $z = 1.432$ to 429,303 km/s, calls it “faster than light,” and compresses the resolution into one clause: “no object is moving *through* space at all, space itself is stretching.” That clause is a chapter’s worth of picture, and this is the chapter it is standing on.

Exercises

Exercise 10.1 — Hubble’s law, both directions

Using $H_0 = 70$ km/s/Mpc, compute the recession speed of a galaxy at $d = 100$ Mpc. Then, without computing anything new, say what distance the same law assigns a recession speed of exactly c , and name it.

Solution

$v = H_0 d = 70 \times 100 = 7000$ km/s. Setting $H_0 d = c$ gives $d = c/H_0$: the Hubble distance, 4283 Mpc, which is *defined* as that distance and so needs no new arithmetic. Nothing physical happens there — recession is not motion.

Exercise 10.2 — The centre test

An astronomer on a galaxy at position $\mathbf{b}$ (as measured by you) observes a third galaxy at $\mathbf{d}$ (also as measured by you). Write down the recession velocity they measure, and say what they conclude about their own location. Then repeat for a hypothetical universe obeying $\mathbf{v} = k|\mathbf{d}|\mathbf{d}$ — speed going as distance *squared* — and say what changes.

Solution

They measure $H_0(\mathbf{d} - \mathbf{b})$ against a relative position of $\mathbf{d} - \mathbf{b}$, i.e. $\mathbf{v}' = H_0\mathbf{d}'$: identical law, identical constant. They conclude exactly what you did, with exactly as much evidence for being the centre as you have.

Under the squared law the subtraction leaves $k(|\mathbf{d}|\mathbf{d} - |\mathbf{b}|\mathbf{b})$, which is *not* $k|\mathbf{d}'|\mathbf{d}'$ for any k : the field measured at $\mathbf{b}$ has a form that depends on $\mathbf{b}$, so comparing enough galaxies lets that observer solve for $\mathbf{b}$ and locate the origin. That universe has a centre and everyone in it can point at it. Linearity is the whole difference. What the argument does *not* say: adding the same constant velocity to everything is not a centre, it is your own motion — real, invisible in relative velocities, and measured instead as a direction-dependent warming and cooling of the microwave background (Ch. 11).

Exercise 10.3 — The naive- cz trap

The Carousel cluster's background reference plane sits at $z = 1.432$. Compute the velocity the naive Doppler reading $v = cz$ assigns it, express it as a multiple of c , and state precisely where the error is: in the measurement, in relativity, or somewhere else.

Solution

$v = cz$ at $z = 1.432$ gives 429,303 km/s, i.e. $1.432c$ — predictable with no arithmetic at all, since cz exceeds c for any $z > 1$.

The measurement is fine: z is a wavelength ratio read off a spectrum. Relativity is fine: it bounds motion *through* space, and nothing here moves through space. The error is the interpretive step $v = cz$, the first-order approximation to the exact $a = 1/(1+z)$ and valid only while z is small. Run any first-order approximation past its regime and the absurd number it returns is the approximation's fault. The always-true reading: the universe was $1/(1+z)$ of its present size when the light left — a size against a size, not a speed.

Exercise 10.4 — Why you cannot do this from your own street

Peculiar velocities run to about 900 km/s and do not grow with distance. Explain why that makes nearby galaxies useless for measuring H_0 , why two of the leftmost points in Figure 10.1 have *negative* recession speeds, and what it implies about the effort a real H_0 measurement takes.

Solution

A measured velocity is $H_0 d$ plus a peculiar term of order 900 km/s, the same size at every distance. The signal grows linearly with d ; the contamination does not. So the fractional error on one galaxy's implied H_0 goes as 900 km/s over $H_0 d$, which blows up as $d \rightarrow 0$. Close enough in, the peculiar term exceeds $H_0 d$ outright and flips the sign: a galaxy whose peculiar motion happens to point at you reads as approaching. Those are the two sub-zero points in Figure 10.1, and it is what Andromeda actually does.

Hence Ch. 9. To measure H_0 you must reach out to where $H_0 d \gg 900$ km/s — compare 7000 km/s at 100 Mpc — which requires *independent* distances to galaxies that far out, which requires standard candles, which requires a ladder, which compounds calibration error at every rung. The wobble is why H_0 is hard, and hard is why Ch. 14 has two answers.

11. The Big Bang and the oldest light

Chapter 10 left the universe expanding: every distant galaxy receding, the ruler itself getting longer, nothing moving through space. This chapter runs that film backwards — the universe was once hot, dense and opaque; it has a finite age you can quote to three digits; and the moment it stopped being opaque left behind a photograph that is still arriving. It also names what the main guide uses without ever naming: the state $a = 0$, which the main guide’s Ch. 14 integrates back to in a single clause and never calls anything. Two numbers below disagree with the ones you will have seen in a magazine, and both disagreements are the point rather than an embarrassment.

What you can skip

You do not need general relativity, nucleosynthesis, inflation, or a Boltzmann code, and you do not need the integral turning $\Omega_m, \Omega_\Lambda, H_0$ into an age — that is Ch. 14’s job and it does it properly. What you already own, and should lean on hard here, is the discipline of knowing where a model’s domain of validity ends. Every claim below is a fitted model extrapolated backwards past the data that constrained it, and the live question is how far back it is still entitled to speak.

Running the expansion backwards

Reverse the sign of time in Ch. 10’s picture. If every comoving separation is growing today, each was smaller yesterday and smaller still the day before. The scale factor a — the single global multiplier Ch. 13 will define, normalised to $a = 1$ today — shrinks as you go back. Fixed matter in a shrinking volume means the density climbs, and compressed gas heats, so the temperature climbs with it.

Push that far enough and you reach a regime with no stars, no atoms, eventually no stable anything: hot, dense, uniform. That is what “the Big Bang” names, and be blunt about what it does *not* name. It was not an explosion: an explosion happens at a place and throws debris into surrounding space, and there is neither here. The objection that killed the centre of the raisin bread in Ch. 10 kills this too — the expansion has no centre, so its reversal has none either. The hot dense state was everywhere, including at the coordinates you are sitting at. What “backwards” contracts is the ruler, not the contents toward a point.

The age. Run the model back until a reaches zero and the elapsed time is finite. Feed this book’s cosmology — `FlatLambdaCDM(H0=70, Om0=0.3)`, the line every campaign in this repository asserts — into `astropy`, ask for `.age(0)`, and you get

$$t_0 = 13.467 \text{ Gyr},$$

Against this book’s ruler that is long without being incomprehensible: light takes about 10^5 years to cross the Milky Way (Ch. 1), so the universe has existed for on the order of a hundred thousand such crossings.

Why this is not 13.8. The number in every popular account is

13.8 Gyr, and this book is going to keep saying 13.467. The gap is

2.47 percent, which is not a rounding error. 13.8 Gyr comes from Planck’s cosmology, whose H_0 is lower than this repository’s 70 and whose Ω_m is very slightly higher. Those push the age in

opposite directions and H_0 wins: a lower H_0 is a slower present expansion rate, and a universe expanding more slowly needs *longer* to have reached its current size. That is why this book will not print 13.8 beside a cosmology that does not produce it. The disagreement between those two values of H_0 is not a choice of convention — it is the open problem this book’s [Ch. 14](#) is about, showing up here in the currency of an age.

What $a = 0$ is, and is not. The model says $a \rightarrow 0$ at a finite time in the past, with the density diverging. A diverging density does not describe an event; it is the standard signature of a theory evaluated outside its domain. General relativity supplies the equation and has no quantum mechanics in it, and conditions near $a = 0$ are exactly where quantum mechanics cannot be left out. Nobody has a tested theory of that regime: **“the Big Bang” does not name a first instant anyone can compute.** It names the hot dense early state the evidence reaches back to — and that evidence stops a long way short of $a = 0$, at the moment the next section is about.

The first light

Temperature and size are locked together: $T \propto 1/a$, so halving the size doubles the temperature. (The reason is [Ch. 13](#)’s, one chapter early: expansion stretches every wavelength by the same factor a , and a blackbody ([Ch. 5](#)) with all its wavelengths stretched by a is still a blackbody, at a temperature $1/a$ times lower.)

Run *forwards* from the hot state, where the physics is easier to state. Above a few thousand kelvin hydrogen is ionised: protons and electrons wander separately rather than bound, and the universe is a plasma. Free electrons scatter photons efficiently, so a photon there travels a short distance, scatters, travels, scatters again. The universe is *opaque* — not dark but fogged, for the same reason you cannot see into the interior of the Sun.

As it expands and cools, photons lose the energy needed to keep knocking electrons off. At around 3000 K, electrons and protons bind into neutral hydrogen and stay bound, and neutral hydrogen does not scatter these photons the way free electrons did. The fog does not thin gradually over billions of years; it clears over a window short compared to everything around it, and the universe goes from opaque to transparent. Every photon bouncing around at that instant stopped, and has travelled in a straight line ever since.

Two names to flag. This event is universally called **recombination**, which is wrong — the electrons and protons had never been combined before, so nothing re-combined. The name is a historical accident from laboratory plasma physics; read it as “combination” and move on. It is also called **last scattering**, which is accurate and more useful: the last time those photons interacted with anything.

3000 K is worth pausing on with [Ch. 5](#) in hand — roughly the photosphere temperature of a cool red star. The entire sky was as hot and bright as a red dwarf’s surface, with no dark between the stars because there were no stars and no dark.

When this happened, and an honest problem. The measured answer is

380,000 years after $a = 0$ — a few times the 10^5 years light needs to cross the Milky Way, and a fraction

2.8×10^{-5} of elapsed history, under three parts in a hundred thousand. On any clock this book has used, the CMB is the universe’s first instant.

That 380 is **measured, not computed here**, and the distinction matters. FlatLambdaCDM(70, 0.3) models matter and dark energy and *no radiation at all* — its Tcmb0 is zero, so the object this book quotes ages from does not know the CMB exists. Ask it for its own age at last scattering and it answers

465,000 years, not 380,000. The error is neither small nor mysterious: the real early universe was radiation-dominated, radiation drives a faster expansion than matter alone, and a model without radiation therefore takes too long to reach any given early size. The model is fine for everything this repository asks of it — all of that sits at z below a few, where radiation is irrelevant — and is outside its domain at $z = 1100$. So this book quotes 380 from measurement, prints the model’s own 465 beside it, and does not blend the two.

The cosmic microwave background

Those free-streaming photons are still arriving. They left a 3000 K blackbody, and expansion has stretched every wavelength by the same factor the universe has grown by since — a measured redshift:

$$z_{\text{CMB}} \approx 1100, \quad a = \frac{1}{1+z} = \frac{1}{1101} \approx 0.000908.$$

The universe was about one part in eleven hundred of its present size. A 3000 K blackbody stretched by 1101 arrives at 3000/1101, and measured from here that is

$$T_0 = 2.725 \text{ K}.$$

Run it back the other way — 2.725×1101 — and you land on the 3000 K this section opened with. Those are not two independent claims but one measurement read in two directions.

A blackbody at 2.725 K peaks at a wavelength [Ch. 5](#) already computed for you, well outside the visible band and squarely in the microwave. Hence the name: the **cosmic microwave background**. It was found by accident, as an unremovable hiss in a radio antenna whose operators spent months hunting for the fault before concluding the noise came from the whole sky at once.

That last part is what the usual shorthand gets wrong. The CMB is not a picture of a distant *object*; it is a picture of a *time*. The photons arriving now were emitted at last scattering by whatever material sat at exactly the right distance for its light to reach us today, and those positions form a sphere around us: the **surface of last scattering**. It is not an edge of the universe and not a wall. Every observer has their own, and observers on ours — where the material long ago cooled into galaxies — see a CMB from a sphere that includes us.

Which is where “the baby picture of the universe” breaks. An infant photograph is of a subject, taken from outside it. This one is taken from inside, of everything, at one moment, and its subject is not a place you could go: that material is not glowing at 3000 K today, it is nearby and cold and made into stars.

What the CMB tells us

Calling this the most informative image ever taken is a statement about how easy the subject is to model, not enthusiasm. Every other astronomical image is of objects whose appearance depends

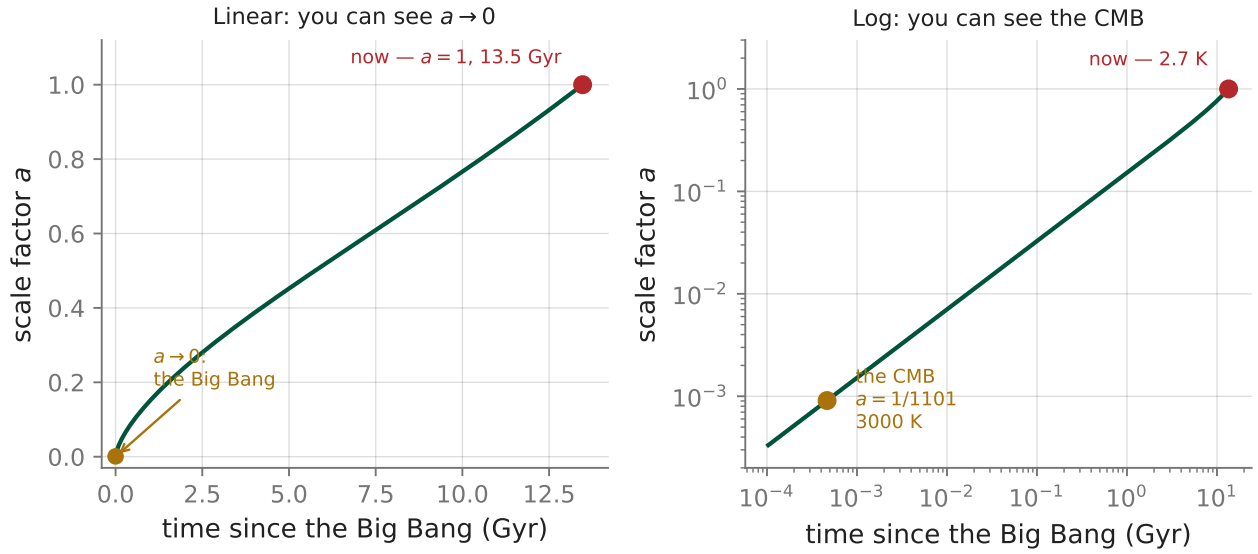


Figure 11: The **same curve** in both panels — $a(t)$ from `FlatLambdaCDM(70, 0.3)`, nothing drawn by hand — because no single pair of axes shows both facts. On linear axes (left) you can see $a \rightarrow 0$, which is what “the Big Bang” names; the CMB marker is there too, crushed against the origin, sitting in the first few thousandths of a percent of the axis. On log axes (right) the CMB is legible at $a = 1/1101$ and $a = 0$ has vanished, because $\log 0$ does not exist. Cosmology is read on log axes for that reason, so a reader who has only seen the right-hand panel has never been shown the thing this chapter is named after. Two honest notes: the “now” marker rounds 13.467 Gyr to 13.5, and the CMB marker’s *time* coordinate is this model’s radiation-free 465 kyr, not the measured 380 — it sits at the CMB’s measured *scale factor*, and the model’s curve puts it in time.

on gas physics, star formation, feedback and mergers, none of which anyone computes from first principles. The CMB predates all of that: nearly uniform, gravitationally linear, governed by physics you can write down completely — the one cosmological dataset whose forward model is under control. Three things come out of it.

The hot dense past is real. The CMB is the most precise blackbody ever measured — departures from the ideal curve are too small to draw at the line width of any published plot. That matters because thermal equilibrium produces a blackbody and effectively nothing else does. A spectrum that good requires matter and radiation to have been thoroughly coupled, which requires the opacity, which requires the density and the temperature. [The first section](#)’s reversal argument is an extrapolation; this is the receipt.

The numbers Ch. 14 of the main guide asks you to type. The CMB is not perfectly uniform: its temperature varies from patch to patch by amounts so small the maps must be rescaled by orders of magnitude before your eye registers anything. Those variations are the seeds gravity grew into galaxies, and their statistics are extraordinarily constraining. Fit the pattern and out come Ω_m , the flatness of space, and a value of H_0 . The $\Omega_m = 0.3$ and $\Omega_\Lambda = 0.7$ you will type into `FlatLambdaCDM` in [Ch. 14](#) without being told where they came from are that fit rounded off — carrying the same slight excess over 0.3 the age gap turned on — and Ω_Λ is not fitted separately at all: it is what flatness leaves once Ω_m is fixed. They came from here. So does the *other* H_0 , the one producing 13.8 Gyr rather than 13.467.

You already know this

Cosmologists rarely report the CMB map itself; they report its **power spectrum**, a curve of variance against angular scale. That is not a compression convenience — it is a claim that the field is very nearly Gaussian, and a Gaussian random field’s entire information content lives in its autocorrelation, equivalently (by [Ch. 7](#)’s Wiener–Khinchin relation) in its PSD. The map has more pixels; it has no more information. This repository leans on the identical fact when it models correlated image noise with a fitted kernel rather than a covariance matrix ([Ch. 24](#)).

Dark matter, independently. The shape of that power spectrum depends on how much gravitating matter was present versus how much of it could feel the photon pressure. Ordinary matter can; dark matter cannot. The fit demands far more of the second kind, with no reference to any galaxy’s rotation or any cluster’s dynamics — making the CMB the fourth of the independent witnesses [Ch. 12](#) lines up, and the one whose systematics have nothing in common with the other three.

What the CMB does not tell you is what came before it: those photons carry no information from earlier than last scattering, because earlier than that they were being scattered. The fog is as opaque looking back through as it was looking forward. Everything this chapter said about $t < 380,000$ years is inference from a model constrained by lighter-touch evidence — chiefly the abundances of the light elements — and the first fraction of a second is not known at all.

Unlocks

What you can now read

- [Ch. 13 — The expanding universe and redshift](#) defines the scale factor as carrying “no units and no meaning in isolation”, normalised so $a(t_0) \equiv 1$ today, then says its

“actual functional form ... is set by how much matter and dark energy the universe holds”. It never mentions the other end of that function. You now have it: $a \rightarrow 0$ is the Big Bang, 13.467 Gyr back in this book’s cosmology, and where the model’s licence expires rather than an event anyone can compute. That also puts Ch. 13’s table of scale factors in proportion — its smallest entry, the Carousel’s reference plane, is a bit under half, while the CMB’s is 1/1101. Everything this repository lenses happened in the most recent, largest, coldest sliver of that history.

- **Ch. 14 — FRW and the Friedmann equations** tells you `.age(0)` is “that same equation integrated back to $a = 0$ ” — one clause, and one of only two mentions of $a = 0$ in the entire guide, both of them in that chapter and both in passing. It then quotes the age at 13.467 billion years and moves on, without a word about the number everyone else quotes being 13.8. You now know what the integration runs back *to*, why the answer is finite, and why this repository’s answer and Planck’s differ by 2.47 percent rather than by a slip. The same chapter’s **density parameters** section says “**flatness is the separate, empirical claim that $\Omega_k = 0$** ”, and that FlatLambdaCDM “doesn’t measure this — it assumes it”. It is right to flag that, and never says who *did* measure it. The CMB did, along with the $\Omega_m = 0.3$ you type beside it.

Exercises

Exercise 11.1 — The temperature that names the background

The CMB is observed today at 2.725 K and comes from a redshift of about 1100. Using $T \propto 1/a$, compute the scale factor at emission and the temperature the radiation had then. Name an astronomical object with roughly that surface temperature.

Solution

$a = 1/(1 + z) = 1/1101 \approx 0.000908$. Since $T \propto 1/a$, the emission temperature is $2.725 \times 1101 \approx 3000$ K — a cool red star’s photosphere. The redshift and the temperature ratio are the same number read twice.

Exercise 11.2 — Two ages, one gap

This book’s cosmology gives 13.467 Gyr; every popular account says 13.8 Gyr. Compute the percentage gap. Then, without doing an integral: Planck’s H_0 is lower than this repository’s 70 — which way does that alone push the age, and why?

Solution

$100 \times (13.8 - 13.467)/13.467 = 2.47$ percent . A lower H_0 is a slower expansion, so the universe needed longer to reach its current size: lower H_0 , longer age. The crude version is the Hubble time $1/H_0$, which Ch. 13 computes and correctly warns is only “roughly the age of the universe”. Planck’s Ω_m is also slightly higher, pushing the other way (more matter, more early deceleration), but H_0 dominates.

Exercise 11.3 — When the model is outside its domain

`FlatLambdaCDM(70, 0.3)` reports its own age at $z = 1100$ as 465,000 years; the measured value is 380,000. Which is wrong, what is the model missing, and why does this not undermine any other number in this book?

Solution

The model is wrong, for a nameable reason: `FlatLambdaCDM(70, 0.3)` carries matter and dark energy and no radiation — its `Tcmb0` is zero. Radiation drives a faster expansion than matter at the same density, so the real radiation-dominated early universe reached $a = 1/1101$ sooner than a matter-only model can. Its 465,000 is an overestimate by construction, not a measurement in tension with anything.

It undermines nothing else because radiation's share of the budget falls as the universe expands, and at the redshifts this repository works at — $z = 0.5$ lenses, $z = 2$ sources — it is negligible. A model can be exactly right where it is used and useless three orders of magnitude outside it. The failure mode is not the model; it is quoting the model's number and the measured one as though they were the same number.

12. Dark matter

[Chapter 3](#) and [Chapter 4](#) handed you galaxies and clusters as objects you can photograph. This chapter is about the part of them you cannot. Four measurements — a spiral disk, a cluster, the light-bending this whole program is built around, and a picture of the universe taken before any galaxy existed — share no instrument, no epoch and no systematics, and report the same thing: most of the matter does not shine. By the end you can expand the name `FlatLambdaCDM` a letter at a time, and know which of its assertions the field can defend and which one it cannot.

What you can skip

You do not need particle physics — nobody has a particle model that works here, which is the honest state of the subject. You do not need the N-body simulations behind the halo profile named below, only that it is a *fit* to simulation output rather than a derived law. You do not need to get $M(<r)$ out of $\rho(r)$ by integrating: [Ch. 10 of the main guide](#) does it in three lines. What is left is the evidence, and the evidence is arithmetic.

Rotation curves

Newton, applied to something on a circular orbit: gravity supplies the centripetal force, so

$$\frac{G M(<r) m}{r^2} = \frac{m v^2}{r} \quad \Rightarrow \quad v(r) = \sqrt{\frac{G M(<r)}{r}}.$$

The test mass m cancels, and — by the shell theorem — only the mass *interior* to the orbit appears. In the solar system essentially all the mass is the Sun, so $M(<r)$ is the same constant for every planet and $v \propto 1/\sqrt{r}$: the outer planets crawl.

A spiral galaxy is not a point mass. Its light is spread through a disk, so moving outward you enclose more stars, $M(<r)$ grows, and v rises. Then the light runs out. Past the outermost stars $M(<r)$ stops growing and you are back in the solar-system case: v should turn over and fall as $1/\sqrt{r}$. That is a prediction with no free parameters — photograph the disk, count the light, assume the mass is where the light is, and the curve is determined.

That prediction is crude in one identified way, and it is worth naming before it is used against anything. The shell theorem is exact for a *sphere*, and a disk is not one: matter outside the orbit does not cancel, and doing the disk properly lifts the predicted outer speed some tens of percent above the spherical reading. Tens of percent is not the discrepancy this chapter is about, which is why the crude version survives — but the curve below is the crude version.

It is wrong. Measure orbital speeds far enough out, using the 21-cm radio line from neutral hydrogen that extends well past the last visible star, and the curve does not turn over. It goes flat and stays flat to the last measurable point. Vera Rubin and Kent Ford established this through the 1970s, galaxy after galaxy.

Read the flat part back through the equation. If v is constant then $M(<r) \propto r$: mass keeps piling up, linearly, past the radius where the light stopped. The main guide derives the density that does this — $\rho \propto r^{-2}$ — from the opposite direction, as the profile of a constant-velocity-dispersion sphere, and notes in passing that it “gives you a disk galaxy’s flat rotation curve for free” ([Ch. 10](#)). Same fact, two kinematic windows.

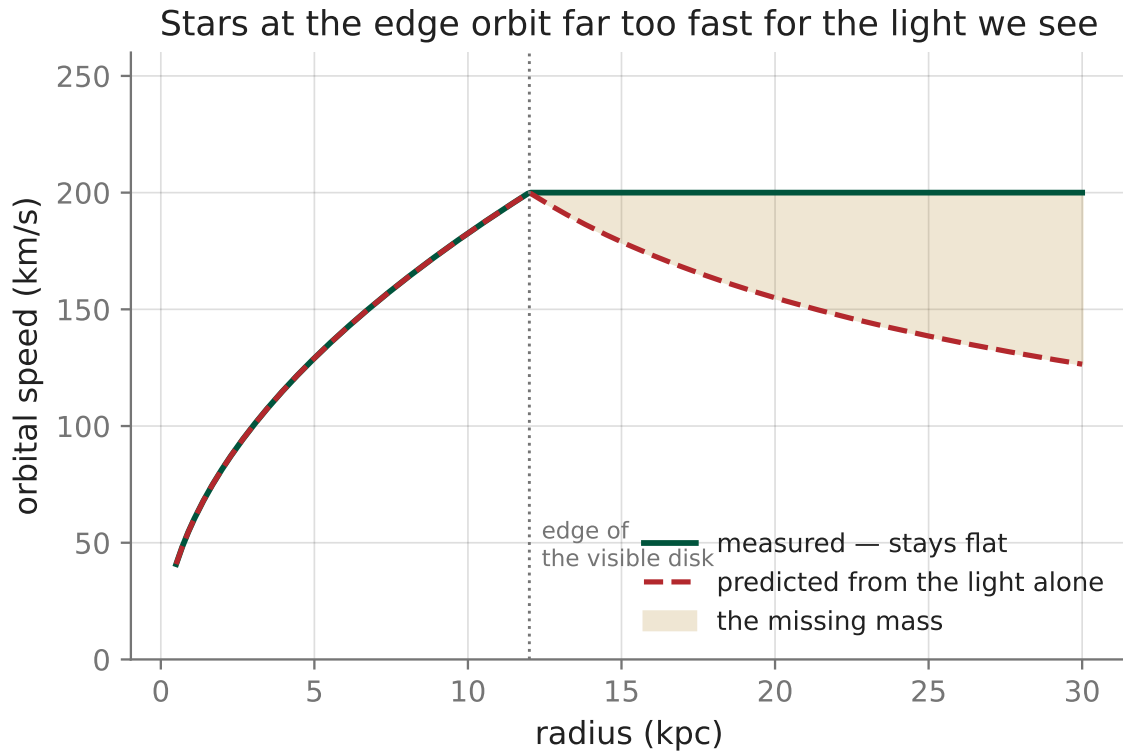


Figure 12: A model spiral: orbital speed against radius, with the visible disk ending at 12 kpc (dotted line). Inside that radius the two curves are *identical* — there is no disagreement where the light is. Outside it, the curve predicted from the light alone falls as $1/\sqrt{r}$ while the measured curve stays flat near 200 km/s, and the shaded gap is mass that has to be there and cannot be seen. A schematic, not data: a real disk’s inner rise has more structure than this smooth ramp, and no real galaxy is flat to four significant figures. The argument lives entirely in the outer half.

There are exactly two ways out. Either there is mass out there you cannot see, or $F = GMm/r^2$ itself fails at the very low accelerations of a galaxy’s outskirts. The second is a real research program — Modified Newtonian Dynamics, MOND — and should not be waved away: on rotation curves MOND is *good*, arguably tighter than the halo picture, predicting the curve from the visible light with one universal constant instead of one fitted halo per galaxy. What sinks it for most of the field is the other three witnesses.

Four independent witnesses

Rotation curves alone would be a suspicious result: one class of object, one technique, one equilibrium assumption, and a proposed fix touching the least-tested regime of the least-tested law in the argument. Three more measurements say the same thing from places that share nothing with it.

Witness	Scale	Epoch	What it weighs
Rotation curves	one spiral disk, tens of kpc	now	$M(<r)$ from orbital speeds
Cluster dynamics	a cluster, several Mpc	now	total mass from the spread of galaxy velocities
Gravitational lensing	galaxy through cluster	halfway back	projected mass, with no equilibrium assumed
The CMB	the whole sky	before any galaxy	ordinary and dark matter, <i>separately</i>

Cluster dynamics. Fritz Zwicky measured how fast Coma’s galaxies moved relative to each other in 1933. Far too fast: a cluster is a bound system, the visible mass could not have held those galaxies, and Coma should have flown apart long before he looked at it. He named the discrepancy *dunkle Materie* — dark matter — and was ignored for forty years, until rotation curves reached the same conclusion from a different kind of object entirely.

Lensing. Your own field, and the cleanest of the four, because it is the only one that does not assume equilibrium: the deflection depends on the mass and nothing else (Ch. 15). The sharpest result is the Bullet Cluster, two clusters that passed through each other. Their hot gas — the majority of the *ordinary* matter, well outnumbering the stars — collided, shocked and lagged behind, while their galaxies, effectively collisionless at those densities, sailed through. The lensing mass went with the galaxies, not the gas: mass and the bulk of visible matter sit in physically separate places on the sky. Changing the law of gravity does not help, because a modified law still needs a *source*, and the source is demonstrably not where the light is.

The CMB. Before atoms formed, ordinary matter was ionised and locked to photons by scattering; the two behaved as one fluid with real pressure, and pressure means sound waves. Dark matter does not scatter light, so it felt none of that pressure — it fell into whatever gravitational wells existed and stayed. Ordinary matter therefore sets the compressions and rarefactions in the temperature pattern of [the oldest light](#), dark matter sets the depth of the wells they oscillate in, and the relative peak heights pull the two apart (Ch. 11) — each as a fraction of the density needed for flatness, the ρ_{crit} of Ch. 14:

$$\Omega_{\text{b}} = 0.049, \quad \Omega_{\text{dm}} = 0.265.$$

Divide them:

$$\frac{\Omega_{\text{dm}}}{\Omega_{\text{b}}} = 5.41.$$

About five and a half times as much dark matter as there are atoms in every star, planet, dust grain and gas cloud in the universe combined. The Milky Way’s 10^{11} stars (Ch. 1), and every other galaxy’s, are a minority component of the matter, and a small one.

You already know this

The force of this section is not any one measurement’s precision — several carry systematics you would not accept in a paper on their own. It is that the four error models are *uncorrelated*: 21-cm radio astronomy plus an equilibrium assumption; optical spectroscopy plus the virial theorem; general relativity plus image deconvolution; microwave photometry plus linear perturbation theory. A shared bias would have to sit in all four and pull them the same way. This is why you ensemble models with decorrelated errors rather than four seeds of one architecture, and why a factor of five no single method could establish to your satisfaction is nevertheless not in doubt.

Halos

The picture that survives all four witnesses: the disk you photograph is the small bright thing at the centre of a much larger, roughly spherical, invisible **halo**. The galaxy sits *in* it, and it keeps going for several times the radius of the visible light. Everything in Chapter 3 — the 10^{11} stars, the 10^5 light-years of disk — is the ornament. The halo is the object.

Everyone models a halo’s density with the **NFW profile**, after Navarro, Frenk and White. One property, stated without calculus: it is shallower than r^{-2} near the centre and steeper far outside, crossing that slope once in between. One property about its *status*: it is not derived from anything, but fitted to what N-body simulations of cold dark matter produce. When Ch. 10 of the main guide sets up the isothermal conspiracy — the puzzle that a Sersic stellar profile plus an NFW halo, neither isothermal, sum to a total mass profile that *is* — the NFW half of that puzzle is this paragraph, and the conspiracy is two fitted forms adding to a third for reasons nobody has derived.

One warning: do not carry the 5.41 into a single galaxy and say the Milky Way has five times more dark matter than stars. That is a global average over the whole universe. What any one galaxy reports depends on where you stop measuring, because the stars stop and the halo does not — which is not a defect of the measurement but what “the light ends and the mass does not” means quantitatively.

That aperture dependence is your problem directly. Strong lensing weighs the mass inside the Einstein radius and nothing else (Ch. 19), and what it weighs there is stars *and* halo summed, inseparable from the image. Every mass model in this program — every γ , every convergence map (Ch. 20) — models the total, and the total is mostly dark. That is not a caveat on the program; it is much of why it is worth running, because lensing weighs mass whether or not it emits (Ch. 16).

What CDM stands for

In [Ch. 14](#) of the main guide you will type this line, or read it in three of this repository’s files:

```
FlatLambdaCDM(H0=70, Om0=0.3)
```

Nobody expands the name. Here it is, a piece at a time.

Flat asserts $\Omega_k = 0$: Euclidean geometry on large scales. The main guide is careful and correct here, and **FlatLambdaCDM** does not measure this — it assumes it, by refusing to expose a curvature argument.

Lambda is Λ , the cosmological constant: dark energy, the thing making the expansion accelerate, and a completely different mystery from this chapter’s ([Ch. 13](#)). Its share,

$$\Omega_\Lambda = 0.685,$$

dwarfs both matter components. Dark matter and dark energy share an adjective and nothing else; do not let the naming suggest otherwise.

CDM is **Cold Dark Matter**, and both words are load-bearing.

Dark does not mean black. Soot is black: it absorbs light, which is a way of interacting with it. Dark matter does not absorb, emit, scatter or reflect — it does not couple to electromagnetism at all, as far as anyone can measure. Transparent is closer, and even that misleads, because glass at least refracts. It has never been observed to do anything except gravitate.

Cold means non-relativistic at the epoch when structure was forming — not a temperature but a statement about the particles’ velocity dispersion, and a falsifiable claim that has so far survived its tests. Were dark matter *hot* (light, fast particles, which is what neutrinos would give you) it would stream out of small overdensities before they could collapse, erasing small-scale structure and forcing the largest structures to form first and fragment downward. Cold dark matter does the opposite: small halos form first and merge upward. The universe we observe is built bottom-up, which rules *hot* out. Be careful how much further you take that: it does not pin the temperature, it bounds it. An intermediate *warm* candidate — heavy enough to keep the bottom-up ordering, light enough to erase the smallest halos — is constrained rather than excluded, and the count of faint satellite galaxies around rulers like the Milky Way is where that bound is currently argued. CDM is the cold end of a range the data allow, not a measured value.

You already know this

Free-streaming is a low-pass filter: the distance fast particles travel before structure collapses is a smoothing kernel on the initial density field, erasing power below it. “Hot” and “cold” name the cutoff wavenumber, and the observable is the surviving power spectrum of galaxies and of the CMB — precisely the object [Ch. 7](#) has you reading as a PSD. A detected cutoff would falsify the **C** and leave **Flat** and **Lambda** untouched: the three letters are separately testable, not a package you take or leave whole.

And what Om0=0.3 contains. Ω_m is *all* matter — ordinary and dark, summed — so the measured total, and the invisible part of it, are

$$\Omega_m = 0.049 + 0.265 = 0.314, \quad \frac{\Omega_{\text{dm}}}{\Omega_m} = 0.844.$$

The argument you type as 0.3 — a rounded stand-in for 0.314 — is 84 percent stuff nobody has identified. The main guide walks through what that line asserts with real care, down to noting that Ω_Λ is computed rather than supplied, and never mentions that its *other* argument is five and a half parts dark matter to one part atoms.

And the sum does not come to one:

$$\Omega_b + \Omega_{\text{dm}} + \Omega_\Lambda = 0.049 + 0.265 + 0.685 = 0.999.$$

Not 1.000 — but this is not a curvature detection and not a mistake. It is three measured values, each rounded to three decimals, being added. `FlatLambdaCDM` sidesteps this by computing Ω_Λ as whatever makes the sum exactly one. The reason to print the 0.001 rather than quietly nudge a digit is that a budget summing to exactly one when its components are quoted to three places is a budget somebody fudged.

Nobody knows what it is. All four witnesses are gravitational. Not one has ever seen a dark matter particle — they have seen mass, and inferred that the mass is not made of the stuff we know. WIMPs led for decades and a generation of underground detectors returned nothing; axions are the current favourite and are likewise undetected; primordial black holes are largely excluded across most of their proposed mass range. The answer may be a particle nobody has thought of, and it remains possible — a minority view, but a live one — that some of the discrepancy is gravity behaving in a way we have not written down. What is *not* in doubt is the observation: gravitating mass exists in quantities the light cannot account for, at four scales, by four methods, at a ratio of 5.41 to one. “Dark matter” names a robust measurement, not an understood substance.

Unlocks

What you can now read

- **Ch. 10 — Galaxies, Sersic profiles, and velocity dispersion** contains the main guide’s entire treatment of dark matter, and it is one subordinate clause: “*The dark matter — inferred, historically, from exactly the flat rotation curves derived above staying flat well past the edge of a disk galaxy’s visible light, which under Newtonian gravity requires unseen mass out there — is typically modeled with an NFW profile.*” That clause compresses this chapter’s first section into a dash-bounded aside, then names a profile it never explains. You now have the rotation curves it gestures at, the three witnesses it does not mention, and the standing of NFW as a fit to simulations — which is what makes the isothermal conspiracy a conspiracy rather than a theorem.
- **Ch. 14 — FRW and the Friedmann equations** promises you can “write down `FlatLambdaCDM(H0=70, Om0=0.3)` ... and say exactly what each of its two arguments asserts, including the third number it fixes that you never typed.” It delivers on H_0 , on $\Omega_k = 0$, and on Ω_Λ being computed — and never expands the letters CDM, nor opens up Ω_m . You now know the argument typed as 0.3 is 0.314 measured, that 0.844 of it is invisible, and that CDM is a falsifiable claim about a free-streaming cutoff which the universe’s bottom-up structure confirmed.

Exercises

Exercise 12.1 — The budget you type without reading

From $\Omega_b = 0.049$, $\Omega_{\text{dm}} = 0.265$ and $\Omega_\Lambda = 0.685$: compute the dark-matter-to-ordinary-matter ratio, the total Ω_m , the invisible fraction of it, and the sum of all three components. Why is the last answer not 1.000, and why is that not evidence of curvature?

Solution

Two divisions and two additions:

$$\frac{0.265}{0.049} = 5.41, \quad 0.049 + 0.265 = 0.314,$$

$$\frac{0.265}{0.314} = 0.844, \quad 0.049 + 0.265 + 0.685 = 0.999.$$

The shortfall is rounding: three measured values quoted to three decimals, added, give a rounded sum, and 0.001 is the error to expect. Curvature is a separate empirical claim, $\Omega_k = 0$, tested against its own error bar and not by this addition. `FlatLambdaCDM` sidesteps the issue by computing $\Omega_\Lambda = 1 - \Omega_m$ rather than accepting a measured value, which is why the code's budget sums to one exactly and the observations' does not.

Exercise 12.2 — What the light alone predicts

Take the model galaxy of Figure 12.1. Outside the visible disk, $M(<r)$ is constant if the mass is where the light is, so $v \propto r^{-1/2}$. By what factor should the orbital speed drop between the edge of the visible disk and a radius four times larger? What does the measured curve do instead, and what does that imply about $M(<r)$ out there?

Solution

With $M(<r)$ fixed, $v(4r)/v(r) = 4^{-1/2} = 1/2$: the speed should halve. The measured curve does not drop at all, so $v(4r)/v(r) = 1$, and holding v constant while r quadruples requires $M(<r)$ to quadruple too — mass still accumulating, linearly, in a region with no light in it. That is the same statement as $\rho \propto r^{-2}$, the profile [Ch. 10](#) of the main guide derives from the constant-velocity-dispersion condition: reached there through stellar kinematics in an elliptical, here through gas orbits in a spiral.

Exercise 12.3 — Why four, and not one

Rotation curves can be fitted by MOND without any dark matter, and fitted well. Explain why the field nonetheless treats dark matter as established, and what specifically a modified-gravity theory would have to reproduce to displace it.

Solution

Because the claim does not rest on rotation curves. Cluster dynamics, lensing and the CMB reach the same conclusion at different scales and epochs with disjoint systematics; a bias explaining away all four would have to sit in radio astronomy, optical spectroscopy, image deconvolution and microwave photometry at once, and pull them the same direction. So a modified-gravity theory must do more than beat dark matter on one witness: it must fit all four at least as well with no more freedom, and in particular explain the Bullet Cluster, where a modified force law still needs a source and the source is not co-located with the light. Nothing has cleared that bar. Note what the argument does *not* do: it does not say what dark matter is made of, and it does not make MOND's rotation-curve success go away as something a complete theory should eventually explain.

13. Dark energy and how it ends

[Chapter 11](#) ran the expansion backwards to a hot, dense start. [Chapter 12](#) established that most of the matter in that universe is not the kind that shines. This chapter adds the result neither prepares you for: matter — dark and ordinary together — is not most of what the universe contains, and the expansion is not slowing down. It is speeding up, and has been for the last 3.39 billion years. By the end you will know what the second argument of `FlatLambdaCDM(H0=70, Om0=0.3)` — the one line this repository’s cosmology reduces to — asserts about the universe, and where the 0.7 that `astropy` computes without your typing it came from.

What you can skip

Nothing here needs general relativity, or the *second* Friedmann equation — the one with pressure in it, which is where accelerated expansion formally comes from. [Ch. 14 of the main guide](#) gets the first out of Newtonian energy conservation and explains, in its Exercise 14.4, why the second has no Newtonian version. That is the whole calculus content of this subject, and this chapter hands it over. If you already hold the concordance budget and can say why a constant ρ_Λ plus $\rho_m \propto a^{-3}$ *forces* a crossover, read only [The energy budget](#), for what this repository’s rounding of Ω_m costs.

The supernova surprise

By the mid-1990s the open question was not *whether* the expansion was slowing down. Gravity is attractive; a universe full of matter pulls on itself; the expansion had to be decelerating. The question was by how much — the same question as how much matter there is. Two teams set out to measure the deceleration with the tool [Ch. 9](#) built: Type Ia supernovae as standard candles, bright enough to see halfway across the observable universe and uniform enough to read a distance off.

The measurement needs two independent numbers per supernova. The **redshift** says how much the universe has expanded since the light left ([Ch. 8](#)); the **brightness** says how far the light actually travelled. Different expansion histories relate those two differently: a universe that decelerated hard was expanding *faster* in the past, so a photon emitted at a given redshift had less far to travel, and its source looks nearer and brighter. One whose expansion was slower in the past puts that same source farther away, and fainter.

Quantify it with this repository’s own cosmology, at its standard lens redshift, $z = 0.5$. The repo’s `FlatLambdaCDM(H0=70, Om0=0.3)` puts a supernova there at a luminosity distance of

$$d_L = 2832.9 \text{ Mpc.}$$

Now build the universe everyone expected: same H_0 , flat, all matter and no dark energy at all, $\Omega_m = 1$ — the model the field called Einstein–de Sitter and used as its default prior. The same supernova lands at

$$d_L = 2357.7 \text{ Mpc,}$$

a factor of 1.2016 nearer. Convert that ratio into the units a telescope reports. The magnitude system ([Ch. 9 of the main guide](#)) is logarithmic in flux, and flux goes as $1/d^2$, so two distances

differ in magnitude by $\Delta m = 5 \log_{10}(d_1/d_2)$:

$$\Delta m = 5 \log_{10} \left(\frac{2832.9}{2357.7} \right) = 0.399 \text{ mag.}$$

Those four tenths of a magnitude are the discovery. The distant supernovae came in fainter than a decelerating universe allowed by about that much — roughly 20% farther than they had any business being. Both teams reported it in 1998, independently, on different pipelines. There is no subtler content to the result: the candles were too dim, and the only way to make them too dim is to have the expansion speed up.

Two honest caveats, because the above is the *shape* of the argument in the repo’s numbers, not a rerun of the published fits. First, the real analyses fit Ω_m and Ω_Λ jointly across many redshifts rather than comparing two fixed models, and the pre-1998 prior was never as clean as $\Omega_m = 1$. Second, and more to the point: four tenths of a magnitude is *not* a large number. It is about what one miscalibrated rung of the distance ladder could produce by itself (Ch. 9 is where errors compound), and the alternatives — distant supernovae being intrinsically dimmer, dust greying them — took years to kill. What settled it was the CMB (Ch. 11) and galaxy clustering independently landing on the same budget, from physics with nothing to do with a light curve. Three unrelated measurements agreeing is the argument; four tenths of a magnitude on its own never was.

What Lambda is

What makes the expansion accelerate is written Λ — the **cosmological constant** — and its definition is one property, not a mechanism: an energy density belonging to space itself that *does not dilute as space expands*. Matter dilutes. Double every distance and the same atoms occupy eight times the volume, so $\rho_m \propto a^{-3}$. Double every distance with Λ in the room and there is more Λ , at the density it always had: $\rho_\Lambda = \text{const.}$ That is the entire model, and Ch. 14 of the main guide states it in one line — “treat the cosmological constant Λ as an extra density component that never dilutes” — before moving on.

You already know this

Two terms with different growth rates cross exactly once. You reason this way whenever you compare an $O(n^3)$ term against an $O(1)$ term: the constants set *where* the crossover is, never *whether*, and which one wins is a question about the regime. Here $\rho_m \propto a^{-3}$ races $\rho_\Lambda \propto a^0$, so at small a matter dominates by an unbounded margin and at large a it loses, guaranteed. The crossover epoch was not fitted to the 1998 data; it is forced the moment you write two components with different exponents. The data only fixed *when*.

The mechanism, as opposed to the bookkeeping, is where general relativity enters and this book stops. In GR the source of gravity is density plus three times pressure, so a component with large *negative* pressure gravitates *repulsively*. The main guide says this exactly once, in the solution to Exercise 14.4, where a negative-pressure component — “exactly what makes something behave like a cosmological constant” — can drive “accelerated expansion — with no Newtonian counterpart at all”.

What nobody knows. “Dark energy” labels the observation; it does not explain it, exactly as “dark matter” does not (Ch. 12). The natural candidate — the vacuum energy of quantum fields — gives the right *kind* of thing, then overshoots the observed value by many tens of orders of

magnitude, depending on how you count. Calling Λ “vacuum energy” in company is a claim, not a synonym. Whether Λ is constant is also open: `FlatLambdaCDM` hard-codes the constant-density case, and recent large spectroscopic surveys report hints that the dark-energy density has evolved — at a significance nobody yet calls a detection, and which may not survive.

The energy budget

Divide every energy density in the universe today by the critical density $\rho_{\text{crit},0}$ that [Ch. 14 of the main guide](#) builds out of H_0 alone. The dimensionless fractions that come back are the **density parameters**, and they are the answer to “what is the universe made of”:

- **Ordinary matter** (atoms — “baryons”, in the field’s loose usage): $\Omega_b = 0.049$.
- **Dark matter**: $\Omega_c = 0.265$.
- **Dark energy**: $\Omega_\Lambda = 0.685$.

Everything you have ever seen is the 4.9%

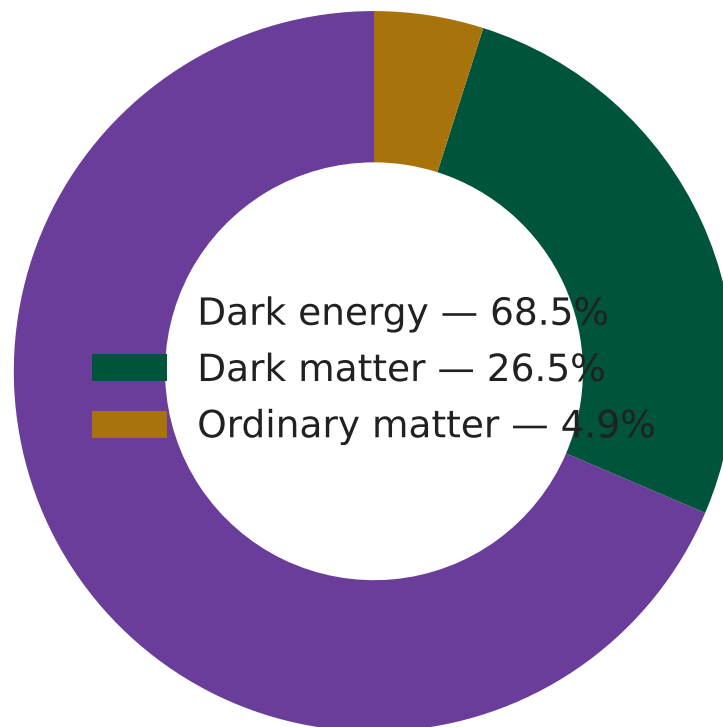


Figure 13: The concordance budget, drawn to scale. The thin gold wedge at the top is everything made of atoms; the green wedge is the dark matter of [Ch. 12](#), the only other thing that clusters and dilutes; the purple two-thirds is Λ , a density with no known constituents, inferred entirely from what it does to the expansion. `FlatLambdaCDM(H0=70, Om0=0.3)` asserts the purple wedge without your typing it, and does not tell the gold from the green at all.

That gold sliver contains this book’s entire ruler. Every star in the Milky Way (Ch. 1), all the gas between them, Andromeda, every galaxy in every image ever taken and every atom of the telescope that took it sit inside the 4.9% — and not even most of it, since the majority of ordinary matter is thin intergalactic gas that was never part of a galaxy. Everything astronomy spent its first three centuries cataloguing is a rounding error on the contents of the universe.

Dark energy today outweighs all matter, dark and ordinary, by a factor of 2.1815. And the three numbers sum to

$$\Omega_b + \Omega_c + \Omega_\Lambda = 0.999,$$

not 1.000. Read nothing into the last digit: these are rounded measured values, and the missing thousandth is the rounding, not a measured curvature. Flatness — $\Omega_k = 0$ — is a separate empirical claim, which the main guide’s Exercise 14.2 is precisely about not confusing with the identity $\Omega_m + \Omega_\Lambda + \Omega_k = 1$, true in any universe.

What `Om0=0.3` actually asserts. The object takes one matter parameter, not two, because the expansion history cannot tell baryons from dark matter: both dilute as a^{-3} , so only their sum enters. That sum is $\Omega_m = 0.049 + 0.265 = 0.314$. The repo types 0.3. Rounding the matter down rounds Ω_Λ up to 0.7, and moves the one epoch this chapter cares about — the crossover where Λ overtakes matter, from setting $\Omega_m(1+z)^3 = \Omega_\Lambda$ — from

$$z = 0.297$$

to

$$z = 0.326.$$

Less matter means Λ wins sooner, so the repo’s universe starts accelerating slightly *earlier* than the measured budget says. That shift is real and costs this program nothing: cosmology enters this repository in three files, none of which asks when the expansion changed gears, and the money number does not know what Ω_m is. Name it anyway — it tells you where a rounded cosmology *would* bite. Two further omissions in the same object: `site/guide_src/cosmo.py` builds it with `Tcmb0` at its default, so it contains no radiation at all (harmless at $z \leq 2$, wrong at the CMB), and it carries no baryon fraction, so Figure 13.1’s gold-versus-green split is invisible to it.

How it ends

Matter keeps diluting and Λ does not, so run the same two-term race forward. In $H(z)^2 = H_0^2 [\Omega_m(1+z)^3 + \Omega_\Lambda]$ — the main guide’s own [Friedmann equation](#), evaluated at a redshift — the matter term goes to zero as the universe expands, leaving a constant:

$$H \longrightarrow H_0 \sqrt{\Omega_\Lambda} = 70 \sqrt{0.685} = 57.935 \text{ km/s/Mpc.}$$

Read that carefully: it contains the trap. H is 70 today and 57.935 in the far future: the Hubble parameter is **decreasing**, monotonically, from here on — while the expansion **accelerates**. The two are not in tension, because H is a *fractional* growth rate, not a speed.

You already know this

This is compound interest, and the confusion dissolves once you say it that way. H is an interest rate in units of inverse time. A balance whose rate is easing toward a floor of 57.935 still pays more interest every year than the last, once the balance has grown enough to outrun the falling rate — the separation between two galaxies grows by more each year while growing by a smaller *percentage* each year. And a rate that stops falling — a *constant* H — is no more a static universe than a fixed APR is a static balance: constant fractional growth is exponential growth, $a \propto e^{Ht}$. Where it breaks: a bank credits interest from somewhere. Nothing is deposited here, and no galaxy moves through space to earn it. The identity is in the arithmetic of growth, not the mechanism.

When the gears changed. The crossover at $z = 0.297$ was, in cosmic time,

10.077 billion years after the beginning, out of the 13.467 billion years the same FlatLambdaCDM(70, 0.3) gives for the age of the universe — the identical number the main guide’s Ch. 14 quotes. So the universe spent

74.8% of its life so far slowing down, exactly as everyone before 1998 assumed it always had, and only the most recent 25.2% speeding up.

That is itself an open problem, and deserves naming. We live near the crossover: the Λ -to-matter ratio is of order one *right now*, at 2.1815, having been arbitrarily small for most of history and headed for arbitrarily large. Whether that is coincidence, selection (the era in which galaxies exist to ask the question need not be uniformly distributed), or a hint that Λ is not constant, nobody knows. It is called the coincidence problem; it is unsolved.

How it ends, given a constant Λ : exponential expansion, forever. Galaxies not already bound to us recede faster and faster, redshift toward invisibility, and cross out of view. This does *not* tear the Milky Way apart, and that is where the expansion picture must not be over-read: Λ ’s density is fixed and small, while inside a bound structure the local matter density is overwhelmingly larger, so the Galaxy, the Local Group and every cluster of Ch. 4 stay bound and drop out of the Hubble flow entirely. The Milky Way and Andromeda will merge; that merger is not in a race with Λ . What is lost is everything else — and with it, the evidence: an astronomer in that merged Local Group would have no distant supernovae and no CMB left to detect, and no way to discover any of this chapter. That holds if Λ is exactly constant. If its density grows, bound structures do come apart, and nobody has ruled that out.

Unlocks

What you can now read

- **Ch. 14 — FRW and the Friedmann equations** instructs you to “treat the cosmological constant Λ as an extra density component that never dilutes”, and never says what Λ is, who measured it, or that anyone was surprised. You now have the measurement: 0.399 magnitudes of unexpected faintness in a $z = 0.5$ Type Ia, in 1998, and the reason it took the CMB to make it stick.
- **Ch. 14’s density parameters** says FlatLambdaCDM(70, 0.3) “asserts three numbers, only one of which you typed”, and that $\Omega_{\Lambda,0}$ “is not independently set, it is *computed*, as $1 - \Omega_{m,0} - \Omega_{k,0} = 0.7$ ”. That 0.7 is not an API convenience: it is a measured 0.685,

rounded, asserting that two thirds of the energy density of the universe has no known constituents. The typed 0.3 likewise rounds $0.049 + 0.265$, moving the crossover from $z = 0.297$ to $z = 0.326$ while touching nothing this repository computes.

- **Ch. 14's Exercises 14.3 and 14.4** are where this chapter's subject surfaces in the main guide, unnamed. 14.3 says H nearly doubles between $z = 0.5$ and $z = 1$ because “the matter term $\Omega_m(1+z)^3$ is starting to dominate over the constant Ω_Λ term as you look back in time” — that sentence *is* the crossover, and the guide never says which side of it we are on. We are on the Λ side, by 2.1815 to one, for the most recent 25.2% of the age it quotes. 14.4 is the only place accelerated expansion appears at all, and you can now read it as a description of the actual universe, not a hypothetical.

Exercises

Exercise 13.1 — The whole discovery, in one division

A $z = 0.5$ Type Ia sits at $d_L = 2832.9$ Mpc in FlatLambdaCDM($H_0=70$, $\Omega_m=0.3$) and at $d_L = 2357.7$ Mpc in a flat, matter-only universe with the same H_0 . Using $\Delta m = 5 \log_{10}(d_1/d_2)$, compute how much fainter the supernova is in the accelerating universe. Then: why did a result of that size need two other kinds of measurement before anyone believed it?

Solution

The distance ratio is $2832.9/2357.7 = 1.2016$, so

$$\Delta m = 5 \log_{10}(1.2016) = 0.399 \text{ mag.}$$

Because every rival explanation makes the same prediction. A bad rung on the ladder, evolution in what a Type Ia *is* at earlier times, and grey dust all say “distant supernovae look too faint”, and no amount of care with supernovae separates them. Only an independent systematic can, which is what the CMB and galaxy clustering supplied.

Exercise 13.2 — Where the crossover has to be

Λ 's density is constant; matter's falls as $(1+z)^3$. Set them equal — $\Omega_m(1+z)^3 = \Omega_\Lambda$ — and solve for z , first with the measured budget ($\Omega_m = 0.049 + 0.265$, $\Omega_\Lambda = 0.685$), then with what this repository actually types (0.3 and 0.7). Which direction does the rounding push the crossover, and why is that the sign you should have expected?

Solution

Rearranging gives $z = (\Omega_\Lambda/\Omega_m)^{1/3} - 1$. Measured: $(0.685/0.314)^{1/3} - 1 = 0.297$.

Repo: $(0.7/0.3)^{1/3} - 1 = 0.326$.

The repo's rounding removes matter and adds Λ , so Λ wins sooner — a higher crossover redshift, meaning earlier in cosmic time. The sign is forced: less to slow the expansion and more to speed it up can only move the switch earlier. Note also how weakly the answer depends on the inputs, because the cube root flattens everything: nobody needs Ω_m to three digits to know which side of the crossover we are on.

Exercise 13.3 — How much of the universe’s life was the surprise?

The crossover at $z = 0.297$ happened when the universe was 10.077 billion years old; its present age in the same cosmology is 13.467 billion years. What fraction of cosmic history was spent decelerating, and how long ago, in lookback time, did the switch happen? Then: why does that number make $z = 0.5$ a well-chosen place to have looked?

Solution

$10.077/13.467 = 0.748$, so 74.8% of the universe’s life so far was spent slowing down and only 25.2% speeding up.

The switch is at lookback time $13.467 - 10.077 = 3.390$ billion years.

$z = 0.5$ is comfortably beyond the crossover, so a supernova there carries light from the decelerating era *and* the accelerating one, and its distance accumulates the difference between the two histories. A survey confined within $z \approx 0.3$ samples a single era, with far less leverage to separate the models. Reaching past the crossover is what made the effect visible at all, and the Type Ia is the candle bright enough to reach.

14. The Hubble tension, and why lensing arbitrates it

Chapter 10 handed you $v = H_0 d$ and left H_0 as a number somebody measures. Two groups measure it. They disagree, by more than either one’s error bar permits, and after two decades of people trying to find the mistake, nobody has found it. This chapter is about that gap: what the two numbers are, why splitting the difference does not settle it, and why the method that could arbitrate it is the one this entire research program is built on. It is the one place in this book where your own field is not a spectator.

What you can skip

You do not need to be walked through combining independent errors in quadrature, or through what “five sigma” asserts. Skip that arithmetic and read for the asymmetry instead: one number is a direct measurement of a local quantity, the other a model’s *prediction* of it, extrapolated from an observation of the very early universe. Those are not the same kind of object, and that is why the gap is interesting rather than merely irritating.

Two answers

The ladder. Chapter 9 built the distance ladder: parallax calibrates Cepheid variables, Cepheids calibrate Type Ia supernovae, and supernovae reach far enough out that the expansion dominates the random peculiar motions of individual galaxies. Measure a distance and a redshift for enough supernovae, fit the slope of v against d , and the slope *is* H_0 . The SH0ES program has been running exactly this measurement, and re-running it against every systematic anyone could name, for two decades. Its answer:

$$H_0^{\text{ladder}} = 73.0 \pm 1.0 \text{ km/s/Mpc.}$$

Every rung is local, in the sense that matters here: nothing in it assumes what the universe was doing before the light left. It assumes only that Cepheids are what we think they are, and that the rungs are calibrated against each other — the error-compounding problem Chapter 9 warned you about.

The CMB. Chapter 11 gave you the other one — a different kind of measurement entirely. The cosmic microwave background is a photograph of the universe from before any galaxy existed, and its temperature fluctuations have a characteristic angular size. That size is not arbitrary: sound waves in the pre-recombination plasma froze out at a physical length the physics predicts, so the map hands you a known ruler, seen at a known epoch, subtending a measured angle. What happens next is the important part: you do **not** read H_0 off the map. You fit a model — flat Λ CDM, whose ingredients Chapter 12 and Chapter 13 named — to that angle, then run it *forward* to today and read off what it says today’s expansion rate must be. Planck’s answer:

$$H_0^{\text{CMB}} = 67.4 \pm 0.5 \text{ km/s/Mpc.}$$

The gap between them is

$$73.0 - 67.4 = 5.6 \text{ km/s/Mpc,}$$

or 8.31 percent of the smaller value. Combine the uncertainties in quadrature — the two sets of errors have nothing to do with each other — and the disagreement is

$$\frac{73.0 - 67.4}{\sqrt{1.0^2 + 0.5^2}} \approx 5.01 \sigma.$$

The error bars do not touch. Somebody is wrong.

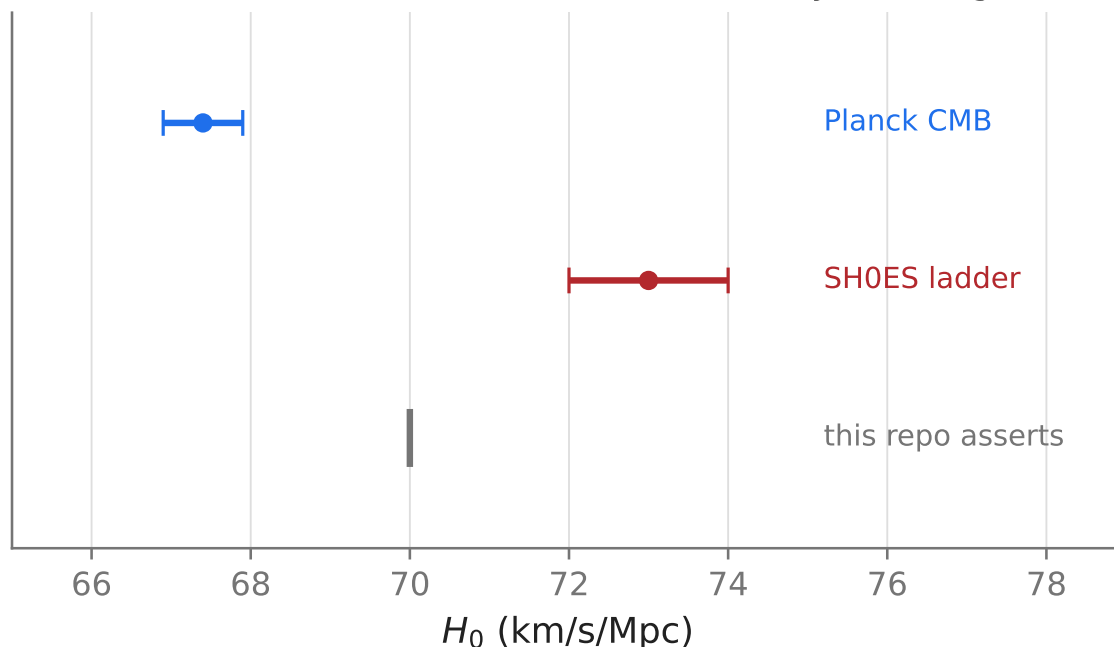


Figure 14: The whole controversy on one axis. Planck’s CMB value (67.4 ± 0.5) on top, SH0ES’s ladder value (73.0 ± 1.0) below it, and — with no error bar at all, because it is an assertion rather than a measurement — the 70 this repository fixes by fiat everywhere cosmology touches it. The bars are 1σ , and they do not come close to touching.

Why it matters

A number without a comparison is not doing its job. Both camps quote km/s/Mpc — extra recession speed per megaparsec of distance — and a megaparsec dwarfs this book’s ruler: the Milky Way’s whole stellar disc fits inside one many times over, with Andromeda well within the first. That is also where the picture breaks. Andromeda is not receding at all; it is falling towards us. The Local Group is bound, and expansion does not operate inside a bound system, so H_0 describes the average sightline in the open field, not the neighbourhood. The ruler that bites here is each camp’s own claimed precision, because that is what the disagreement has to be measured against.

The ladder claims to know its own answer to 1.37 percent of itself. The CMB claims 0.74 percent — better than one part in a hundred, on a property of the entire universe. Against those, the 8.31 percent gap between them is 6.07 times the ladder’s own quoted precision and 11.2 times the CMB’s. Each side claims a tolerance many times tighter than the thing they disagree about. At most one can be right, and nobody can rule out that neither is. This is also why averaging is not

an option: the disagreement is itself evidence that at least one *error model* is wrong, and averaging does not repair one.

You already know this

The asymmetry between the two numbers is one you have debugged before. The ladder measures H_0 where H_0 lives — in-distribution, at $z \approx 0$. The CMB fits a model to data from one regime, *extrapolates* it across the whole history of the universe, and reports the prediction with the fit’s internal uncertainty attached. When a model with excellent in-sample residuals makes a confident out-of-distribution prediction that a direct measurement contradicts, the first hypothesis is not “the measurement is broken” — it is that the model class is misspecified somewhere between the regimes. That is not a metaphor here — it is one of the two live explanations, and it is why “new physics before recombination” is a phrase people say out loud in seminars.

The stakes are concrete. H_0 sets the age of the universe and the size of the observable one, so [Chapter 11](#) already showed you this disagreement in different clothes: a gap between the age you get by running the expansion backwards and the age the CMB fit reports. Age and expansion rate are two readings of one dial, and a five-sigma disagreement about the dial is a five-sigma disagreement about how old everything is.

The third method

Suppose you had a way to measure H_0 that uses neither ladder nor CMB — no Cepheids, no supernova calibration, no assumption about what the universe was doing at recombination. It would inherit neither camp’s systematics, so it could not be wrong in either of the ways they might be. That is what “independent” has to mean here, and it is a hard bar: most methods fail it by standing on a rung of the ladder.

Strong gravitational lensing clears it. When a lens produces multiple images of one background source ([Chapter 16](#) shows the geometry), each image’s light took a different route through the lens’s gravitational field, and the routes differ in length and in the depth of well they cross. So the images do not arrive at the same time. If the source flickers — a quasar, or a supernova — you see that flicker replayed in each image, weeks or months apart. The delay is a *measured time*, in days, on a wall clock.

The delay is a length divided by c , and the length is set by the geometry of the whole configuration — how far away the lens and source are, and how the two relate. The main guide calls that combination the **time-delay distance**, and the single fact you need is that it scales as $1/H_0$: a faster-expanding universe is a smaller one, so its distances shrink together and its delays get shorter. Measure the delay, model the lens well enough to predict what the delay *would* be at a reference distance, take the ratio, and H_0 falls out. [The main guide’s Ch. 21](#) sets it up in one line — “measure Δt , model $\Delta \tau$, read off $D_{\Delta t}$, read off H_0 ” — and then spends the rest of the section on why that line is harder than it looks.

The method touches no Cepheid and no CMB: one lens, general relativity, and a stopwatch.

And now the honest part. Time-delay lensing has not ended the argument, because of a problem this program’s own guide devotes a chapter to: the **mass-sheet degeneracy**. Add a uniform sheet of mass to the lens and rescale the source, and every image position, every arc shape, every flux ratio comes out *identical*. The imaging cannot see it — not poorly, not noisily, but exactly not at

all, because the transformation is a symmetry of what the image data constrain (the main guide’s Ch. 21 proves this). The time delays are *not* invariant under it: they scale with the sheet. So an unmodelled sheet biases the inferred H_0 by exactly its own factor, and the bias runs in the direction that matters — a sheet you failed to model makes H_0 come out too high.

Ch. 21’s time-delay section works a representative case, and for a sheet strength nobody would call implausible the bias exceeds the entire 8.31 percent gap this method is meant to arbitrate. Read that as an order of magnitude, not a result: it is one illustrative sheet, and nobody knows the true sheet for any individual lens — which is the problem. The point survives the crudeness. The systematic lensing must control is larger than the effect it is measuring.

Lensing teams handle it by importing an external constraint that *is* sensitive to mass rather than light-bending geometry — most often the lens galaxy’s stellar velocity dispersion, which that same time-delay section identifies as the thing that breaks the degeneracy. How much you assume about the lens’s mass profile then decides how tight your answer is, and that choice is not a detail. Tighten the assumptions and lensing lands near the ladder with a small error bar. Marginalise the mass-sheet freedom fully and the bar swells until the answer is consistent with both camps and arbitrates nothing.

So the third method is real, genuinely independent, and without the last word. Its precision is a function of how much you are prepared to believe about elliptical galaxies — an uncomfortable place for a measurement to sit, and where this one sits.

Where it stands

Nobody knows. That is the accurate summary, and it should be said before the list of candidate explanations rather than after, because the list makes the field look closer to an answer than it is.

Neither side has cracked under scrutiny. The ladder’s most-suspected rung — whether Cepheids in crowded fields are measured correctly — has been re-examined with a better telescope, and the ladder still lands near 73.0 . The CMB’s map has been independently remade by other experiments, which agree with Planck. Methods avoiding one rung or another tend to land between the two camps with error bars wide enough to be consistent with everything, which is not a resolution.

What is left is uncomfortable in an interesting way. Either one of two extremely well-audited measurements carries a systematic that twenty years of adversarial re-analysis has missed, or Λ CDM — the model Chapter 13 told you accounts for the universe’s contents — is missing something between recombination and now. Extra relativistic species; dark energy that mattered briefly early and then went away; dark energy whose density is not constant after all. Each proposal fixes H_0 and breaks something else.

And this repository asserts 70.

The midpoint of the two camps is 70.2, so the value hard-coded everywhere cosmology touches this program sits a fifth of a unit below exactly halfway, committed to neither side. That is not evasion, and it is worth seeing why. The main guide’s Ch. 13 and Ch. 14 establish that this program’s money number never touches H_0 : the lens-modelling likelihood runs entirely in arcseconds, and cosmology enters only where a physical scale is genuinely required — a short, enumerated list. Ch. 14 puts it flatly, that the money number “would come out identical if this repository had asserted $H_0 = 67$ instead.”

Read that sentence again now that you know what 67 is. It is not a rhetorical alternative pulled

from the air — it is the CMB camp’s number, rounded. The main guide names one side of the biggest open argument in cosmology without mentioning that there is an argument. The scoping claim is true and worth making; the number it makes the point with is the tension itself.

Unlocks

What you can now read

- **Ch. 13 — The expanding universe and redshift** states, as a bare repo convention, that “*everywhere this repository touches cosmology, it fixes $H_0 = 70$ km/s/Mpc*”, and derives the Hubble time and Hubble distance from it. What it never says is that 70 is nobody’s measurement: the camps sit at 73.0 ± 1.0 and 67.4 ± 0.5 and miss each other by 5.01σ . You can now read that chapter’s H_0 for what it is — a placeholder in an argument it does not join.
- **Ch. 14 — FRW and the Friedmann equations** closes its scoping argument with “*it would come out identical if this repository had asserted $H_0 = 67$ instead.*” That is the Hubble tension, named by its value and nothing else, in a book where the phrase never appears. You now know which camp 67 belongs to, why the two do not overlap, and — the part that makes this your problem rather than a cosmologist’s — that the strong lenses that chapter is a prerequisite for are the third method that could tell them apart, if the mass-sheet degeneracy (Ch. 21) can be controlled.

Exercises

Exercise 14.1 — The sigma, by hand

The two measurements are 73.0 ± 1.0 and 67.4 ± 0.5 km/s/Mpc. Compute the tension in sigma. Then state the two assumptions that arithmetic makes, and say which one you would attack first.

Solution

The difference is 5.6 km/s/Mpc. The errors are independent — nothing in a CMB map shares a systematic with a Cepheid photometry pipeline — so they add in quadrature, and the tension is

$$\frac{73.0 - 67.4}{\sqrt{1.0^2 + 0.5^2}} \approx 5.01$$

sigma. Two assumptions are buried in it. First, that both error bars are Gaussian, so “five sigma” carries its usual tail probability; real systematic budgets are sums of terms whose distributions nobody knows, and the tails are exactly the part you cannot check. Second, that each quoted error is *complete*. Attack the second: the five-sigma result is itself the strongest available evidence against it. A disagreement this size is not primarily a claim about which value is right — it is a claim that at least one error model is incomplete.

Exercise 14.2 — The gap against each camp’s own ruler

Express each camp’s quoted uncertainty as a percentage of its own central value, and the gap

between the camps as a percentage too. What is the gap in units of each camp's own claimed precision, and what does that tell you that the raw sigma does not?

Solution

The ladder: $1.0/73.0 = 1.37$ percent . The CMB: $0.5/67.4 = 0.74$ percent . The gap: $5.6/67.4 = 8.31$ percent . In units of each camp's own precision, that gap is 6.07 and 11.2 respectively. The sigma tells you the disagreement is significant; this ratio tells you what the sigma hides — *how far outside each camp's own competence claim* the missing systematic would have to live. Whoever is wrong is not wrong by a little. They are wrong by many times the precision they publish, on the quantity they have spent careers measuring.

Exercise 14.3 — What the repo's number is doing

This repository fixes $H_0 = 70.0$ km/s/Mpc. Compute the midpoint of the two camps and compare. Then answer the harder question: is choosing 70 a scientific decision, an evasion, or neither?

Solution

The midpoint is $(73.0+67.4)/2 = 70.2$, and the repo asserts 70.0 — a fifth of a unit below dead centre, and a value no published measurement claims. It is neither, for the reason the main guide's Ch. 14 argues: this program's money number comes from a likelihood that runs entirely in angles and never multiplies by a distance, so H_0 has no path into it. When a parameter cannot affect the result, picking a round number near the middle of a fight you are not in is correct engineering with no scientific content at all. It *would* be an evasion in a program that used H_0 to set a physical scale — which is why the guide enumerates the three files where it does.

Exercise 14.4 — What disqualifies a third method

A “third method” is only useful if it is independent of both existing ones. For each of the following, say whether it would break the tie, and why or why not: (a) a new supernova sample twice the size, calibrated on the same Cepheids; (b) a better CMB map from a different satellite; (c) a time-delay measurement of a lensed quasar; (d) that same time-delay measurement, where the lens's mass profile was calibrated against a Cepheid-based distance.

Solution

- (a) No. More supernovae shrink the statistical error on a rung that was never the suspect; the systematic under suspicion lives in the Cepheid calibration, which the new sample inherits unchanged. Doubling N on a biased estimator buys a tighter wrong answer.
- (b) No, for the mirror reason. A better map re-measures the same angle and feeds it through the same Λ CDM extrapolation. If the model is what is broken, a perfect map does not help — and this has been done, by more than one experiment, all agreeing with Planck.
- (c) Yes, in principle: no Cepheid, no supernova calibration, no assumption about the universe before the light left. It can be wrong, but not *in either of the two ways currently under suspicion* — the whole point of an independent method. Its own vulnerability is separate, and it is the mass-sheet degeneracy (Ch. 21).
- (d) No, and this is the trap. The moment the lens model's scale is pinned by anything standing on a ladder rung, the method inherits the ladder's systematic and stops being a third opinion — it becomes a correlated restatement of the first. Independence is a property of the whole inference chain, not of the headline technique, and auditing that chain for a smuggled dependency is the same discipline as checking that no validation feature leaked in from training.

15. Gravity bends light

Every arc, ring and multiple image this program hunts is one physical fact applied at galactic scale: mass bends light, by a specific and computable amount. The main guide’s [Ch. 16](#) derives that amount — $\alpha = 4GM/c^2b$ — in about a page, using an integral this book does not have. This chapter’s job is to make that page worth reading: what the claim physically says, why the coefficient is *twice* the Newtonian one rather than approximately twice, and how a single factor of two in a single amplitude became one of the most famous measurements in astronomy. The derivation belongs to Ch. 16. The reason to care belongs here.

What you can skip

You do not need general relativity, geodesics, the metric, or the integral that produces the coefficient — the main guide does that integral and this book never will. Take “mass curves spacetime and light follows the curve” as the claim and read on for what it buys. If you already know that GR predicts exactly twice the Newtonian deflection and that a 1919 eclipse was the test, skip to [Why a factor of two mattered](#) — the honest version of what 1919 established is less than the story you have probably been told.

The idea

Start with Newton, who has two answers, and both matter.

The first is zero. Newtonian gravity is a force between masses, $F = GMm/r^2$, and light has no mass. Put $m = 0$ in and nothing happens: light passes a star in a straight line and no eclipse is worth mounting. That was a real position, not a strawman — it is the first thing the formula says.

The second answer is not zero, and comes from noticing that m cancels. Divide the force by the mass being pulled and you get an *acceleration*, $a = GM/r^2$, with no reference to the falling body at all: a feather and a cannonball fall alike. In the eighteenth-century picture light was a stream of tiny corpuscles travelling at speed c , and corpuscles are bodies, and bodies fall. Henry Cavendish did this calculation privately around 1784 and Johann Georg von Soldner published it in 1801; both got $\alpha = 2GM/c^2b$, where b is the **impact parameter** — the closest approach the undeflected ray would have made to the mass’s centre. For light grazing the Sun’s visible edge, that is

$$\alpha_N = \frac{2GM_\odot}{c^2R_\odot} \approx 0.876'',$$

an arcsecond being 1/3600 of a degree, the unit [Ch. 9 of the main guide](#) builds.

Now Einstein, whose claim is not “gravity pulls on light harder than Newton thought”. It is a different sentence in a different vocabulary. Mass changes the *geometry* of spacetime, and light travels the straightest available path through whatever geometry it finds. Nothing pulls on the ray; the ray does not turn. It goes straight, and “straight” in a geometry that is not flat does not look straight to someone drawing flat pictures from far away. The word “bends”, in this chapter’s title and in every paper you will read, describes what happens to the ruler, not to the light.

Where the analogy breaks, and it does break. You can carry “the Sun tugs the photon sideways” around as an intuition, and it will get you the right functional form and half the right amplitude — which is exactly why it is dangerous. The Newtonian answer is not a rough approximation that GR nudges; it is precisely half, for a reason worth holding onto. In the weak-field

limit mass warps two things: the flow of time, and space itself. A slowly moving object weights the spatial part by something of order $(v/c)^2$ and never notices it, which is why Newtonian mechanics, built for slow particles, is right about planets and captures only the time half. Light, at $v = c$, feels both halves equally — two equal contributions, one of which Newtonian physics has no vocabulary for. So

$$\alpha_{\text{GR}} = \frac{4GM_{\odot}}{c^2 R_{\odot}} \approx 1.752'',$$

and the ratio of the two predictions is

$$\frac{\alpha_{\text{GR}}}{\alpha_{\text{N}}} = 2$$

exactly: the second contribution *equals* the first, it does not resemble it. [Ch. 16](#) writes the weak-field metric down and shows the two terms contributing equally, which is where that “exactly” is earned. One caveat: both formulas are leading-order results, dropping corrections suppressed by a further factor of $GM/(c^2 b)$ that is negligible at the Sun’s limb. The exactness lives inside a stated limit, not in the sky.

How much bending is on offer. $1.752''$ is the largest gravitational deflection anywhere in the solar system, and not by a close margin: the Sun holds essentially all the local mass, and its limb is the nearest a sightline can be brought to that mass without passing through it. To do better you need more mass, and the next rung up is a galaxy — where this book’s ruler earns its keep. A Milky Way holds roughly 10^{11} stars ([Ch. 1](#)), so you might expect 10^{11} times the bending. You do not get it: the ray does not graze a stellar surface, and the impact parameter grows by very nearly the same factor the mass does. The deflection sees only the combination M/b , the two enormous factors cancel, and a whole galaxy bends light by something in the neighbourhood of the Sun’s own $1.752''$ — [Exercise 15.3](#) is that arithmetic. This near-cancellation, and nothing deeper, is why every Einstein radius in this program is around an arcsecond rather than a microarcsecond or a degree.

Treat that as an order-of-magnitude count, not a calculation: a real lens galaxy is heavier than the Milky Way, holds most of its mass as dark matter rather than stars ([Ch. 12](#)), and — being extended rather than a point — only puts the mass *inside* the impact parameter to work. [Ch. 16](#) gives the real number.

Eddington 1919

Starlight passing close to the Sun arrives in company with sunlight, and the daytime sky beside the Sun is overwhelmingly brighter than any star in it. A total eclipse removes the sunlight for a few minutes and leaves the star field standing. That is the entire reason this is an eclipse experiment, and why the test had to wait for a date the sky picked.

The measurement is differential, which is where the difficulty lives. You photograph the star field around the eclipsed Sun; months later, at night, with the Sun elsewhere on the sky, you photograph the same field through the same instrument; then you subtract the two sets of plate positions. If light bends, the stars nearest the limb will have shifted outward during the eclipse — radially away from the Sun — by an amount falling off as $1/b$.

29 May 1919 was chosen because the eclipsed Sun would sit in front of the Hyades, a cluster dense enough to put several usable stars near the limb; the eclipses either side of it offered a nearly empty field. The Astronomer Royal, Frank Dyson, organised two expeditions against the weather: Arthur Eddington to Príncipe, off West Africa, and Andrew Crommelin and Charles Davidson to Sobral, in Brazil. The number that went into the textbooks came from Sobral's 4-inch telescope:

$$\alpha_{\text{measured}} = 1.98'' \pm 0.16''.$$

That is not the only number the expeditions produced, and here the popular telling stops. Sobral's *other* instrument, the astrographic telescope, was the primary one; its mirror appears to have distorted in the heat of the day, its images came out defocused, and its plates gave a deflection close to the Newtonian value. Those plates were set aside. Príncipe was clouded out through most of totality; Eddington came away with a handful of usable plates whose value was lower than Sobral's and whose error bar was far wider. Taken by itself, Príncipe excluded nothing.

Was setting the astrographic plates aside legitimate? Most likely yes: the defocus was a real fault with an identified physical cause rather than a post-hoc rationalisation, later re-analyses of those plates support the decision, and a century of better measurements agrees with the answer the surviving data gave. But the cut was made after the numbers were in hand, by people who could see which hypothesis each subset favoured and who had a preferred answer, with no pre-registered criterion to point at. That is a fact about the result, not an accusation against anyone in it — and it is the sort of thing you would flag in a colleague's ablation table, so there is no reason to give it a pass here.

Why a factor of two mattered

Look at what the two theories actually disagree about. Newton's corpuscular prediction is $2GM/c^2b$; Einstein's is $4GM/c^2b$. Same variables, same linear dependence on M , same $1/b$ falloff: the functional form is *identical*, and the entire disagreement between two theories of gravity is one multiplicative constant.

That is why the test was hard, and the difficulty should be familiar. Had the two predictions differed in *shape* — a different power of b , a dependence on wavelength, anything — you could have measured a spread of stars at a spread of impact parameters and let the shape decide, no absolute calibration required, because a shape survives a wrong scale factor. They do not differ in shape. Divide one by the other and every variable cancels: the ratio is 2 for every b , every M , every star on the plate, so on a log axis the two predictions are parallel lines offset by a constant $\log 2$. No curve fit separates them; only an absolute amplitude does.

You already know this

A constant that multiplies every prediction equally is perfectly confounded with any multiplicative error in your instrument. The plate scale — how many arcseconds one millimetre of emulsion stands for — enters the measurement in the same place the physics does, so a plate scale wrong by 2 and a theory wrong by 2 produce identical data. The information you need is not in the shape of the curve; it is in the units of the axis. Eddington's real adversary was not Newton. It was the possibility that his telescope had changed its own units between Brazil in daylight and Brazil at night.

And the amplitude on offer is small. The gap between the two theories — the whole content of the argument, the entire reason for two ships and a year of preparation — is $0.876''$, the Newtonian prediction over again, as it must be once the ratio is 2. Under one arcsecond, to be resolved on a plate exposed for a few seconds through a telescope shipped to the tropics, against a comparison plate taken months earlier at a different temperature.

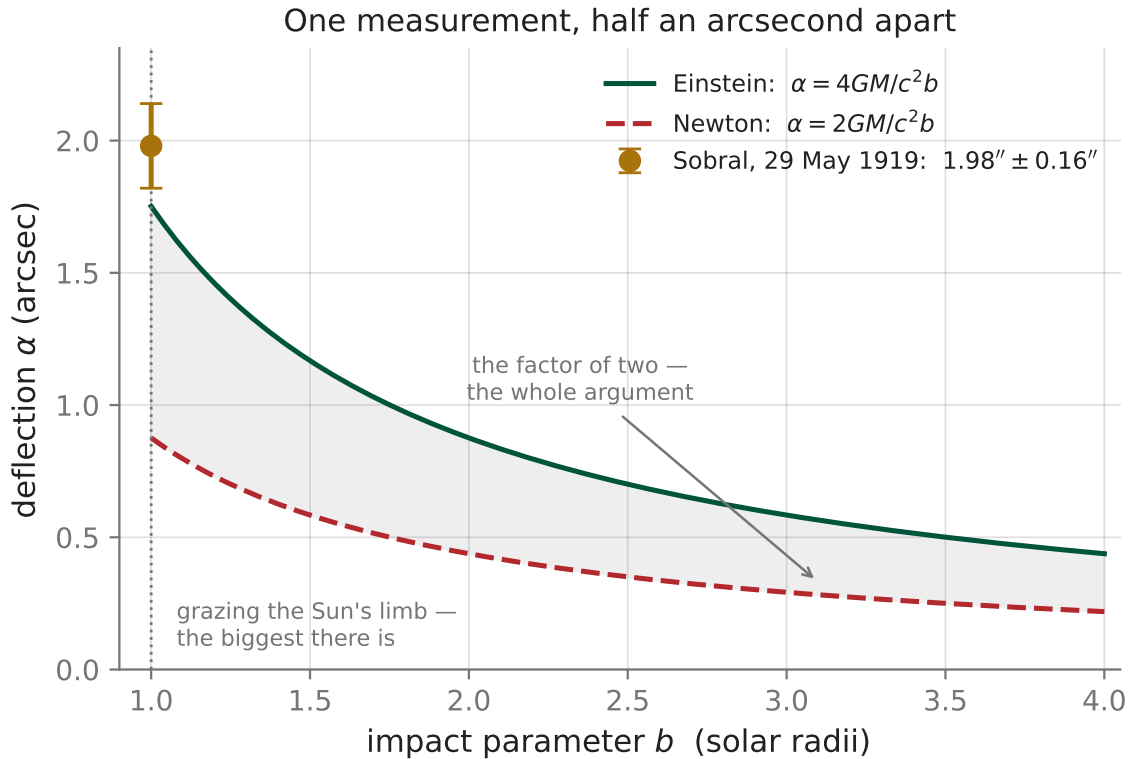


Figure 15: Both theories give the same $1/b$ falloff, so the two curves have one shape and stand everywhere in the ratio 2: no fit to the *shape* tells them apart, and the shaded region is the whole argument. These axes are linear, so the constant *ratio* is not a constant gap: the curves converge as b grows, and only on a log axis would they be parallel. The gap is widest where the mass is closest, at the Sun's limb ($b = 1 R_{\odot}$), and even there it is Newton's $0.876''$ against Einstein's $1.752''$. The point with the error bar is Sobral's 4-inch result of 29 May 1919, $1.98'' \pm 0.16''$: far above the dashed line, and a little above the solid one — the honest shape of the result.

Now the arithmetic a century of retelling usually skips. Against Newton:

$$\frac{1.98'' - 0.876''}{0.16''} \approx 6.9,$$

standard deviations. Under the Newtonian hypothesis that is not a fluctuation anyone has to entertain. Newton was excluded — that is the claim 1919 actually established.

Now the same subtraction against Einstein's $1.752''$ — carefully, because this is where the story turns into a victory parade. The gap is much smaller, but it is not zero: Sobral's central value sits above the GR prediction by more than one standard deviation and less than two, which in the language you would use on your own results is *consistent with* general relativity and nothing stronger. *Príncipe*, lower and wider still, excluded neither theory on its own.

So the honest summary is: **1919 excluded Newton. It did not confirm Einstein to any precision.** Those are different statements, and it was the second that made the newspapers. Modern confidence in $4GM/c^2b$ does not rest on those plates and never should have — radio interferometry measures the same deflection for quasars near the Sun with no eclipse required, the glare not being a radio problem, and space astrometry has since pinned the coefficient far beyond anything 1919 could approach. The eclipse was decisive about Newton, and it was famous, and those are not the same fact.

Why this program needs the factor of two. The coefficient does not stay in this chapter. [Ch. 16 of the main guide](#) carries the 4 — as a $2/c^2$ in the definition of the lensing potential — into Σ_{cr} , then into the convergence κ , then into θ_{E} , the Einstein radius every strong-lensing paper quotes in its abstract. Its Exercise 16.4 works out what follows from using the Newtonian amount instead: κ reads at half strength for the same real mass, and every mass this program measures comes out a factor of 2 wrong. That is what the eclipse bought — not a story about a physicist becoming famous, but one multiplicative constant that every mass in the field is still divided by.

Unlocks

What you can now read

- [Ch. 16 — How much light bends, and the factor of two](#) does the derivation this chapter refused to do, and does it well — but it assumes you already know *why anyone should care about a factor of two*. It says the ratio is “exactly 2 — nothing lets it be anything else, since α_{GR} was *defined* as double α_{N} above. The physics is in which of the two numbers ... the sky actually shows”, and then settles that question in a single clause. Its whole account of the *result* is that Dyson, Eddington and Davidson “reported results close to the GR prediction and clearly separated from the Newtonian one” — no number, no error bar, no mention that the primary instrument’s plates were cut. You now have $1.98'' \pm 0.16''$, excluding Newton by 6.9σ while agreeing with Einstein only to within its own wide error bar — so you can read that sentence as the compression it is.
- Its “**You already know this**” box says the eclipse “had three sharply separated candidates to rule between”, and its **skip box** offers to send you straight to [The thin-lens approximation](#) “if you already know the GR light-bending result and where the factor of two comes from”. You have all three candidates — zero from putting $m = 0$ into $F = GMm/r^2$, the Newtonian value from treating light as an ordinary falling body, the GR value from letting space curve as well as time — so that skip is a real option, and the thin-lens section — the machinery every later chapter leans on — is where your attention should go.

Exercises

Exercise 15.1 — Why the verdict is asymmetric

The text computes that $1.98'' \pm 0.16''$ sits 6.9σ from Newton. Do the same subtraction against Einstein’s $1.752''$, in error-bar units, and explain why “excluded Newton” and “confirmed Einstein” are not two readings of one result.

Solution

The gap to Einstein is more than one error bar and less than two — so the measurement is *consistent with* GR, which is the weakest thing a measurement can say about a hypothesis it does not reject. Exclusion and confirmation are asymmetric because rejection needs only distance, while confirmation needs precision, and this measurement had distance without precision: it kills Newton outright and pins nothing.

Exercise 15.2 — Why no curve fit could have settled it

Both predictions have the form $\alpha = kGM/c^2b$, differing only in $k \in \{2, 4\}$. Suppose you measure many stars at many impact parameters, but your plate scale — arcseconds per millimetre of emulsion — carries an unknown multiplicative error s . Show that the shape of the falloff tells you nothing about k , and name the quantity you must know to have an experiment at all.

Solution

What you record is $s \cdot kGM/(c^2b)$: every point is the true prediction times the constant s , so the *shape* — a $1/b$ falloff, slope -1 on log-log axes — is identical for $k = 2$, $k = 4$, and any s whatever. Fitting the shape recovers only the product sk , and $s = 1, k = 4$ is indistinguishable from $s = 2, k = 2$: the theories disagree by 2, a pure number with no b in it, and s is a pure number with no b in it too. What you must know independently is the plate scale — the absolute calibration of your angular axis. Hence the *differential* design: the night-time comparison plate is the calibration, and every worry the expeditions had (mirror temperature, focus, emulsion shrinkage) is a worry about s drifting between the two exposures, not about physics.

Exercise 15.3 — Why a galaxy bends light about as much as the Sun

A Milky Way holds about 10^{11} stars, so about 10^{11} times the Sun's mass. Why does it not bend light 10^{11} times as far as the Sun's limb does — and what does the answer say about why every Einstein radius in this field is quoted in arcseconds?

Solution

The deflection sees M and b only through M/b , and b grows by nearly the factor M does. A lensed ray passes tens of thousands of light-years from a galaxy's centre while an eclipse ray grazes the Sun's surface — and the Sun's radius is a couple of light-seconds, while ten thousand light-years is a few times 10^{11} light-seconds. Numerator and denominator both go up by roughly 10^{11} ; they cancel to within an order of magnitude; the galaxy lands within a small factor of $1.752''$. Two enormous numbers dividing out is the whole reason strong lensing is an arcsecond-scale phenomenon, and therefore the reason it is *measurable* at all.

16. What a strong lens is, and where to go next

Fifteen chapters have been building one object. [Chapter 15](#) established the physics: mass curves spacetime, light follows the curve, and Eddington measured the bend. This chapter puts that physics into the only configuration that makes it useful — a massive galaxy sitting on the sightline between you and something far behind it. That is a **strong gravitational lens**, which the main guide names on its first page and never draws. By the end you will have the geometry, a number for how rare the accident is, the vocabulary for what it looks like on a plate, the reasons anyone spends a telescope night on one, and a map into the main guide. This is the last chapter, and it is the handoff.

What you can skip

You do not need to be told that finding a rare class in a huge candidate set is a base-rate problem, or why excellent specificity still delivers mostly false positives at a prior this low. You have shipped that argument. What is new is the *physical* source of the base rate: where the 10^{-12} comes from, why it is a solid angle rather than a population fraction, and why it is not the number that governs a survey's yield. No deflection algebra appears here.

The geometry

Three things sit on one line: you, a galaxy, and a second galaxy far behind the first. Nothing else is required, and nothing else is available — this is not an experiment anyone set up.

Fix the redshifts, the pair the main guide uses as its standard test case throughout Part IV without ever saying what they mean physically. The near galaxy — the **lens**, or **deflector** — sits at $z_{\text{lens}} = 0.5$; the **source** sits at $z_{\text{source}} = 2.0$. [Chapter 8](#) turned those into distances and lookback times: the source's light left billions of years before the lens's did, and the gap between them along the sightline dwarfs either galaxy's width.

That gap is the first thing to get right, because the everyday sense of “in front of” is wrong here. The lens and the source are not neighbours: not interacting, not bound, not part of the same structure, neither aware the other exists. They share exactly one property, a **direction**. Light from the source happens to pass close to the lens on its way to your detector, and close is an angle on the sky, not a distance in space. Two galaxies “on one sightline” is a fact about your position, not theirs — and [Chapter 5](#) has the sharper version: an image is never a report of *now*, and these two are not even reports of the same *then*.

The figure's last claim is worth naming. A real galaxy has depth — its mass spreads over a hundred thousand light-years along the sightline, not a plane. Treating it as a **thin lens**, a flat sheet of projected mass that bends light once and instantly, is an approximation, and a good one for the reason Figure 16.1 makes visible: the galaxy's own depth is negligible against the distances to it and past it. [Ch. 16 of the main guide](#) does that integral and marks the exact line where “thin” enters. You need only know that the flat sheet in every equation from there on is a real 3-D galaxy, squashed on purpose, with a stated justification.

One more distinction, since “strong” is doing work. Every mass on every sightline bends light a little; a background galaxy's shape is distorted by a percent or so, and measuring that statistically across millions of galaxies is **weak** lensing — a different discipline with different tools. A lens earns **strong** only when the alignment is close enough to produce *more than one image*, or a visibly stretched arc. That threshold has a number, and the number is why any of this is rare.

A strong lens is an accident of alignment

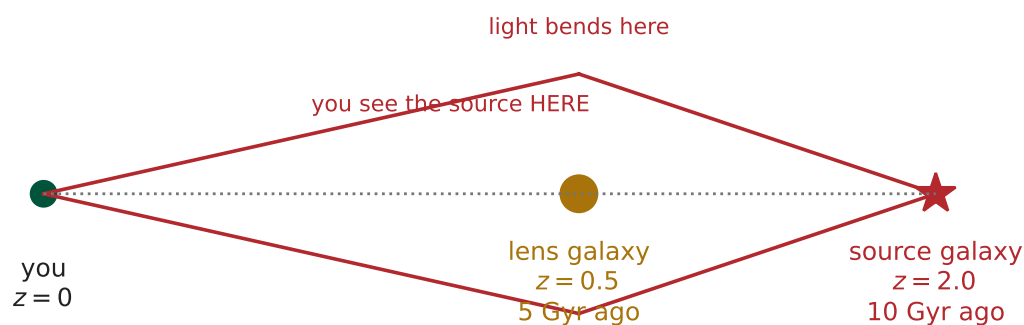


Figure 16: The whole geometry: you on the left, the lens galaxy at $z = 0.5$, the source galaxy at $z = 2.0$. The dotted line is where the source actually is. The two solid rays are paths light really takes — each bends at the lens plane and arrives from a direction that is *not* the dotted line, so you see the source displaced, twice. The figure lies about two things on purpose. The rays leave you some eight degrees off the dotted line and kink through about twenty at the lens plane; both real angles, computed below, are of order one arcsecond, so a true-to-scale version would show two lines indistinguishable from the dotted one. And the horizontal axis is not a distance scale — the lens sits where the labels fit, not where $z = 0.5$ falls. What it gets right is the topology: two paths, one source, and a deflection that happens at one plane rather than gradually along the way.

Why it is rare

The threshold is an angle called the **Einstein radius**, θ_E : the source must lie within roughly θ_E of the lens's axis, as seen by you, for multiple images to appear at all.

θ_E depends on the lens's mass and on the two redshifts. Mass, for an elliptical, is read off its **velocity dispersion** σ_v — the spread of speeds of its stars, which a spectrum measures directly (Ch. 7 is how, Ch. 12 of the main guide is the arithmetic). A fiducial massive elliptical, the kind Ch. 3 identified as this program's quarry and heavier than the Milky Way, has $\sigma_v = 250$ km/s, and at $z_{\text{lens}} = 0.5$, $z_{\text{source}} = 2.0$ that gives

$$\theta_E = 1.145''$$

The formula is Ch. 19's; one structural fact is worth carrying there. θ_E goes as σ_v^2 — mass enters quadratically — and that exponent is what separates a galaxy's ring from a cluster's, larger by about an order of magnitude (Ch. 4 puts the two side by side).

Now turn 1.145'' into a rarity. An arcsecond is 1/3600 of a degree, so the disc of sky inside the Einstein radius has area

$$\pi \theta_E^2 = 3.18 \times 10^{-7} \text{ deg}^2$$

against a whole sky of 41,253 deg^2 — 4π steradians in square degrees, a definition rather than a measurement. Divide, and one lens galaxy's disc covers 7.71×10^{-12} of the sky: one part in 1.30×10^{11} . Point somewhere at random and those are the odds it lands inside a given lens galaxy's disc — near enough the odds of picking one particular star out of the Milky Way's 10^{11} (Ch. 1). That is the target strong lensing has to hit, and nothing aims it.

Two caveats, because this number is easy to overread.

It is a per-pair figure, not a lens rate. It is the chance one direction lands inside one galaxy's disc — not the fraction of galaxies that are lenses, and not a survey's yield. This book does not compute a yield, and Exercise 16.3 is where you work out why you cannot get one from the figure above.

A $\sigma_v = 250$ km/s galaxy is not a typical galaxy. It is a fiducial massive elliptical, chosen because it is the kind that lenses. Most galaxies are far less massive, and since $\theta_E \propto \sigma_v^2$ and disc area goes as θ_E^2 , the target shrinks as the *fourth* power of the dispersion. Halve the dispersion and the disc is sixteen times smaller, with an Einstein radius no ground-based survey could resolve even if the alignment happened. Rarity is not one number; it is a steeply mass-weighted one.

So how does anyone find any? Brute force. The DESI Legacy Surveys' DR11 sweep scored 5.38×10^7 galaxies — one classifier decision each — for a few thousand candidates. That is what Ch. 6 meant by a survey: not a telescope pointed at something interesting, but a telescope pointed at everything, because the interesting thing is an accident nobody can predict.

You already know this

A prior this low is a precision problem before it is a physics problem. At 5.38×10^7 decisions and a positive class of order one in ten thousand, a finder with a false-positive rate of even one

in a thousand returns a list that is overwhelmingly contamination, and no AUC on a balanced validation set tells you so. This is the half of the program needing no lensing at all: [Ch. 27](#) derives where the threshold has to sit and why the survey’s own resolution bounds it; [Ch. 28](#) asks whether the human grades those finders train against carry any information about truth. Both are yours already. The physics here supplies their base rate.

What you see

The alignment is invisible. What you see is the light, rearranged, and it comes in a small vocabulary.

An arc. For a circular lens the deflection points straight at the centre: it changes how far a ray passes from the lens, not its direction around it. The image therefore keeps the source’s azimuth but sits at a larger radius — and the same azimuthal span is a longer arc the farther out it sits. So the source’s radial extent comes through roughly intact while its tangential extent is stretched, the more so the tighter the alignment. A small round blob becomes a curved sliver, concentric with the lens galaxy. Arcs are what a lens finder — human or convolutional — actually keys on.

A ring. Put the source exactly on the axis and the configuration is rotationally symmetric: every direction around the lens is equally valid, so the image is a complete circle of radius θ_E . This is the **Einstein ring**, which this program’s reproductions are named after. Note the cost: exact alignment is measure-zero, so a perfect ring never happens. Near-rings do, and they are the best systems in any catalogue, because a ring displays θ_E as a radius you can measure with a caliper. Figure 16.1 is a slice through this case — its source is on the axis — and its two rays are two points on a circle the 2-D picture would show as continuous.

A double. Move the source off-axis but keep it inside θ_E and the ring breaks into two images on opposite sides of the lens. Push it outside and the second image does not fade — it ceases to exist. [Ch. 17](#) proves that by trying to find the root and failing, and derives a result you can use without its algebra: for an isothermal lens — the profile real ellipticals sit close to, for reasons the next section admits nobody has explained — the two images of a double are separated by $2\theta_E$, and where inside the Einstein radius the source sits cancels out completely. A separation is a ruler measurement; it converts to θ_E by dividing by two, and θ_E converts to a mass.

A quad. Four images. A circular lens cannot make one — the mass has to be elliptical, which real ellipticals are. The extra pair appears when the source lands inside a region called a **caustic**, the thing the main guide’s landing page admits it assumes you already know ([Ch. 18](#) is where you meet it). Quads are prized because four images constrain a model far harder than two.

[Ch. 17](#) says those three words “cover almost every strong lens actually observed”, and derives all three as root counts of one equation. You now know what they are as things in a picture, which is the half that chapter takes for granted.

Why anyone cares

Three reasons, in ascending order of what they are worth.

Mass, including the mass that does not shine. Deflection responds to *all* the mass on the sightline, whether or not it emits light. So a lens weighs a galaxy without asking what it is made of — and the answer comes out consistently larger than the stars account for. That is the fourth of the independent witnesses in [Ch. 12](#), and the hardest to argue with, because it assumes nothing about how stars move or how gas heats. The main guide has a clean identity for it: the mean

projected mass density inside θ_E , in the natural units of lensing, is exactly 1 — not approximately — which turns a measured ring radius into an enclosed mass in one step (Ch. 19 proves it; Ch. 19 cashes it).

The shape of the mass, not only its total. A ring radius gives the mass inside one circle; an arc gives more, because its brightness and curvature respond to how the density *falls off* with radius. That slope is the main guide’s destination number, γ in $\rho \sim r^{-\gamma}$ (Ch. 20 defines it; all of Part V extracts one galaxy’s value from pixels). Why an exponent is worth a campaign: the stars in an elliptical follow one profile, the dark matter another, and the *total* lands very close to a specific slope for reasons nobody has explained. The main guide calls that the isothermal conspiracy (Ch. 10) and is candid that it is an unexplained regularity, not a derived one.

H_0 . This one reaches outside the field. When the source varies — a quasar, a supernova — the images vary too, but not at the same moment, because the light took paths of different lengths. The **time delay** between images is measured in days; a delay is a length, a length across a known redshift is a distance, and a distance against a redshift is H_0 . That chain is independent of the distance ladder and of the CMB, which is why it matters: Ch. 14 laid out the argument between the ladder’s answer and the CMB’s, and the reader’s own field is the independent third voice (Ch. 21 works it).

Now the part that should not be sold. A strong lens does not measure mass unambiguously. Add a uniform sheet of mass across the image and rescale the source to compensate, and the result is a *pixel-identical* image — same arcs, same positions, same brightnesses, different mass. This is the **mass-sheet degeneracy**; it is exact rather than approximate, no better sampler removes it, and it is the largest single reason lensing’s H_0 values move when the assumptions behind them move. Ch. 21 states it; Ch. 21 reframes it as something you recognise from anywhere a model has a direction the data cannot see. Every claim above is real; each is also conditioned on an assumption that breaks that degeneracy, and the assumption is where the arguing happens.

Where to go next

The book is finished. Here is the map.

The destination was always one sentence: **you can open the main guide at Ch. 9 and not be lost**. That chapter’s first line is “Nothing astronomical comes with a ruler” — and you now know what it is failing to rule.

Main guide part	Open at	What this book gave you
I — The mathematical spine (2–8)	Skim, or skip	Nothing. It is your own field’s neighbour. Read it when a later chapter cites it.

Main guide part	Open at	What this book gave you
II — The physical universe (9–12)	Ch. 9 — start here	Almost the whole book: Ch. 1 (the scale ladder), Ch. 2 (why old means red — the premise under Ch. 10’s stellar populations), Ch. 3 (which galaxies lens), Ch. 5 (photons, blackbodies, inverse square), Ch. 6 (HST, DESI and Euclid, none of which the guide introduces), Ch. 7 (lines), Ch. 8 (what z means).
III — Cosmology (13–15)	Ch. 13	Ch. 10 (the raisin bread Ch. 13 formalises as $a(t)$), Ch. 11 (what $a = 0$ is; the CMB), Ch. 12 (what the CDM in FlatLambdaCDM stands for — you type it in Ch. 14), Ch. 13 (what $\Omega_m = 0.3$ asserts), Ch. 14 (why H_0 is contested at all).
IV — Gravitational lensing (16–21)	Ch. 16	Ch. 15 (why the factor of two was worth an eclipse) and this chapter (what the thing being modelled <i>is</i>).
V — This repository’s science (22–26)	Ch. 22	This chapter’s “why anyone cares” — the only reason to model an arc at all. The rest is inference, and it is yours.
VI — Discovery (27–28)	Ch. 28 — the guide says read it second	Ch. 6 (what a survey is) and this chapter’s base rate. Ch. 28 needs no physics; it is pure label epistemics.
VII — Synthesis (29)	Ch. 29	Read it last, as intended.

Two things to carry across the boundary.

The first: this book gave you nouns, not derivations. When the main guide integrates something you will not be able to check the integral from anything here — but you will know what it is integrating, which was never the part that was missing. **The handoff is the structure**, and every Unlocks box was a receipt for it.

The second is a habit rather than a fact. The main guide’s culture is to retract its own numbers — its final report carries a “ $\sim 17\sigma$ ” claim that reconciles with none of the uncertainties it quotes, and **Ch. 25** makes you do the division yourself instead of telling you the answer. This book ran the same gate: every number on every page is tagged, recomputed by a script, and fails the build if it drifts. Nothing here asked to be believed. Take that to Ch. 9.

Unlocks

What you can now read

- **Ch. 1 — What this repository does, and what one number costs** describes discovery as combing “through tens of millions of galaxy images for the rare few whose light has been bent by something massive sitting in front of them” — then never says why they are rare, or what “in front of” means. It means one sightline and nothing else: two galaxies billions of light-years apart sharing a direction and no other property. “Tens of millions” is 5.38×10^7 cutouts; “rare” is 7.71×10^{-12} of the sky per lens. Ch. 1 also puts $\gamma = 1.103 \pm 0.008$, “the density slope of one galaxy’s mass profile”, on its first page as the book’s destination — and you now know what galaxy, and why a slope rather than a mass.
- **Ch. 9 — Arcseconds, magnitudes, and the units of the sky** is this book’s destination. It notes in a parenthesis that $1''$ is “already large by lensing standards — most Einstein radii are $1-2''$ ” and moves on, ten chapters before an Einstein radius exists. You now have $\theta_E = 1.145''$ for a $\sigma_v = 250$ km/s elliptical at $z_{\text{lens}} = 0.5$, $z_{\text{source}} = 2.0$, and the arithmetic that makes that scale the whole game. Its opening — “Nothing astronomical comes with a ruler” — is where Ch. 9 of this book stops being a problem statement and becomes a four-rung ladder.
- **Ch. 16 — How much light bends, and the factor of two** proves that general relativity’s deflection is exactly twice Newton’s, and assumes throughout that you know why anyone wants a deflection angle. You do: Ch. 15 gave you the eclipse that settled the factor, and this chapter gave you the geometry that makes its central approximation reasonable on sight — a galaxy’s own hundred-thousand-light-year depth is nothing against the gap between $z = 0.5$ and $z = 2.0$, exactly the condition Ch. 16 marks as “where ‘thin’ actually enters”.
- **Ch. 17 — The lens equation** states that “three words cover almost every strong lens actually observed: a **double**, a **ring**, or a **quad**”, and derives all three as root counts of $\beta = \theta - \alpha(\theta)$ without ever saying what one looks like. You now have them as objects. And Ch. 17’s β — “where the source would be if nothing were in the way” — is a real galaxy at $z = 2$, ten billion years back, with no idea it is being modelled.

Exercises

Exercise 16.1 — rebuild the rarity

Start from $\theta_E = 1.145''$. Convert to degrees, compute the area of a disc of that radius in square degrees, divide by the $41,253 \text{ deg}^2$ of the whole sky, and invert.

Solution

$1.145/3600 = 3.181 \times 10^{-4}$ degrees; disc area $\pi (3.181 \times 10^{-4})^2 = 3.18 \times 10^{-7} \text{ deg}^2$; divided by the sky, 7.71×10^{-12} , or one in 1.30×10^{11} . Note what did no work: no cosmology, no dark matter, no general relativity. Once θ_E is a number, the rarity is geometry — a small disc against a sphere.

Exercise 16.2 — which one is the lens?

Two galaxies appear at essentially the same position on the sky; spectra give redshifts 0.5 and 2.0. Which is the deflector, and how do you know? Is there anything in the *image alone* that tells you the same thing?

Solution

The $z = 0.5$ galaxy is the lens; the $z = 2.0$ one is the source. Bigger z means the light left when the universe was smaller, so longer ago, so farther away (Ch. 8) — and light from behind is what gets bent by what is in front. Nothing about their *positions* decides it: position is an angle and both are at the same angle, which is the entire point of a sightline. From the image alone you can guess — the deflector is the big red smooth one at the centre (Ch. 3), the source the blue stretched sliver around it. Guessing is what a lens finder does, which is why a candidate stays a candidate until somebody takes a spectrum.

Exercise 16.3 — what the number does not say

A colleague reads “one in 1.30×10^{11} ”, multiplies by the 5.38×10^7 galaxies DESI scored, and concludes the survey should have found no lenses at all. It found a few thousand. Where is the reasoning wrong?

Solution

Two errors, pushing the same way. First, the alignment fraction is the fraction of the *sky* one lens’s disc covers — the chance a randomly thrown direction lands in it. Multiplying it by a count of *galaxies* multiplies a solid-angle fraction by a population; the two have nothing to divide out, and the product is a rate of nothing. Second, the correct calculation asks how many *background sources* are available to fall into each foreground disc, and the sky behind a lens is not empty — it is crowded with faint high-redshift galaxies. The real yield is a tiny per-pair probability times a very large number of pairs, and this book does not compute it. What the 7.71×10^{-12} is good for is intuition about the *target*: the alignment tolerance is absurdly tight. It says nothing about how often nature has hit it.

Exercise 16.4 — where do you open it?

For each question, name the chapter of the main guide you would open and the chapter of *this* book that prepared you for it.

1. What does `pixscale=0.262` in a line of survey code physically assert?
2. Why does the repository’s lens-modeling likelihood contain no cosmology, even though its two galaxies are at wildly different redshifts?
3. Is the human grade on a lens candidate worth training against?
4. What is $\gamma = 1.103 \pm 0.008$ a measurement *of*, and should you believe it?

Solution

1. [Ch. 9](#) — pixel scale in arcsec/pixel. Prepared by [Ch. 6](#) (why the ground/space split is a resolution argument) and [Ch. 1](#) (why an angle is the only thing a telescope measures).
2. [Ch. 17](#) — every distance factor folds once into the scaled deflection, after which the problem is purely angular. Prepared by [Ch. 8](#) (what the redshifts mean) and this chapter's geometry (why there are exactly three planes to carry distances between).
3. [Ch. 28](#) — the most striking result in the repository. Prepared by nothing in this book, deliberately: it needs no astronomy. Read it second, as [Ch. 1](#) says.
4. [Ch. 20](#) for what it measures — the exponent in $\rho \sim r^{-\gamma}$ — and [Ch. 25](#) for whether to believe it. Prepared by [Ch. 12](#) (most of that mass is dark, so the profile is not the starlight's) and this chapter's third section (why an arc constrains a slope, and what the mass-sheet degeneracy costs when it does). Do not skip to the verdict: [Ch. 1](#) asked you to write down a guess first, and that exercise works once.